自抗扰控制的思想与基本结构

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概要

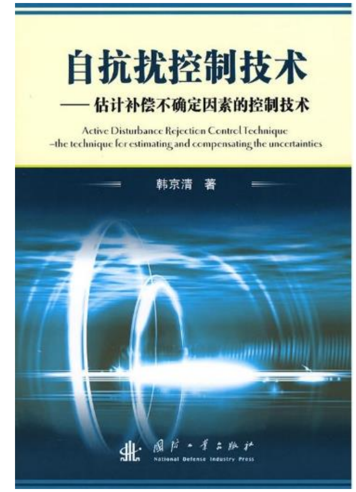
- 自抗扰控制的思想
- 自抗扰控制的非线性设计方法
- 自抗扰控制的线性设计方法
- 自抗扰控制的一些应用实例

自抗扰控制的提出者

韩京清：1937-2008

- 1960s：线性最速控制理论中的“等时区”方法
- 1970s：最优控制用于拦截问题中的导引律设计
- 1980s：线性系统理论的构造性方法、中国控制系统计算机辅助设计工程化软件系统(CADCSC)
- 1995-2008：自抗扰控制

韩京清：1937-2008



- 宋健，韩京清. 线性最速控制系统的分析与综合理论. 数学进展, 1962, 5(4): 264-282.
- 韩京清. 拦截问题中的导引律. 北京: 国防工业出版社, 1977.
- 韩京清. 控制理论-模型论还是控制论. 系统科学与数学, 1989, 9(4): 328-335.
- 韩京清. 一类不确定对象的扩张状态观测器. 控制与决策, 1995, 1: 85-88.
- 韩京清. 自抗扰控制器及其应用. 控制与决策, 1998, 1: 18-23.
- 韩京清. 自抗扰控制技术. 北京: 国防工业出版社, 2008.
- J. Han. From PID to Active Disturbance Rejection Control. IEEE Trans. Ind. Electron. , 2009, 56(3): 900-906.

控制设计中的不变性原理(invariant principle), 1950s-60s

扰动调节

——扰动补償和不变性——

〔苏联〕Г. М. 烏兰諾夫 著

胡保生 譯 徐俊荣 校

在以后，所謂不变性，我們將理解为扰动补償，即所研究的自动調节系統（包括測量系統和仪表）能达到与作用在其上的扰动完全无关或部分无关。

扰动調节原理发展的历史道路是比較長的，它的产生还在上世紀中，毫无疑問是自动調节中的最古老原理。在我們这一世紀前的几千年，由阿拉伯人創造的第一批风力磨的控制装置，已經包含了按負載来控制的元件。在这种风力磨中，磨子的轉速是按外部負載力矩的作用来改变风帆而进行調节的。

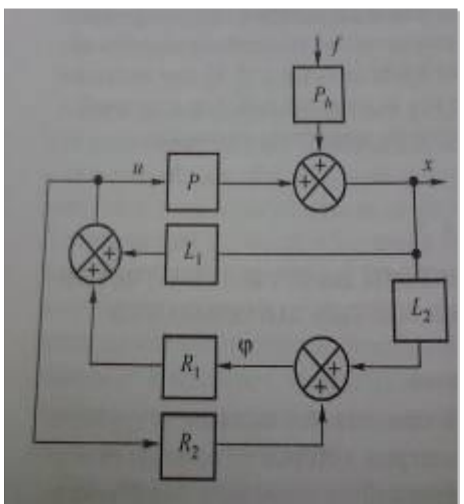
扰动控制和調节的現代景况，往往与法国学者龐雪莱[23]和俄国学者契柯列夫的名字有关，他們給出了按被控系統外部負載作用而工作的調节器的描述和技术設計。历史上之所以注意龐雪莱装置（1830 年）和契柯列夫装置（1847 年）是由于按偏差原則构成的調节器（瓦特調节器）在应用方面沒有成效（在 19 世紀中叶），这些調节器一时还不很完善。在此时，出現了在原則上不同于瓦特調节器，并直接根据所測負載变化来进行調节的新型調节器，这对整个自动控制和自动調节产生了重要的影响。后来，理論証实了將按偏差原則（瓦特原理）的調节与按負載的調节原則（龐雪莱-契柯列夫原理）相結合，在許多情况下是最好的解决办法。

Grigoriy Shipanov (1903-1953)


ОТДЕЛ I.
Оригинальные исследования.

Расчет авиационных уклономеров маятникового типа

Г. В. Щипанов.



- Soviet Engineer, Professor.
- **Problem of automatic control: making output "invariant" in the presence of disturbance.**
- 1938, dissertation, not understood by the university committee members: 6 voted "No," 4 voted "Yes," 20 didn't vote.
- The same committee awarded Shipanov the title of Professor.
- 1938, hired by Kulebyakin, head of the Institute of Automatics and Telemechanics, finished his research there.
- 1939, a short publication of his completed research provoked heated debate on the ideological basis.
- 1941, publicly accused of creating a "fantastic" controller by mathematical speculations instead of meeting demands of a growing Soviet economy.

ENGINEERING CYBERNETICS

H. S. TSIEN

Daniel and Florence Guggenheim Jet Propulsion Center
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内外扰动的补偿问题

- **ataraxia (disturbance free)**

15.6 Design for Stable Operation. Stability of any control system means that the design performance of the system will be obtained even with the presence of internal and external disturbances. We have seen how this requirement is satisfied in the case of conventional servomechanisms and other more general control systems in the preceding chapters. For optimalizing control systems, the essential part of the operation is the proper coordination of the input drive with the output behavior, so that the output stays within a close neighborhood of the optimum. This operation must not be influenced by internal and external disturbances. When this is achieved by a good design of the system, we have stable operation.

Engineering Cybernetics, H.S. Tsien, 1954

最优控制中的扰动补偿问题

Accommodation of disturbances in optimal control problems†

C. D. JOHNSON

University of Alabama in Huntsville, Huntsville, Alabama 35807

[Received in revised form 19 May 1971]

Modern theories of automatic control have virtually ignored one of the oldest and most fundamental problems of control—the problem of controlling systems subjected to persistently acting, unknown external disturbances.

In this paper the problem of control in the face of unknown disturbances is studied from the optimization point of view. It is first shown that a straightforward approach to the problem leads to control laws that are hopelessly unrealizable. A novel alternative approach is then proposed which leads to physically realizable control laws which perform optimally in the face of a broad class of unknown external disturbances. The proposed method is illustrated by posing and solving a modified version of the classical linear–quadratic regulator problem, in which persistently acting disturbances have been added.

伺服系统中外扰补偿问题

Accommodation of External Disturbances in Linear Regulator and Servomechanism Problems

C. D. JOHNSON, MEMBER, IEEE

III. AN EFFECTIVE MATHEMATICAL MODEL OF DISTURBANCES

Disturbances are by definition plant inputs $w(t) = (w_1(t), \cdots, w_p(t))$ which cannot be manipulated by the designer and are not completely known beforehand. Various schemes have been proposed for mathematically modeling partially unknown functions $w(t)$ of this type. The traditional approach consists of using classical Fourier series methods on representative experimental recordings of $w(t)$ to estimate the discrete harmonic content of the disturbance. Another method consists of treating the disturbance as a random process and experimentally computing the mean, covariance, power spectral density, and other statistical properties of $w(t)$. This method is frequently used to model disturbances $w(t)$ as a continuous fusillade of various hues of colored noise. A less common, and conceptually different, way to model a partially unknown function $w(t)$ is to give a differential equation which $w(t)$ is known to satisfy. This latter technique, called disturbance "state" modeling, is based on the idea of characterizing the possible waveform modes of disturbances and will be used exclusively in this paper. We will show that this ap-

$$w(t) = H(t)z; \qquad w = (w_1, \cdots, w_p)$$

$$\dot{z} = D(t)z + \sigma(t); \qquad z = (z_1, \cdots, z_\rho),$$

two parts:

$$\boxed{u = u_c + u_R} \tag{4}$$

where the component u_c is assigned the task of counteracting the disturbance $w(t)$ and the component u_R is responsible for regulating the state $x(t)$. In particular, u_R should choose the counteraction component of $u(t)$ as

$$u_c(t) = -\Gamma(t)\hat{w}(t) \tag{8}$$

and the regulating component of $u(t)$ as

$$u_R(t) = K(t)\hat{x} \tag{9}$$

where $\hat{w}(t)$, $\hat{x}(t)$ are accurate estimates of $w(t)$, $x(t)$ which we will obtain from $y(t)$ by an on-line real-time state reconstructor device, and $K(t)x(t)$ is the control which would

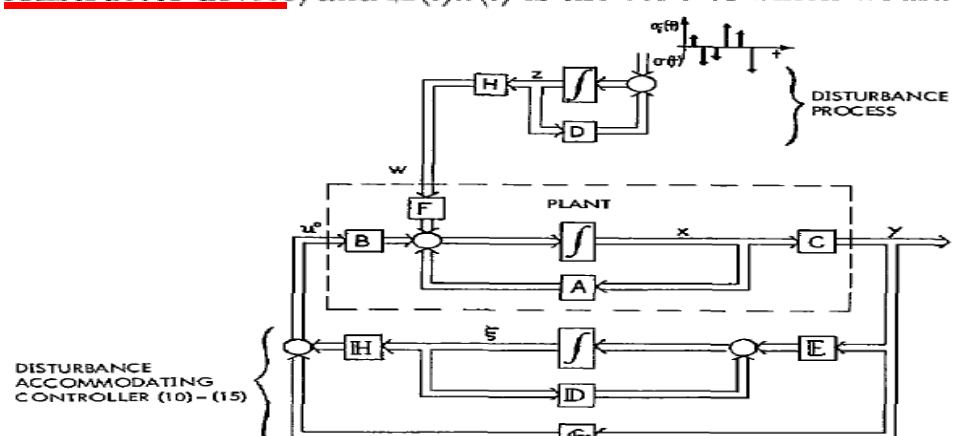


Fig. 1. Optimal linear regulator with disturbance accommodating controller (counteraction mode).

具有未知输入系统的观测器

Observers for systems with unknown and inaccessible inputs†

J. S. MEDITCH‡ and G. H. HOSTETTER§

Observer theory for constant-coefficient linear systems is extended to systems with unknown inaccessible inputs. Two types of observers for such systems are defined, and simplified criteria for their existence are developed. A theorem regarding the structure of these observers is given which allows the application of results for systems with single unknown inputs to systems with multiple unknown inputs.

Observer theory (Luenberger 1964, 1966, 1971, Anderson and Moore 1971) generally requires that all inputs to the system whose state is being estimated be known. In practice, there exist many situations in which the system inputs are unknown (even in a statistical sense) and inaccessible (i.e. they cannot be measured). These typically take the form of internal and/or external disturbances in the plant, and unknown errors in the output measurements.

vable. Davison (1970, 1972) and Johnson (1968, 1970) have obtained some results for feedback controllers in this case, but not from the standpoint of observer theory. The relation of their results to observer theory has been commented on elsewhere (Hostetter and Meditch 1973 a).

系 统 科 学 与 数 学
J. Sys. Sci. & Math. Scis.
9 (4) (1989), 328—335

控制理论——模型论还是控制论[1)]

韩 京 清

(中国科学院系统科学研究所)

Journal of System Sciences and Mathematical Sciences, **9**(4) (1989), 328-325

Control Theory: the Doctrine of Model or the Doctrine of Control?[1]

Jing Qing Han

(Institute of System Sciences, Chinese Academy of Sciences)

(Translated by ZhiqiangGao, Cleveland State University, Feb. 19th, 2013)

控制理论：控制论还是模型论?

在导引理论中，控制律是依靠对象实时状态的某些信息来建立的．她不大去考虑对象数学模型的细节，不把系统分成线性的和非线性的而采用不同的研究方法．根据视线信息有“三点法”，根据位置信息有“平行接近法”，根据视线和视线角速信息有“比例导引”，根据位置和速度信息有“前置量法”、“预测导引法”等．在作理论分析时，常常采用最简单的线性控制系统模型：

$$\begin{cases}\dot{x}=v,\\ \dot{v}=a(t)+u(t),\end{cases} \tag{2.1}$$

其中，$u(t)$ 是控制量，$a(t)$ 是所谓的“固有加速度”，是许多非线性效应的随时变化量．在这里，通常不把 $a(t)$ 写成状态变量的非线性函数．有时为了提高精度，对 $a(t)$ 作出适当估计是必要的，但不一定要知道 a 与状态变量之间非线性函数关系本身．在这里，并不大关心非线性对象的全局形态，关心的焦点是如何根据一个过程提供的实时信息来搞好这个过程的控制问题．这是导引理论提供给我们的主要思想方法．

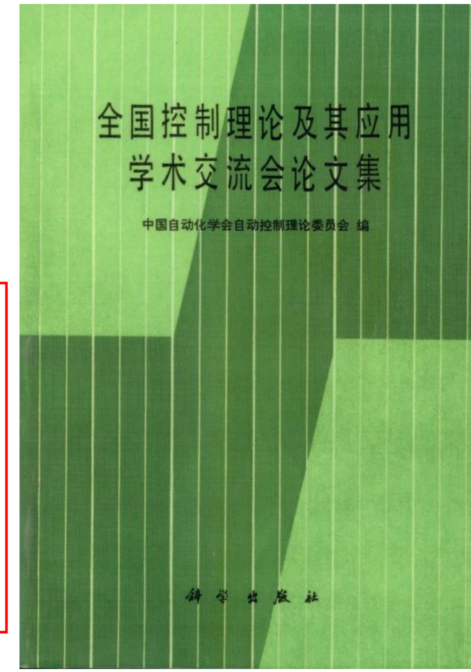
根据系统对信号的某些响应特征或过程的某些实时信息来确定控制好一个过程的控制律，是与靠系统的数学模型去找控制律的方法完全不同的思考方式．这种思考方式，相对于模型论方法，可以称为控制理论中的“控制论”方法．

线性和非线性系统的标准型(Canonical form)

线性系统的结构与反馈系统计算

韩 京 清

（科 学 院 数 学 所）



自抗扰控制源头：韩京清，第1届中国控制会议，1979年

The systems, linear or nonlinear, under some conditions, can be transformed into the canonical form of cascade integrators via feedback.

-Han J. The structure of linear system and computation of feedback system, in: Proc. National Conference on Control Theory and Its Application, Academic Press, 1980

对应于这个阵的系统叫做积分器串联系统．这是系统中最简单的系统．任意完全能控系统经适当坐标变换和状态反馈均可化成这种形式．由于反馈不改变系统的能控性，因此研究与系统的能控性有关问题，都可用对应的积分器串联形式来讨论．这种现象不仅对线性系统存在，而且在一类非线性控制系统中也成立．

积分串联型：线性和非线性系统的标准型

The systems, linear or nonlinear, under some conditions can be transformed into the canonical form of cascade integrators via feedback.

$$\boxed{\begin{aligned}\dot{x} &= Ax + Bu,\\ y &= Cx\end{aligned}}$$

$$A = \begin{bmatrix} 0 & 1 & \cdots & 0 & 0 \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ & & & & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

积分串联型系统：线性/非线性系统的标准型

积分串联型的获得：通过微分几何精确线性化系统

1980s-1990s: 非线性系统的理论分析是热点

- **R. W. Brockett, Feedback invariants for nonlinear systems, 1978 IFAC.**
- **B. Jakubczyk and W.Respondek, On linearization of control systems, 1980.**
- **B. Charlet, J.Levine, On dynamic feedback linearization, 1989.**

differential geometry

- **M. Fliss, Generalized controller canonical forms for linear and nonlinear dynamics, 1990.**

differential algebra

通过微分几何的工具精确线性化系统

Isidori A., Nonlinear Control System, Springer-Verlag

积分串联型的获得：通过微分几何精确线性化系统

当假设系统的非线性函数光滑并且精确已知时：

SISO nonlinear system $\dot{x} = f(x) + b(x)u,\ x \in R^n$

$$y = h(x)$$

如果系统相对阶为n, 则存在一个等价变换 $z = T(x)$

以及一个状态反馈控制律 $u = \alpha(x) + \beta(x)v$

$$\alpha(x) = -\frac{L_f^n h(x)}{L_b L_f^{n-1} h(x)},\ \beta(x) = \frac{1}{L_b L_f^{n-1} h(x)}$$

使得系统变化为如下纯积分串联型系统

$$\boxed{\begin{aligned}\dot{z} &= Az + Bv,\\ y &= Cz\end{aligned}}$$

$$A = \begin{bmatrix} 0 & 1 & \cdots & 0 & 0 \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ & & & & 0 \end{bmatrix},\ B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix},\ C = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

问题：系统具有不确定性时，如何使系统变为纯积分串联型?

扩张状态观测器：估计纯积分串联型之外的总扰动

The Extended State Observer for A Class of Uncertain Systems, Control and Decision, 1995, 10(1): 85-88.

设有受到未知外扰作用的非线不确定对象

$$\begin{cases} \dot{x}_1 = x_2 \\ \cdots \\ \dot{x}_{n-1} = x_n \\ \dot{x}_n = f(x_1, x_1, ..., x_n, w(t), t) + bu \end{cases}$$

其中 $f(x_1, x_1, ..., x_n, w(t), t)$**为未知函数，**$w(t)$ **为未知干扰。若** $x_1(t)$**为量测量，那么能否构造出不依赖于** $f(\cdot)$ **的非线性系统，使得它能由量测量** $x_1(t)$ **估计出被扩张的系统状态变量呢?**

扩张状态观测器：估计纯积分串联型之外的总扰动

设有受到未知外扰作用的非线性不确定对象

$$\begin{cases} \dot{x}_1 = x_2 \\ \cdots \\ \dot{x}_{n-1} = x_n \\ \dot{x}_n = f(x_1, x_1, \ldots, x_n, w(t), t) + bu \end{cases}$$

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状态空间的形式

$$\begin{cases} \dot{X} = AX + B\left(f(\bullet) + u\right), \\ y = CX \end{cases}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}, A = \begin{bmatrix} 0 & 1 & \cdots & 0 & 0 \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ & & & & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

扩张状态观测器：估计纯积分串联型之外的总扰动

韩京清. 一类不确定对象的扩张状态观测器. 控制与决策, 1995, 1: 85-88.

扩张状态观测器

Extended State Observer (ESO):

$$\begin{cases} \dot{\hat{x}}_1 = \hat{x}_2 - G_1(\hat{x}_1(t) - y(t)) \\ \cdots \\ \dot{\hat{x}}_n = \hat{x}_{n+1} - G_n(\hat{x}_1(t) - y(t)) + bu \\ \dot{\hat{x}}_{n+1} = -G_{n+1}(\hat{x}_1(t) - y(t)) \end{cases} \tag{2}$$

$$\hat{x}_1(t) \to x_1(t), \ldots,\ \hat{x}_n(t) \to x_n(t),\ \ \hat{x}_{n+1}(t) \to f(x(t), w(t), t) \tag{3}$$

$G_1(\hat{X}), \ldots, G_n(\hat{X}), G_{n+1}(\hat{X})$ ⟹ **待设计的非线性函数**

$f(x(t), w(t), t)$ ⟹ **扩张状态：总扰动**

例子：运动控制系统

- **对象** $\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2, w(t), t) + bu \end{cases}$

  $y = x_1$

- **标准型** $\ddot{y} = bu$

- **总扰动估计值** $\hat{f}(t) \approx f(x_1, x_2, w(t), t)$

总扰动的估计：扩张状态观测器

$$x_3 = f(y, \dot{y}, w, t)$$

$$\ddot{y} = f(x_1, x_2, w(t), t) + bu \quad \Longrightarrow \quad \begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 + bu, \\ \dot{x}_3 = \dot{f} \\ y = x_1 \end{cases}$$

Extended State Observer (ESO)

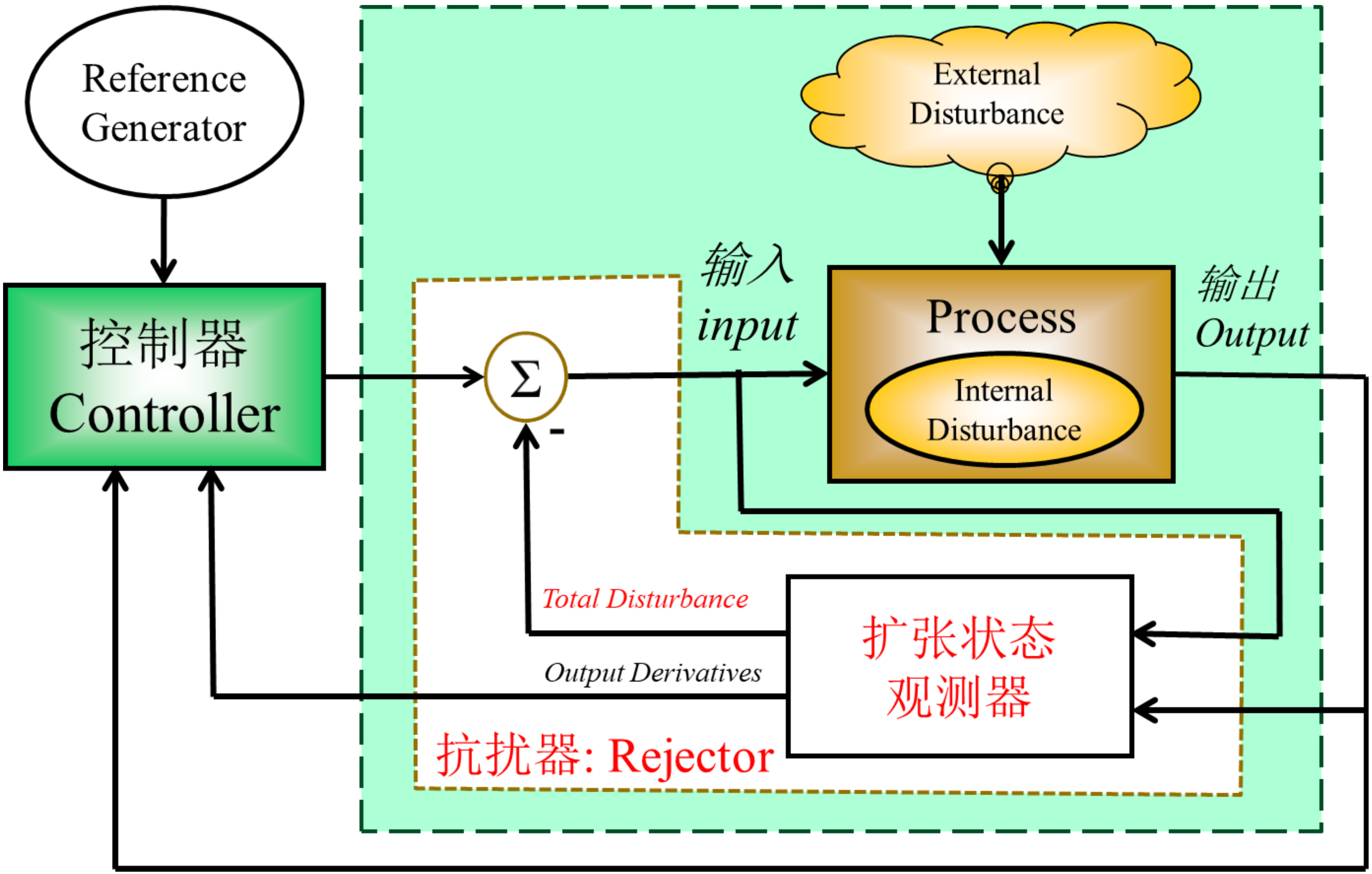
$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - \beta_1 G_1(\hat{X}_1 - y) \\ \dot{\hat{X}}_2 = \hat{X}_3 - \beta_2 G_2(\hat{X}_1 - y) + bu \\ \dot{\hat{X}}_3 = -\beta_3 G_3(\hat{X}_1 - y) \end{cases}$$

$$\hat{X}_1 \approx x_1 \quad \hat{X}_2 \approx x_2 \quad \hat{X}_3 \approx x_3 = f$$

自抗扰设计

$$\ddot{y} = f(x_1, x_2, w(t), t) + bu \xrightarrow{u = u_0 - \frac{\hat{f}}{b},\ \hat{f} \approx f} \ddot{y} \approx bu_0$$

自抗扰控制的核心部分：扩张状态观测器



Motion Control Test, 1997, Cleveland State University

自抗扰控制的思想与基本结构

- 自抗扰控制的思想
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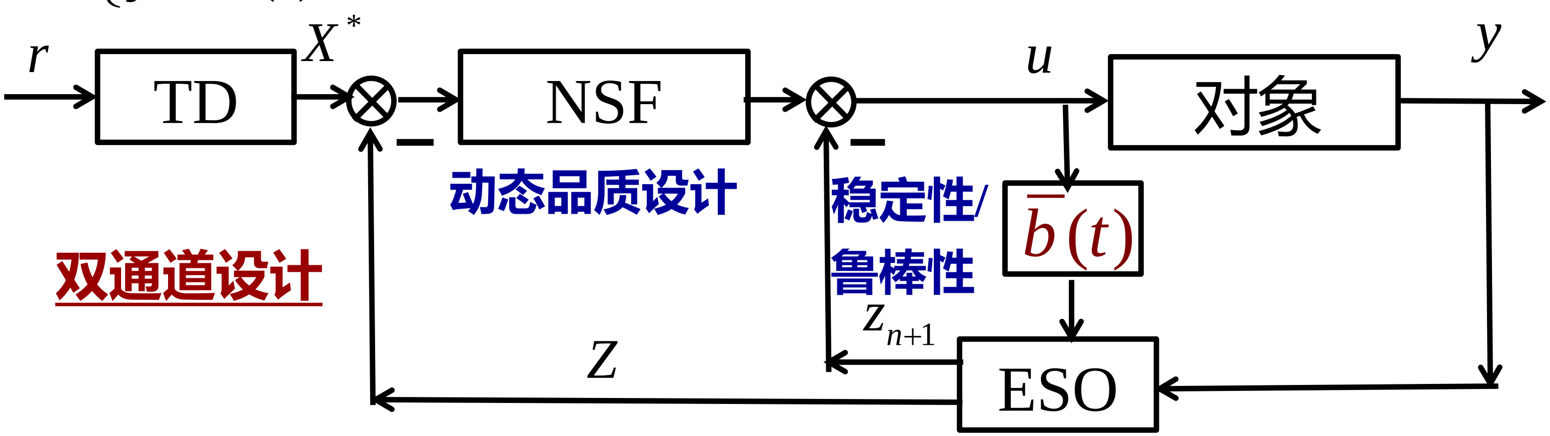
- 自抗扰控制的线性设计方法
- 自抗扰控制的一些应用实例

自抗扰控制 (ADRC)的早期设计框架

- (1) Tracking Differentiator (TD) : 跟踪微分器
- (2) Nonlinear State Feedback (NSF): 非线性状态反馈
- (3) Extended State Observer (ESO)：扩张状态观测器

$$\begin{cases} x^{(n)}(t) = f(X,t) + d(t) + b(X,t)u(t) \\ y = x(t) \end{cases}, X = \left[x, \cdots x^{(n-1)}\right]^T$$



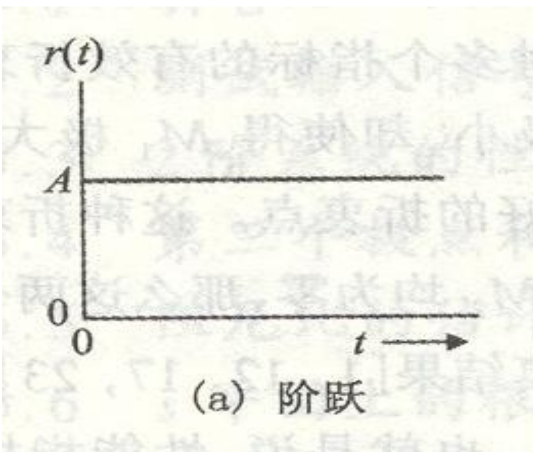
跟踪控制

$$u = \frac{-\varphi(Z - X^*) - z_{n+1}}{\bar{b}(t)}, \quad Z = \left[\hat{X}_1, \cdots \hat{X}_n\right]^T, \quad X^* = \left[x^*, \cdots x^{(n)*}\right]$$

自动控制原理：系统对阶跃信号的响应

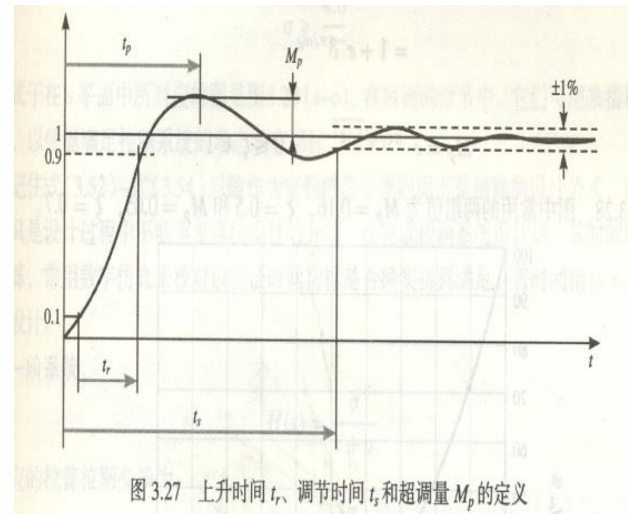
$$Y(s)=\frac{a_2}{s^2+a_1 s+a_2}R(s),\ a_1>0, a_2>0$$

稳定的系统！



线性定常系统

$Y(s) \rightarrow G(s) \rightarrow R(s)$



控制输入为阶跃信号幅值时，

容易使得系统输入瞬态等达到饱和，控制性能变差！

ADRC第1部分：跟踪微分器(Tracking Differentiator, TD)：安排过渡过程

TD的形式：

$$
\begin{cases}
\dot{z}_1 = z_2 \\
\cdots \\
\dot{z}_{n-1} = z_n \\
\dot{z}_n = r^n h\left(z_1 - y(t), \dfrac{z_2}{r}, \ldots, \dfrac{z_n}{r^{n-1}}\right)
\end{cases}
$$

调节的参数

被跟踪的信号

- 输入信号的约束条件:

$y(t)$ 在任意有界区间 $t \in [t_0, t_0 + L], 0 < t_0 < t_0 + L < \infty$ 上有界并可积。

- 输出信号的特性:

$$\lim_{r \to \infty} \int_{t_0}^{t_0+L} |z_1(t) - y(t)| dt = 0$$

当r足够大时，$z_1(t)$可以充分接近于$y(t), t \in [t_0, t_0 + L]$

$z_i(t), i \in \underline{n}$ 视为$y^{(i-1)}(t), t \in [t_0, t_0 + L]$的估计值. 当$y^{(i-2)}(t)$不可导时，$y^{(i-1)}(t)$为其广义导

- 韩京清, 王伟. 非线性跟踪 - 微分器. 系统科学与数学, 1994, 14(2): 177-183.
- XUE Wenchao, HUANG Yi, YANG Xiaoxia, What Kinds of System Can Be Used as Tracking-Differentiator, 2009 CCC, 2009.

ADRC第1部分：跟踪微分器(Tracking Differentiator, TD)：安排过渡过程

TD：
$$\begin{cases} \dot{z}_1 = z_2 \\ \cdots \\ \dot{z}_{n-1} = z_n \\ \dot{z}_n = r^n h(z_1 - y(t), \frac{z_2}{r}, \ldots, \frac{z_n}{r^{n-1}}) \end{cases}$$

原始的系统：
$$\begin{cases} \dot{x}_1 = x_2 \\ \cdots \\ \dot{x}_{n-1} = x_n \\ \dot{x}_n = h(x_1, x_2, \ldots, x_n) \end{cases}$$

被跟踪的信号：假设条件

- 设输入信号$y_1(t) \triangleq y(t)$在$[t_0, t_0+L]$上只含有第一类不连续点$\{t_i\}_{i=1}^{q_L}$，且满足:
$$\sup_i |y_1(t_i^-) - y_1(t_i^+)| \le W_1 < \infty.$$
- 在$t \in [t, \infty) \backslash \{t_i\}_{i=1}^{q_L}$时$y_2(t) \triangleq \dot{y}_1(t)$存在, 并且$y_2(t)$ 在间断点$\{t_i\}_{i=1}^{q_L}$的左右极限$y_2(t_i^-), y_2(t_i^+)$存在以及 $\sup\limits_{t \in [t_0, t_0+L]} \{|y_2(t^+)|, |y_2(t^-)|\} \le W_2 < \infty.$

XUE Wenchao, HUANG Yi, YANG Xiaoxia, What Kinds of System Can Be Used as Tracking-Differentiator, 2009 CCC, 2009.

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$$\begin{cases} \dot{x}_1 = x_2 \\ \cdots \\ \dot{x}_{n-1} = x_n \\ \dot{x}_n = h(x_1, x_2, \ldots, x_n) \end{cases}$$

定理：设原始系统原点在 $(0,\infty)\times \mathscr{B}(0,\rho), \rho > w_1$ 上一致渐近稳定，则当 $[z_1(t_0)-y(t_0), z_2(t_0),\ldots, z_n(t_0)]^T \in \mathscr{B}(0,\rho)$ 时，TD满足：

$$\lim_{r\to\infty}\int_{t_0}^{t_0+L} |z_1(t)-y(t)|dt = 0.$$

XUE Wenchao, HUANG Yi, YANG Xiaoxia, What Kinds of System Can Be Used as Tracking-Differentiator, 2009 CCC, 2009.

最速控制形式的TD

对于二阶线性定常系统：

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = u, |u| \le r \end{cases}$$

以原点为终点的快速最优控制综合函数为：

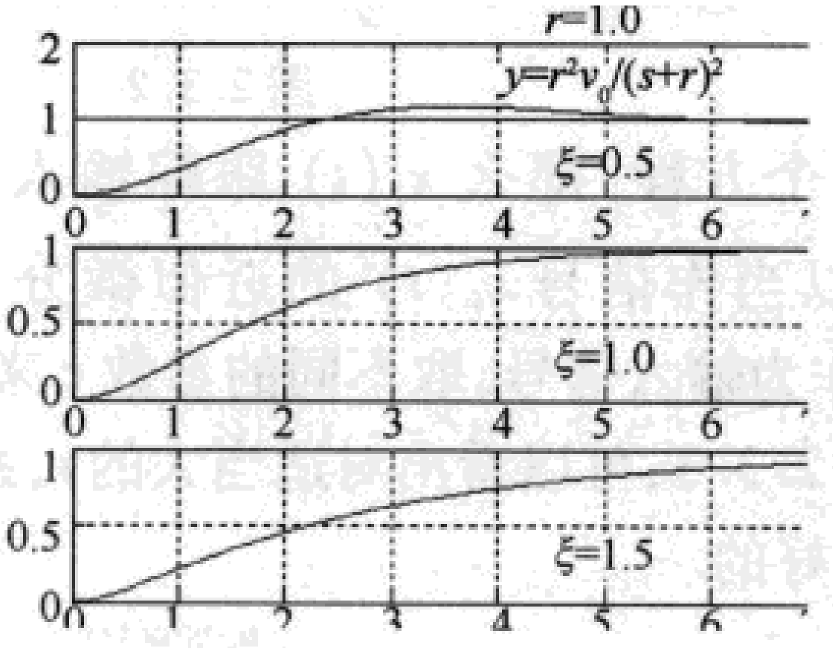
$$u(x_1, x_2) = -r\ sign(x_1 + \frac{x_2 \,|\, x_2 \,|}{2r})$$

快速最优控制对应的TD为：

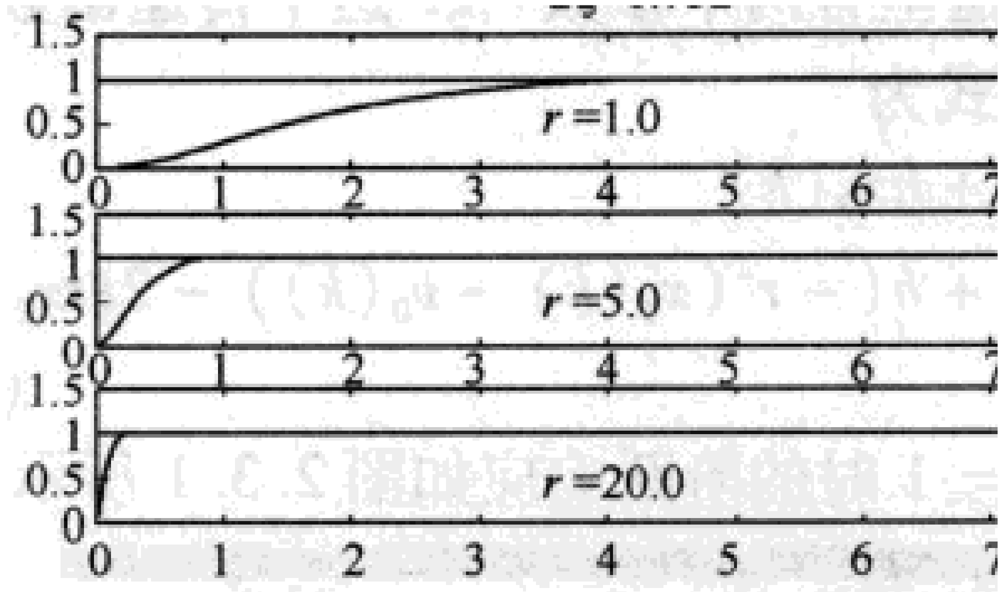
$$\begin{cases} \dot{z}_1 = z_2 \\ \dot{z}_2 = -r\ sign(z_1 - y(t) + \frac{z_2 \,|\, z_2 \,|}{2r}) \end{cases}$$

这个系统的解的分量 $z_1(t)$ 在加速度限制 $|\ddot{z}_1| \le r$ 之下，将最快地跟踪输入信号 $y(t)$，而且 r 越大，跟踪得越快。显然，当 $z_1(t)$ 充分地接近 $y(t)$ 时，可以把解的另一分量 $z_2(t) = \dot{z}_1(t)$ 当作输入信号 $y(t)$ 的近似微分。

二阶惯性环节安排
过渡过程



TD
安排过渡过程



韩京清，自抗扰控制技术，国防工业出版社，2008.

ADRC第2部分：非光滑状态反馈

先考察一个简单例子：设有一阶受控对象

$$\dot{x} = w(x,t) + u$$

对它施以状态的线性反馈

$$u = -kx, k > 0.$$

得闭环系统

$$\dot{x} = -kx + w(x,t)$$

今取上述状态反馈,使其状态变量x在扰动$w(x,t)$的作用之下，尽可能地收敛到零，这里的k称作**反馈增益**。

进一步假定：存在两常数m, w_0,满足 $|w(x,t)| < \left|\frac{x}{m}\right| + w_0, \quad \forall x \in R, \forall t > 0$

线性反馈

$$\dot{x} = -kx + w(x,t) \qquad (2.1)$$

$$|w(x,t)| < |\frac{x}{m}| + w_0, \forall x, t > 0$$

对于系统(2.1)，有

(1)　若 $w_0 = 0$, **而取反馈增益** $k > \frac{1}{m}$, **那么对所有** x **都有** $\frac{dx^2}{dt} < 0$, **系统(2.1)渐近稳定，所有解都趋近于0.**

线性反馈

$$\dot{x} = -kx + w(x,t) \qquad (2.1)$$

$$|w(x,t)| < |\frac{x}{m}| + w_0, \forall x, t > 0$$

对于系统(2.1)，有

(1)　若 $w_0 = 0,$ **而取反馈增益** $k > \frac{1}{m},$ **那么对所有** x **都有** $\frac{dx^2}{dt} < 0,$ **系统(2.1)渐近稳定，所有解都趋近于0.**

(2)　若 $w_0 \neq 0,$ **而取反馈增益** $k > \frac{1}{m},$ **那么当** $|x| > \frac{m}{m-1/k}\frac{w_0}{k}$ **时，就有** $\frac{dx^2}{dt} < 0,$ **因此，闭环系统(2.1)的解满足**

$$\lim_{t\to\infty} x(t) \in \left[-\frac{m}{m-1/k}\frac{w_0}{k}, \quad \frac{m}{m-1/k}\frac{w_0}{k}\right]$$

非光滑反馈

非光滑控制反馈： $u = -k|x|^{\alpha} sign(x), 0 < \alpha < 1$

闭环系统： $\dot{x} = -k|x|^{\alpha} sign(x) + w(x,t)$ (2.2)

进一步假定：存在两常数 m, w_0**,满足** $|w(x,t)| < \left|\frac{x}{m}\right|^{\alpha} + w_0, \forall x, t > 0$

非光滑反馈

非光滑控制反馈： $u=-k|x|^{\alpha}\ sign(x), 0<\alpha<1$

闭环系统： $\dot{x}=-k|x|^{\alpha}\ sign(x)+w(x,t) \qquad (2.2)$

进一步假定：存在两常数 m, w_0 **，满足** $|w(x,t)|<\left|\frac{x}{m}\right|^{\alpha}+w_0, \forall x, t>0$

对于系统(2.2)，可以得出

$$\frac{d|x|^2}{dt}=-2k|x|^{\alpha+1}\ xsign(x)+2xw(x,t)$$

$$=-2k|x|\left(|x|^{\alpha}-sign(x)\frac{w(x,t)}{k}\right)$$

$$<-2k|x|\left(|x|^{\alpha}-\frac{1}{k}\left(\left|\frac{x}{m}\right|^{\alpha}+w_0\right)\right)$$

非光滑反馈

非光滑控制反馈： $u = -k|x|^{\alpha}\ sign(x), 0 < \alpha < 1$

闭环系统： $\dot{x} = -k|x|^{\alpha}\ sign(x) + w(x,t)$ (2.2)

$$|w(x,t)| < \left|\frac{x}{m}\right|^{\alpha} + w_0, \forall x, t > 0$$

选取反馈增益 $k > 1/m^{\alpha}$. **进一步因为**

$$|x|^{\alpha} > \frac{m^{\alpha}}{m^{\alpha} - 1/k}\frac{w_0}{k} \Rightarrow \left(|x|^{\alpha} - \frac{1}{k}\left(\left|\frac{x}{m}\right|^{\alpha} + w_0\right)\right) > 0 \Rightarrow \frac{d|x|^2}{dt} < 0$$

所以(2.2)满足：

$$\lim_{t\to\infty} x(t) \in \left[-\left(\frac{m^{\alpha}}{m^{\alpha} - 1/k}\right)^{\frac{1}{\alpha}}\left(\frac{w_0}{k}\right)^{\frac{1}{\alpha}}, \left(\frac{m^{\alpha}}{m^{\alpha} - 1/k}\right)^{\frac{1}{\alpha}}\left(\frac{w_0}{k}\right)^{\frac{1}{\alpha}}\right].$$

线性反馈与非光滑反馈

在通常情形都有 $k > |w_0|$，因此只要

$$k > |w_0|, 0 \le \alpha < 1, (\frac{1}{\alpha} > 1),$$

就有

$$(\frac{w_0}{k})^{\frac{1}{\alpha}} \ll \frac{w_0}{k}$$

稳态误差远小于线性反馈留下的稳态误差。

$$u = -kx, k > 0.$$

从以上的讨论可以看出，在反馈系统中有意识的引入经合适的非线性结构，特别是非光滑结构，将显著性改善闭环系统的动态特性。因此开发利用非线性（特别是非光滑）特性是反馈系统设计的很有意义的新课题。

ADRC的核心部分：非线性ESO设计

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - G_1(\hat{X}_1(t) - y(t)) \\ \cdots \\ \dot{\hat{X}}_n = \hat{X}_{n+1} - G_n(\hat{X}_1(t) - y(t)) + bu \\ \dot{\hat{X}}_{n+1} = -G_{n+1}(\hat{X}_1(t) - y(t)) \end{cases} \tag{2}$$

此系统各状态分别跟踪被扩张的状态变量，即有

$$\hat{X}_1(t) \to y(t) = X_1(t), \ldots, \ \hat{X}_n(t) \to y^{(\mathrm{n}-1)}(t) = X_n(t), \ \hat{X}_{n+1}(t) \to y^{(\mathrm{n})}(t) - bu = X_{n+1}(t) \tag{3}$$

需要通过适当选取的非线性函数 $G_1(\hat{X}),\ldots,G_n(\hat{X}),G_{n+1}(\hat{X})$ **，实现上述跟踪目的。**

二阶系统的常用非线性ESO

对非线性系统

$$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = f(x_1, x_2, w(t), t) + bu, \\ y = x_1, \end{cases} \qquad (2.3)$$

我们把作用于开环系统的加速度 $f(x_1, x_2, w(t), t)$ 的实时作用量扩充成新的状态变量 x_3,

$$x_3(t) \triangleq f(x_1, x_2, w(t), t)$$

并记 $w_0(t) \triangleq \dot{x}_3(t)$, 那么系统(2.3)可扩张成新的线性控制系统：

$$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = x_3 + bu, \\ \dot{x}_3 = w_0(t), \\ y = x_1, \end{cases} \qquad (2.4)$$

二阶系统的常用非线性ESO

对上述被扩张的系统(2.4)建立状态观测器：

$$\begin{cases} e_1 = \hat{X}_1 - y, \\ \dot{\hat{X}}_1 = \hat{X}_2 - \beta_{01} e_1, \\ \dot{\hat{X}}_2 = \hat{X}_3 - \beta_{02} \left| e_1 \right|^{1/2} sign(e_1) + bu, \\ \dot{\hat{X}}_3 = -\beta_{03} \left| e_1 \right|^{1/4} sign(e_1), \end{cases} \tag{2.5}$$

$$\hat{X}_1(t) \to x_1(t),$$

$$\hat{X}_2(t) \to x_2(t),$$

$$\hat{X}_3(t) \to x_3(t) = f(x_1, x_2, w(t), t)$$

二阶系统的常用非线性ESO

对上述扩张状态观测器(2.5)来说，适用于如下三种类型系统

$$\begin{cases}\dot{x}_1 = x_2\\ \dot{x}_2 = f(t) + bu;\\ y = x_1\end{cases}\begin{cases}\dot{x}_1 = x_2\\ \dot{x}_2 = f(x_1, x_2) + bu;\\ y = x_1\end{cases}\begin{cases}\dot{x}_1 = x_2\\ \dot{x}_2 = f(x_1, x_2, w(t)) + bu.\\ y = x_1\end{cases}$$

估计误差方程：
$$\begin{cases}\dot{e}_1 = e_2 - \beta_{01}e_1, & e_1 = \hat{X}_1 - x_1\\ \dot{e}_2 = e_3 - \beta_{02}|e_1|^{1/2}sign(e_1), & e_2 = \hat{X}_2 - x_2\\ \dot{e}_3 = w_0 - \beta_{03}|e_1|^{1/4}sign(e_1), & e_3 = \hat{X}_3 - x_3\end{cases}$$

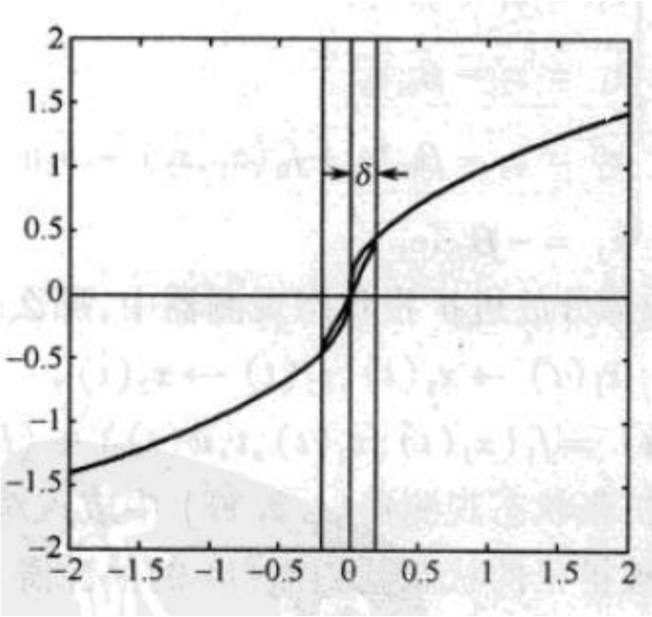
假设 $\lim\limits_{t\to\infty} w_0(t) = w^*$**，则系统进入稳态时，即方程右端全收敛于零时，得到稳态误差为：**

$$\lim_{t\to\infty}|e_1(t)| = \left(\frac{w^*}{\beta_{03}}\right)^4, \quad \lim_{t\to\infty}|e_2(t)| = \beta_{01}\left(\frac{w^*}{\beta_{03}}\right)^4, \quad \lim_{t\to\infty}|e_3(t)| = \beta_{02}\left(\frac{w^*}{\beta_{03}}\right)^2$$

二阶系统的常用非线性ESO

在进行数值仿真时，为了避免高频颤振现象的出现，我们把函数 $|e|^{\alpha}\ sign(e)$ 改造成原点附近具有线性段的连续的幂次函数

$$\mathrm{fal}(e,\alpha,\delta)=\begin{cases}\dfrac{e}{\delta^{n-1}}, & |e|\le\delta\\ |e|^{\alpha}\ sign(e), & |e|>\delta\end{cases}$$



式中：δ 为线性段的区间长度，如图4.3.3所示。

图4.3.3连续的幂次函数

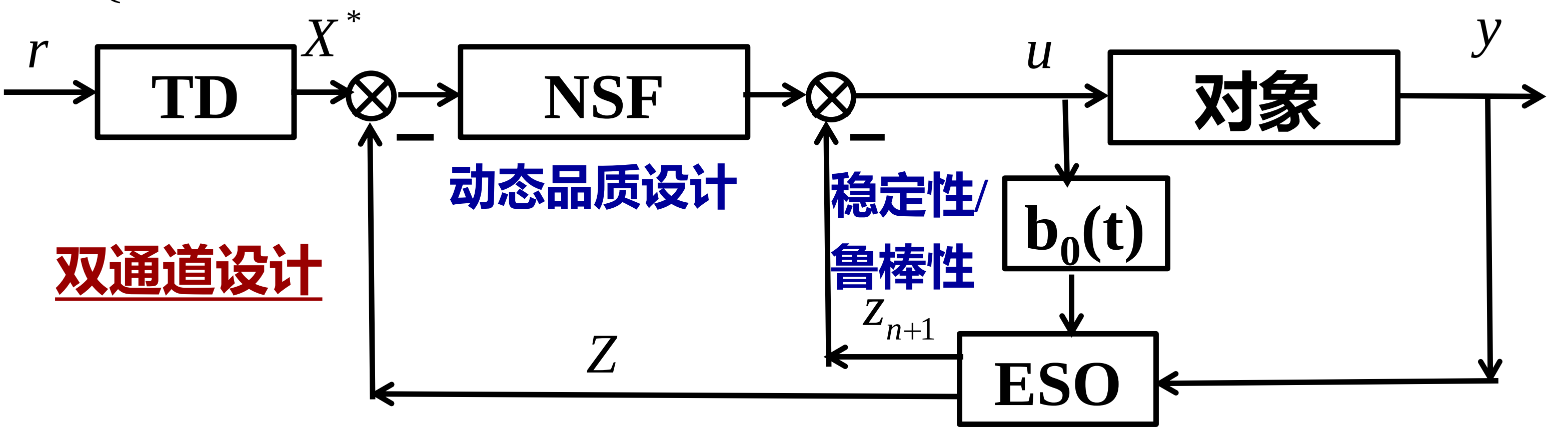
常用的非线性ESO设计

$$\begin{cases}\dot{z}_1=\hat{X}_1+h(\hat{X}_2-\beta_{01}e)\\ \dot{z}_2=\hat{X}_2+h(\hat{X}_3-\beta_{02}f_e)\\ \dot{z}_3=\hat{X}_3+h(-\beta_{03}f_{e1})\\ e=\hat{X}_1-y,\quad f_e=\mathrm{fal}(e,\dfrac{1}{2},\delta),\quad f_{e1}=\mathrm{fal}(e,\dfrac{1}{4},\delta)\end{cases}$$

自抗扰控制 (ADRC)的早期设计框架

非线性设计较多

$$\begin{cases} x^{(n)}(t) = f(X,t) + d(t) + b(X,t)u(t) \\ y = x(t) \end{cases}, X = \left[x, \cdots x^{(n-1)}\right]^T$$



双通道设计

跟踪控制

$$u = \frac{-\varphi(Z - X^*) - z_{n+1}}{\bar{b}(t)}, \quad Z = \left[\hat{X}_1, \cdots \hat{X}_n\right]^T, \quad X^* = \left[x^*, \cdots x^{(n)*}\right]$$

自抗扰控制的思想与基本结构

- 自抗扰控制的思想
- 自抗扰控制的非线性设计方法
- 自抗扰控制的线性设计方法
- 自抗扰控制的一些应用实例

线性ADRC

A parameterized linear ADRC (LADRC) was proposed in Gao Z. (2003, CDC)

使得控制器的参数具有物理意义，更加易于调节

非线性ADRC

ESO

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - G_1(\hat{X}_1 - Y) \\ \cdots \\ \dot{\hat{X}}_n = \hat{X}_{n+1} - G_n(\hat{X}_1 - Y) + \bar{B}(t)U(t) \\ \dot{\hat{X}}_{n+1} = -G_{n+1}(\hat{X}_1 - Y) \end{cases}$$

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} + U_0(Y, \hat{X}_2, \cdots \hat{X}_n)$$

线性ADRC

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - \beta_1(\hat{X}_1 - Y) \\ \cdots \\ \dot{\hat{X}}_n = \hat{X}_{n+1} - \beta_n(\hat{X}_1 - Y) + \bar{B}(t)U(t) \\ \dot{\hat{X}}_{n+1} = -\beta_{n+1}(\hat{X}_1 - Y) \end{cases}$$

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} - \bar{B}^{-1}(t)\sum_{i=1}^{n} k_i \hat{X}_i$$

闭环动态设计：极点配置

增益：极点配置

预期系统的响应

带宽参数化的线性ADRC

A parameterized linear ADRC (LADRC) was proposed in Gao Z. (2003, CDC)

使得控制器的参数具有物理意义，更加易于调节

NADRC

ESO

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - G_1(\hat{X}_1 - Y) \\ \cdots \\ \dot{\hat{X}}_n = \hat{X}_{n+1} - G_n(\hat{X}_1 - Y) + \bar{B}(t)U(t) \\ \dot{\hat{X}}_{n+1} = -G_{n+1}(\hat{X}_1 - Y) \end{cases}$$

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} + U_0(Y, \hat{X}_2, \cdots \hat{X}_n)$$

LADRC

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - \beta_1(\hat{X}_1 - Y) \\ \cdots \\ \dot{\hat{X}}_n = \hat{X}_{n+1} - \beta_n(\hat{X}_1 - Y) + \bar{B}(t)U(t) \\ \dot{\hat{X}}_{n+1} = -\beta_{n+1}(\hat{X}_1 - Y) \end{cases}$$

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} - \bar{B}^{-1}(t)\sum_{i=1}^{n} k_i \hat{X}_i$$

闭环动态设计：极点配置

$$s^{n+1} + \sum_{i=1}^{n+1} \beta_i s^{n-i+1} = (s + \omega_e)^{n+1}, \quad \omega_e > 0$$

LESO带宽

$$s^n + \sum_{i=1}^{n} k_i s^{i-1} = (s + \omega_c)^n, \quad \omega_c > 0$$

预期动态的带宽

ESO (1995)

$$\begin{cases} \dot{X}_1 = X_2 \\ \cdots \\ \dot{X}_{n-1} = X_n \\ \dot{X}_n = X_{n+1} + \bar{B}(t)U(t) \end{cases}$$

标准型 + 总扰动

$$\begin{aligned} &\hat{X}_1 \to X_1 = Y \\ &\cdots \\ &\hat{X}_n \to X_n = Y^{(n-1)} \\ &\hat{X}_{n+1} \to X_{n+1} \end{aligned}$$

ADRC (Han,1998)

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} + U_0(Y, \hat{X}_2, \cdots \hat{X}_n)$$

抵消总扰动的效应

对标称系统的控制

$$\begin{cases} \dot{X}_1 = X_2 \\ \cdots \\ \dot{X}_{n-1} = X_n \\ \dot{X}_n = \bar{B}(t)U_0(\bullet) \end{cases}$$

鲁棒性

预期的性能

双自由度设计

自抗扰控制的思想与基本结构

- 自抗扰控制的思想
- 自抗扰控制的非线性设计方法
- 自抗扰控制的线性设计方法
- 自抗扰控制方法的一些应用实例

ADRC in U.S.: Milestones

- **1997: made the 1st successful ADRC hardware test on a servo mechanism**
- **2008: $1M venture capital, grew by $5M in 2012.**
- **2010: 1st factory implementation, 10 Parker extrusion lines (挤压机生产线) at a Parker Hannifin Extrusion Plant in North America**

  (cpk: from 2.3 to >8; avg. 节能 57%)

- **2011: implemented ADRC in several high energy particle accelerators (高能粒子加速器) in the National Superconducting Cyclotron （超导回旋加速器） Lab in the U.S.**

- **2011: Texas Instrument adopts ADRC; 3 patents granted.**
- **2013: Texas Instrument released the ADRC based motion control chips (德州仪器的运动控制芯片)**

从学术到工业的跨越

ADRC as a powerful solution has been explored in many domain of control engineering:

- **电机控制** (speed control of induction motor and permanent-magnet synchronous, noncircular turning process, etc)
- **飞行控制** (attitude control of space ship, high-speed vehicles, UAV, morphing aircraft, etc)
- **机器人** (force control, uncalibrated hand-eye coordination, AUV, etc)
- **热工过程** (unstable heat conduction systems, boiler-turbine-generator systems, ALSTOM gasifier, fractional-order system, etc)
- **电力电子装置** (DC-to DC power converters, rectifiers, inverters, HVDC SMC systems, etc)
- **飞行器**
- **电力系统、发动机、汽车、内燃机** ……

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2020年12月22日12时37分，由中国航天科技集团一院抓总研制的我国新一代中型运载火箭长征八号，从文昌航天发射场点火升空，将新技术验证七号、海丝一号、元光号、天启星座零八星、智星一号A星5颗卫星送入预定轨道，圆满完成首次飞行任务。

/s/8DVNOcND7t3ppXM2gcReTA

多项新技术让火箭更聪明

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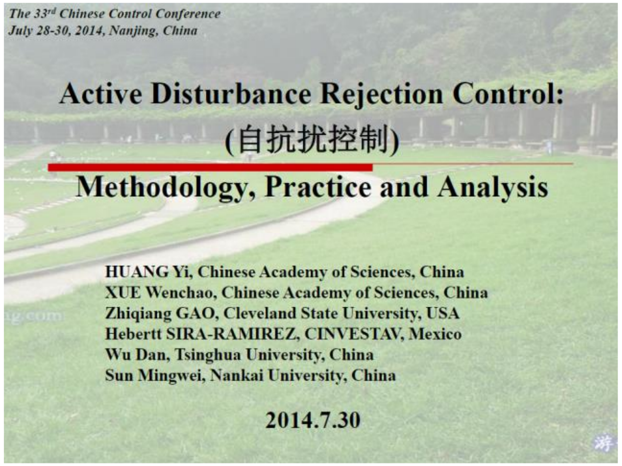
近年来，我国运载火箭高密度发射已呈常态化，但也偶尔发生飞行故障甚至飞行失利的案例。如果火箭具备故障诊断和自主飞行能力，在发生故障时，自主调整，飞行结果便能大幅改善。

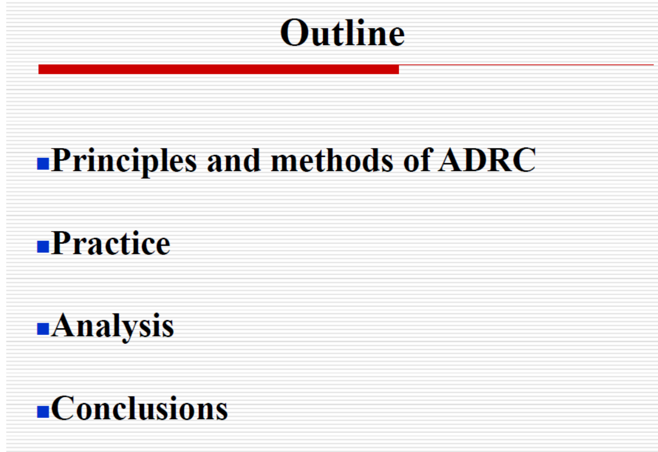
科研人员在长征八号身上，通过研究分析各种减载稳定控制方法，并采用自抗扰技术进行实时补偿控制，提高主动减载的效果，解决了大整流罩带来的静不稳定度大的难题。

同时，基于控制效果的喷管极性辨识和控制重构算法，能让运载火箭更聪明。通过该技术的应用，长征八号具备了滑行段飞行故障在线辨识能力，能够在特定故障工况下自主进行姿态控制重构，提升了飞行控制适应性和智能化水平。

针对当前我国运载火箭测试周期长、在发射区占位时间长的问题，科研团队积极探索火箭快速发射路径，为长征八号设计了"两平一垂"发射模式，即水平组装、水平状态整体运输、星罩组合体垂直转场对接。**预计在2022年左右，长征八号将实现"两平一垂"发射。届时发射区将不再需要规模庞大、组成复杂的塔架，可减少建设成本。**

Yi Huang, plenary talk of the 33rd Chinese Control Conference, 2014.





brilliant idea:

treating the unknown model as a special state (named extended state by Han) using a special state observer, the extended state observer (ESO), to estimate it in real time

In such a brave and elegant stroke:

a system, which may be uncertain, nonlinear, and time varying →

reduced to the canonical form of the linear chain of integrators

http://tcct.amss.ac.cn/newsletter/2014/201408/#plenarytalk

Zhiqiang Gao, Cleveland State University, U.S.A

- **Texas Instrument,**
- **Freescale,**
- **Danfoss,**

- **Rockwell,**
- **Parker,**
- **Power plant in china,**
- **…**


2013 年，美国德州仪器公司推出一系列基于自抗扰控制算法的运动控制芯片


2017 年 ADRC 在山西同达电厂 300MW 循环流化床机组协调控制回路投运


ADRC 用于多用途服务机器人的运动规划和避障控制中

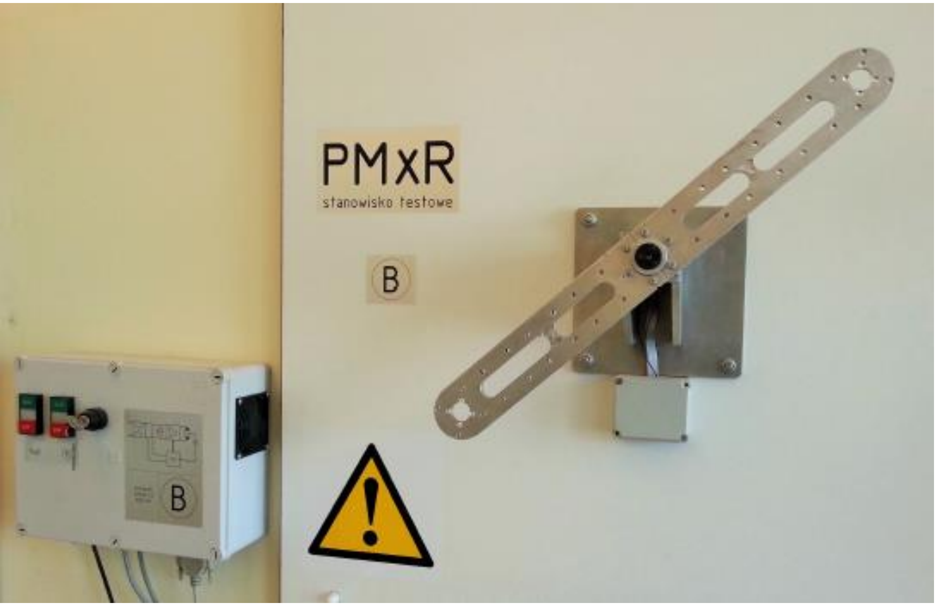
自抗扰控制应用（二）

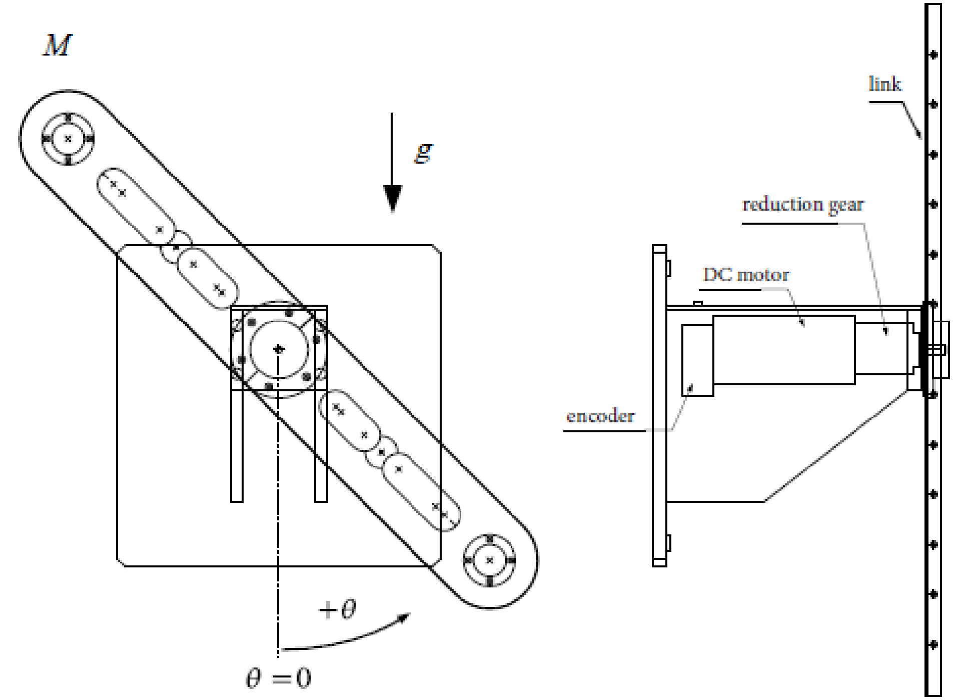
刚体机械臂转动控制

薛文超

中国科学院数学与系统科学研究院

刚性机械臂





数学模型： $\ddot{\theta} = -\frac{RC_b + K_E K_I}{JR}\dot{\theta} + \frac{K_I}{JR}u(t) + \bar{\delta}(\theta, \omega, M, u, d(t))$

θ is the angular position, u is the control input, n is the sensor noise

J is the system inertia, C_b is the friction coefficient, M is the mass of the manipulator,

R is the armature resistance, K_I is the torque constant, K_E is the speed motor constant,

$\bar{\delta}(\theta, \omega, M, u, d(t))$: unmodeled dynamics, unknown mass changes, external disturbance

Wenchao Xue, etc., Add-on Module of Active Disturbance Rejection for Set-Point Tracking of Motion Control Systems, IEEE Transactions on Industry Applications, vol. 53, no. 4, pp. 4028-4040, 2017.

无速度传感器下的位置定点跟踪控制

数学模型:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = a_2 x_1 + a_1 x_2 + b(t)u + \delta(x,t), \\ y(t) = x_1(t) + n(t), \quad t \ge 0, \end{cases}$$

x_1：位置　　x_2：速度

u：控制输入　　y：位置量测量

$\delta(x,t)$：总扰动 (不确定动态 + 外部扰动)

n：量测噪声

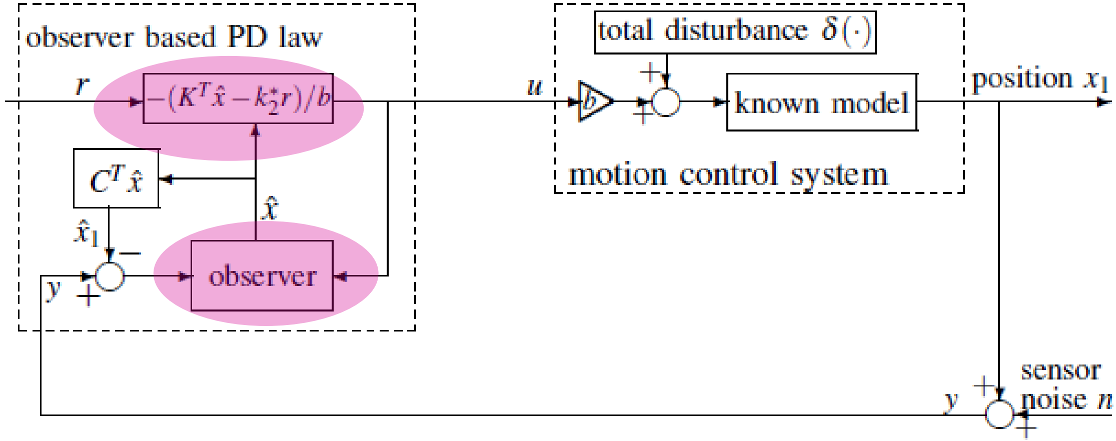
目标：在参数不确定及扰动下达到预期的动态及问题性能

$$\begin{cases} \dot{x}_1^* = x_2^* \\ \dot{x}_2^* = -k_2^*\left(x_1^* - r\right) - k_1^* x_2^*, \end{cases}$$

r：位置的设定点

$$x_1^*(0) = x_1(0),\ x_2^*(0) = x_2(0)$$

经典的基于观测器的PD控制方法



◆**龙贝格观测器:**

$$\begin{cases}\dot{\hat{x}}_1=\hat{x}_2+l_1(y-\hat{x}_1)\\ \dot{\hat{x}}_2=a_2\hat{x}_1+a_1\hat{x}_2+l_2(y-\hat{x}_1)+b(t)u\end{cases}$$

提供了速度的估计值，并用于反馈设计

◆**PD 反馈:**

$$u=-\left(\left(k_2^*+a_2\right)\hat{x}_1+\left(k_1^*+a_1\right)\hat{x}_2-k_2^*r\right)/b(t)$$

通过极点配置达到预期响应

$$\begin{cases}\dot{x}_1^*=x_2^*\\ \dot{x}_2^*=-k_2^*\left(x_1^*-r\right)-k_1^*x_2^*\end{cases}$$

预期的位置响应

预期的动态

$$\begin{cases} \dot{x}_1^* = x_2^* \\ \dot{x}_2^* = -k_2^*\left(x_1^* - r\right) - k_1^* x_2^* \end{cases}$$

真实的位置响应

基于经典观测器的闭环系统

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -k_2^*\left(x_1 - r\right) - k_1^* x_2 + \underbrace{k_2 e_1 - k_2 e_2}_{\text{状态估计误差}} + \delta(x,t), \\ \dot{e}_1 = e_2 - l_1 e_1 - l_1 n \\ \dot{e}_2 = \left(a_2 - l_1\right)e_1 + \left(a_1 - l_2\right)e_1 + \delta(x,t) - l_2 n \end{cases}$$

“总扰动”：

导致真实位置响应与预期位置响应之间的偏差

存在的问题：如何对付“总扰动”？

自抗扰控制设计方法：加入ESO模块

被控对象 $\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = a_2 x_1 + a_1 x_2 + b(t)u + \delta(x,t), \end{cases}$

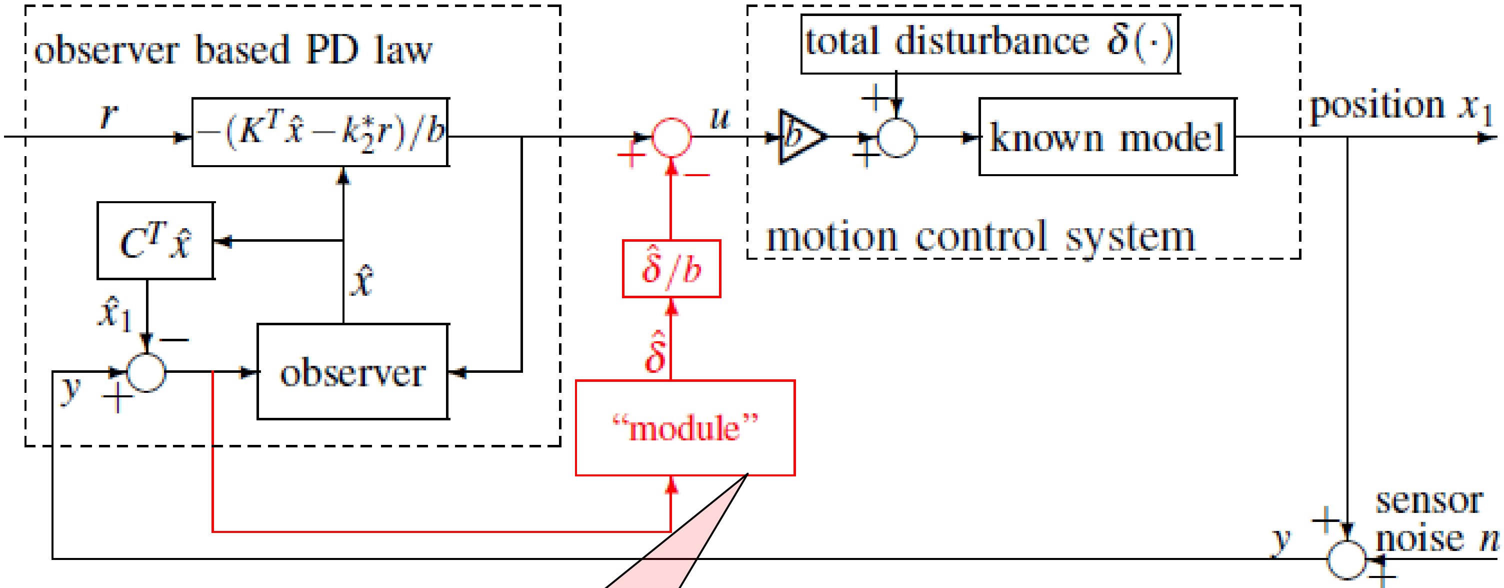
ESO $\begin{cases} \dot{\hat{x}}_1 = \hat{x}_2 + l_1(y - \hat{x}_1) \\ \dot{\hat{x}}_2 = a_2 \hat{x}_1 + a_1 \hat{x}_2 + l_2(y - \hat{x}_1) + b(t)u \\ \dot{\hat{\delta}} = l_3(y - \hat{x}_1) \end{cases}$

$\hat{\delta}(t)$: $\delta(x,t)$ 的估计值

控制输入：$u = \underbrace{-\left(\left(k_2^* + a_2\right)\hat{x}_1 + \left(k_1^* + a_1\right)\hat{x}_2 - k_2^* r\right)/b(t)}_{\text{PD 反馈}} \underbrace{-\hat{\delta}(t)/b(t)}_{\text{补偿 } \delta(t)}$

通过在线估计和补偿“总扰动”使得
实际系统达到预期的动态

自抗扰控制设计方法：加入ESO模块



观测器
跟踪误差的积分

$$\dot{\hat{\delta}} = l_3(y - \hat{x}_1)$$

$l_3 = 0$ 基于观测器的**PD**反馈

$l_3 \neq 0$ 增加**ESO**模块

增加的ESO模块
不用改变原有的控制回路及参数

频率分析

◆ 闭环系统的位置（基于观测器的PD反馈控制）

$$X_1(s) = \bar{G}_{YR}(s)R(s) + \bar{G}_{Y\Delta}(s)\Delta(s) + \bar{G}_{YN}(s)N(s)$$

◆ 闭环系统的位置（加入ESO模块之后的控制）

$$X_1(s) = \bar{G}_{YR}(s)R(s) + G_{Y\Delta}(s)\Delta(s) + G_{YN}(s)N(s)$$

$\Delta(s)$：Total disturbance　　　$N(s)$：Sensor noise

$\bar{G}_{Y\Delta}(s)$ 或 $G_{Y\Delta}(s)$ 越小，➡ 抗扰能力越强

$\bar{G}_{YN}(s)$ 或 $G_{Y\Delta}(s)$ 越小，➡ 滤噪能力越强

频域分析

定理：假设两个闭环系统都稳定，则两者的扰动-输出的传递函数满足：

1) $$\left|\frac{G_{Y\Delta}(j\omega)}{\bar{G}_{Y\Delta}(j\omega)}\right| < 1, \quad \forall \omega \in \left[0, \frac{\sqrt{l_3}}{\sqrt{2(l_1 - a_1)}}\right)$$

低频时，提高了抗扰性能

2) $$\lim_{\omega \to 0}\left|\frac{G_{Y\Delta}(j\omega)}{\bar{G}_{Y\Delta}(j\omega)}\right| = 0$$

频率越低，性能提高越大

3) $$\lim_{\omega \to \infty}\left|\frac{G_{YN}(j\omega)}{\bar{G}_{YN}(j\omega)}\right| = \left|1 + \frac{l_3}{k_2 l_1 + k_1 l_2}\right|$$

对量测噪声的滤波效果处于同一量级

Wenchao Xue, etc., Add-on Module of Active Disturbance Rejection for Set-Point Tracking of Motion Control Systems, IEEE Transactions on Industry Applications, vol. 53, no. 4, pp. 4028-4040, 2017.

机械臂系统中的总扰动

$$\begin{cases} \dot{x} = Ax + B(\delta(\cdot) + bu), \\ y(t) = C^T x(t) + n(t), \quad t \geq 0, \end{cases}$$

$x = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}, A = \begin{bmatrix} 0 & 1 \\ a_2 & a_1 \end{bmatrix}, a_2 = 0, a_1 = -60.72, b = 724,$

设定点: $r = 20[dec]$

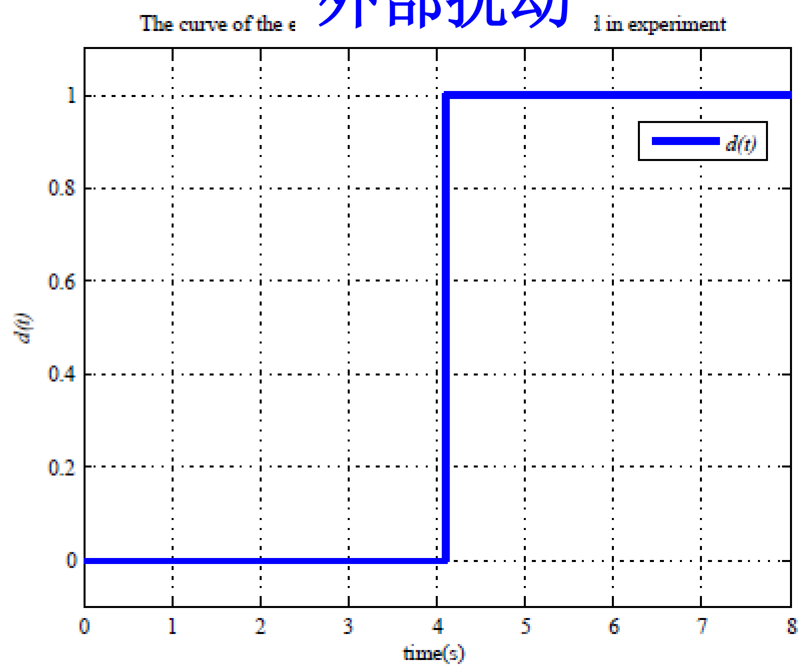
总扰动: $\delta(\cdot) = \left(-\dfrac{RC_b + K_E K_I}{JR} + 60.72 \right) \dot{\theta} + \left(\dfrac{K_I}{JR} - 724 \right) u + \bar{\delta}(\cdot)$

3种情况下的机械臂质量

$$\begin{cases} C1: M = M_0 \\ C2: M = M_0 + M_\delta \\ C3: M = M_0 + 2M_\delta \end{cases}$$

Nominal case

外部扰动



实验结果

$$\begin{cases} \dot{\hat{x}} = A\hat{x} + L(y - \hat{x}_1) + B(-K^T\hat{x} + k_2^* r), \\ \dot{\hat{\delta}} = l_3(y - \hat{x}_1), \\ u = -\dfrac{K^T\hat{x} - k_2^* r}{b(t)} - \dfrac{\hat{\delta}}{b(t)}, \end{cases}$$

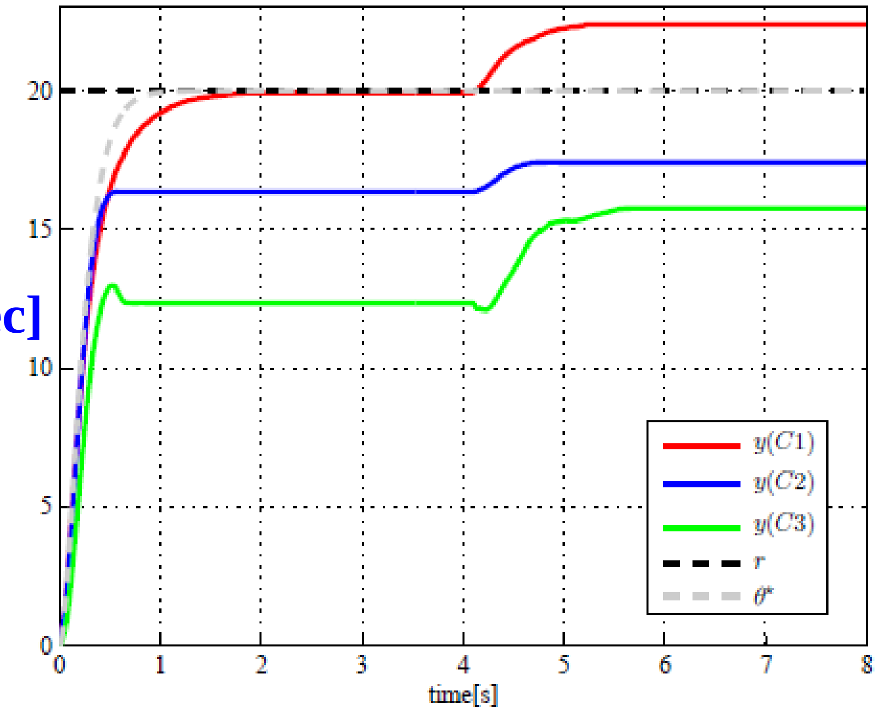
$k_1^* = 5852.72,\ k_2^* = 23424,\ l_1 = 60,\ l_2 = 1200$

$l_3 = 0$ ：正常观测器及 PD反馈

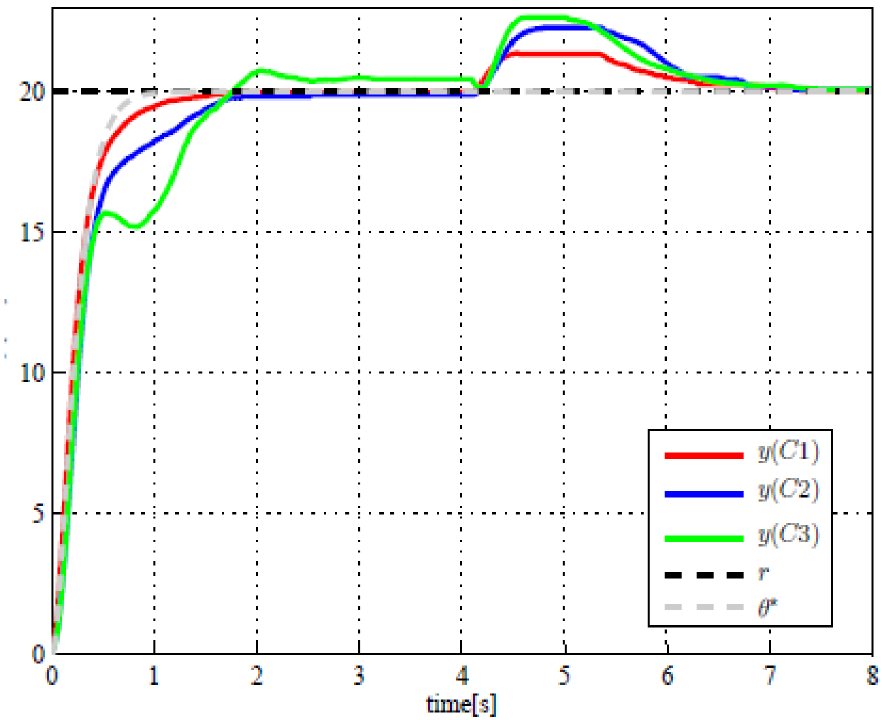
$l_3 = 8000$ ：加入ESO模块

正常观测器及 PD反馈

Y[dec]



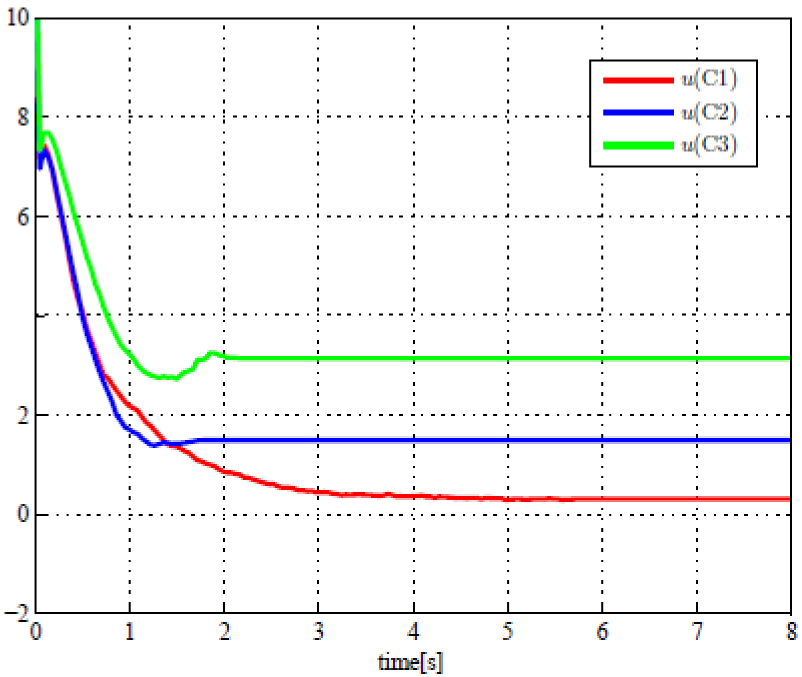
加入ESO模块后



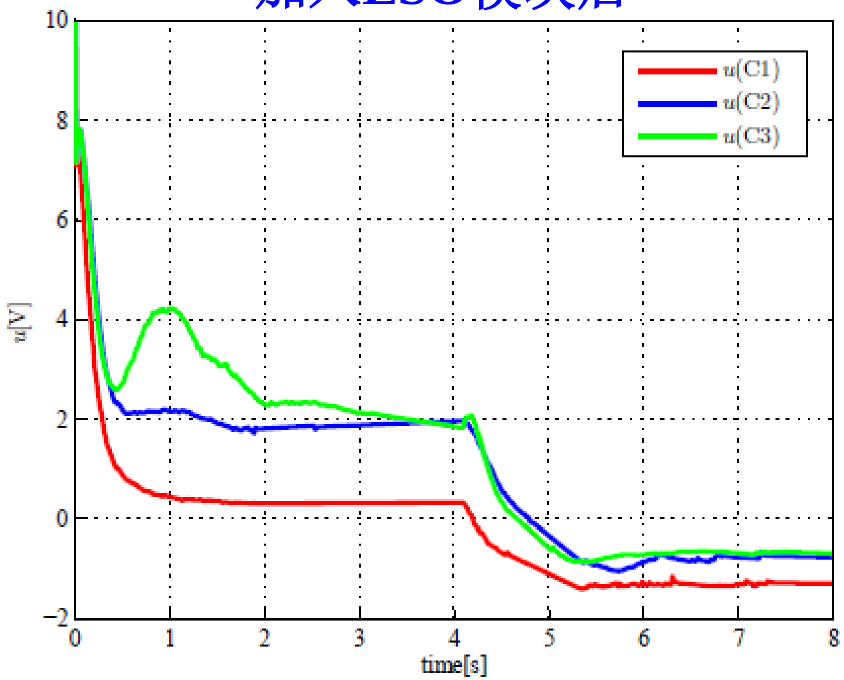
实验结果

正常观测器及 PD反馈

u[V]

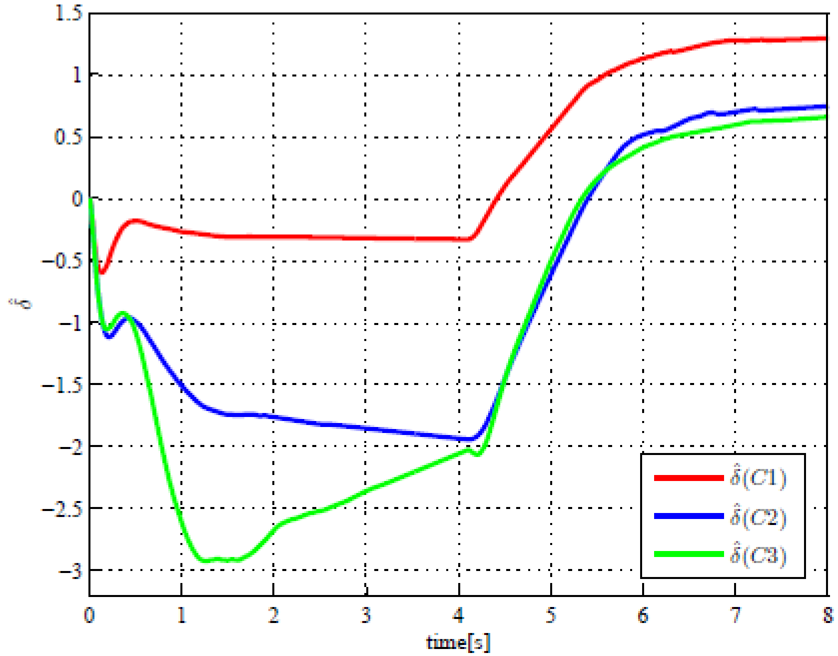


加入ESO模块后



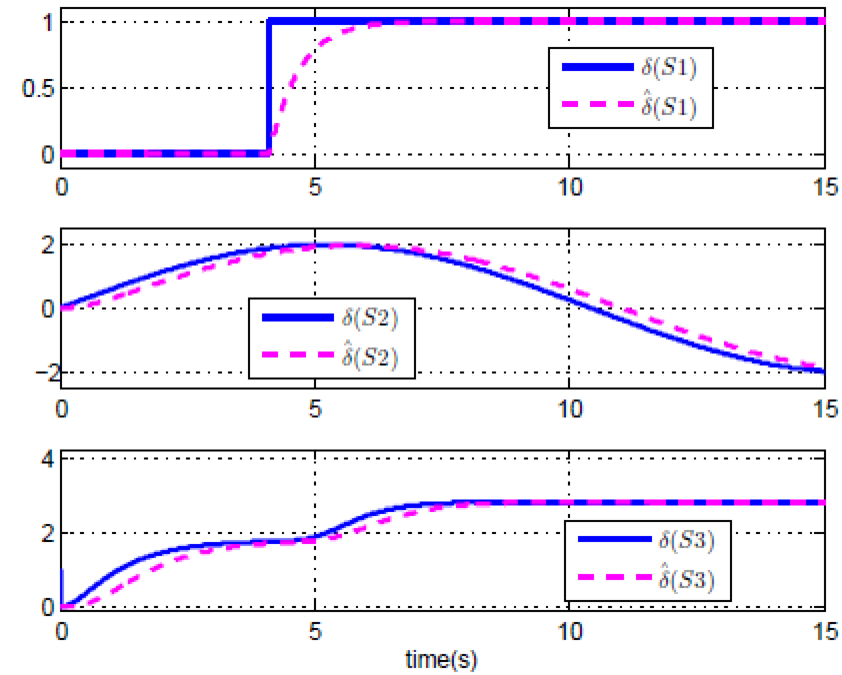
实验结果

“总扰动”的估计值



仿真结果

“总扰动”的估计值



$$\begin{cases} S1: \delta(\cdot) = d(t) \\ S2: \delta(\cdot) = 2sin(0.3t) \\ S3: \delta(\cdot) = 2sin(0.01\theta)^2 + exp(-\dot{\theta}^2) \end{cases}$$

自抗扰控制的典型应用
飞行器角速度控制

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内容

一、典型飞行器角速度控制模型介绍

二、全通道自抗扰控制设计

三、参数调节方法及仿真结果

一、飞机控制模型介绍

以F16飞机控制仿真模型为例

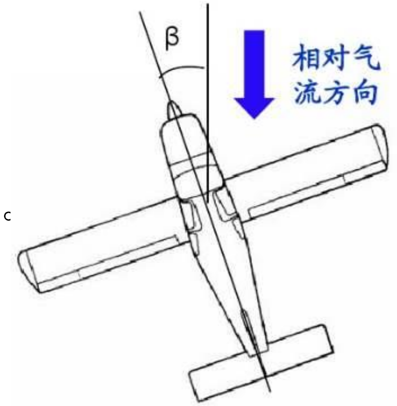
物理条件：

- ➢飞行器是刚体
- ➢地面是平坦的，忽略地球的自转作用
- ➢飞机质量不变，即忽略燃料消耗对飞行器质量的改变
- ➢飞行器关于 $X_B O Z_B$ 平面对称，即转动惯量 $I_{yz} = I_{xy} = 0$
- ➢水平尾翼的舵面是对称的
- ➢不使用气动减速装置

一、飞机控制模型介绍

飞机机体系（B 系）

原点为飞行器的质心，X_B 指向机头，Y_B 指向右侧机翼，Z_B 指向飞机下方。（前右下）

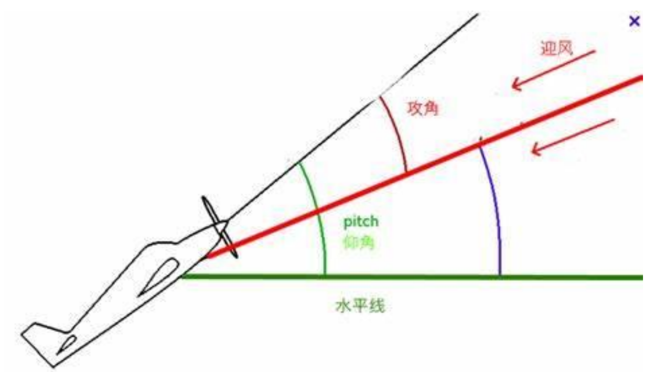


气流坐标系（W 系）

（X_W 指向飞机相对于空气的速度矢量的方向，Z_W 位于飞机对称面 X_BOZ_B 内，指向飞机下方）

气流系到机体系的转换矩阵

$$C_B^W = \begin{bmatrix} \cos\alpha\cos\beta & \sin\beta & \sin\alpha\cos\beta \\ -\cos\alpha\sin\beta & \cos\beta & -\sin\alpha\sin\beta \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix}$$



控制系统模型介绍（续）

角速度的方程（机体系）

典型控制目标：

横向：滚转角速度**跟踪**

航向：侧滑角**稳定**

纵向：俯仰角速率**稳定**

气动力矩：可以改变的力矩

$$I\begin{bmatrix}\dot{p}\\ \dot{q}\\ \dot{r}\end{bmatrix}+\begin{bmatrix}p\\ q\\ r\end{bmatrix}\times I\begin{bmatrix}p\\ q\\ r\end{bmatrix}=\begin{bmatrix}0\\ -h_E r\\ h_E q\end{bmatrix}+\begin{bmatrix}\bar{L}\\ \bar{M}\\ \bar{N}\end{bmatrix}$$

控制系统模型介绍（续）

角速度的方程（机体系）

气动力矩：可以改变的力矩

$$I\begin{bmatrix}\dot{p}\\ \dot{q}\\ \dot{r}\end{bmatrix}+\begin{bmatrix}p\\ q\\ r\end{bmatrix}\times I\begin{bmatrix}p\\ q\\ r\end{bmatrix}=\begin{bmatrix}0\\ -h_E r\\ h_E q\end{bmatrix}+\begin{bmatrix}\bar{L}\\ \bar{M}\\ \bar{N}\end{bmatrix}$$

气动力矩系数

$$\begin{bmatrix}\bar{L}\\ \bar{M}\\ \bar{N}\end{bmatrix}=\frac{1}{2}\rho V_T^2 S\begin{bmatrix}bC_{l_T}(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\ldots)\\ \bar{c}C_{m_T}(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\ldots\\ bC_{n_T}(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\ldots\end{bmatrix}$$

复杂的非线性函数
或者表格形式

控制设计面临的关键难点：

气动力矩系数：非线性函数、不确定函数、多通道耦合

$$\begin{bmatrix} \bar{L} \\ \bar{M} \\ \bar{N} \end{bmatrix} = \frac{1}{2}\rho V_T^2 S \begin{bmatrix} bC_{l_T}\left(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\ldots\right) \\ \bar{c}C_{m_T}\left(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\ldots\right) \\ bC_{n_T}\left(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\ldots\right) \end{bmatrix}.$$

表格形式

标称表与真实表有偏差

$$C_{l_T} = C_l\left(\alpha,\beta,\delta_e\right) + \left[\delta C_{l_{\delta_a}} + \delta C_{l_{\delta_{aLEF}}}\left(1-\frac{\delta_{LEF}}{25}\right)\right]\cdot\left(\frac{\delta_a}{20}\right) + \delta C_{l_\beta}\left(\alpha\right)\beta$$
$$+\delta C_{l_{\delta r}}\cdot\left(\frac{\delta_r}{30}\right) + \delta C_{l_\beta}\left(\alpha\right) + \frac{rb}{2V_T}\left[C_{l_r}\left(\alpha\right) + \delta C_{l_{rLEF}}\left(\alpha\right)\left(1-\frac{\delta_{LEF}}{25}\right)\right]$$
$$+\frac{pb}{2V_T}\left[C_{l_p}\left(\alpha\right) + \delta C_{l_{pLEF}}\left(\alpha\right)\left(1-\frac{\delta_{LEF}}{25}\right)\right] + \delta C_{l_{LEF}}\cdot\left(1-\frac{\delta_{LEF}}{25}\right),$$

$$C_{m_T} = C_m\left(\alpha,\beta,\delta_e\right) + C_{Z_T}\left[x_{cgr} - x_{cg}\right] + \delta C_{m_{LEF}}\cdot\left(1-\frac{\delta_{LEF}}{25}\right)$$
$$+\frac{q\bar{c}}{2V_T}\left[C_{m_q}\left(\alpha\right) + \delta C_{m_{qLEF}}\left(\alpha\right)\right]\left(1-\frac{\delta_{LEF}}{25}\right) + \delta C_m\left(\alpha\right) + \delta C_{m_{ds}}\left(\alpha,\delta_e\right),$$

$$C_{n_T} = C_n\left(\alpha,\beta,\delta_e\right) + \left[\delta C_{n_{\delta_a}} + \delta C_{n_{\delta_{aLEF}}}\cdot\left(1-\frac{\delta_{LEF}}{25}\right)\right]\cdot\left(\frac{\delta_a}{20}\right)$$
$$+\delta C_{n_{LEF}}\cdot\left(1-\frac{\delta_{LEF}}{25}\right) + \frac{rb}{2V_T}\left[C_{n_r}\left(\alpha\right) + \delta C_{n_{r_{LEF}}}\left(\alpha\right)\cdot\left(1-\frac{\delta_{LEF}}{25}\right)\right]$$
$$+\frac{pb}{2V_T}\left[C_{n_p}\left(\alpha\right) + \delta C_{n_{p_{LEF}}}\left(\alpha\right)\left(1-\frac{\delta_{LEF}}{25}\right)\right] + \delta C_{n_{\delta_r}}\cdot\left(\frac{\delta_r}{30}\right) + \delta C_{n_\beta}\left(\alpha\right)\cdot\beta - C_{Y_T}\left[x_{cgr} - x_{cg}\right]\frac{\bar{c}}{b},$$

非线性函数形式

$$\begin{cases} \delta C_{l_{LEF}} = C_{l_{LEF}}\left(\alpha,\beta\right) - C_l\left(\alpha,\beta,\delta_e = 0^\circ\right) \\ \delta C_{l_{\delta_a}} = C_{l_{\delta_a}}\left(\alpha,\beta\right) - C_l\left(\alpha,\beta,\delta_e = 0^\circ\right) \\ \delta C_{l_{\delta_{aLEF}}} = C_{l_{\delta_{aLEF}}}\left(\alpha,\beta\right) - C_{l_{LEF}}\left(\alpha,\beta\right) - \delta C_{l_{\delta_a}} \\ \delta C_{l_{\delta r}} = C_{l_{\delta r}}\left(\alpha,\beta\right) - C_l\left(\alpha,\beta,\delta_e = 0^\circ\right) \end{cases}.$$

$$\delta C_{m_{LEF}} = C_{m_{LEF}}\left(\alpha,\beta\right) - C_m\left(\alpha,\beta,\delta_e = 0^\circ\right)$$

$$\begin{cases} \delta C_{n_{LEF}} = C_{n_{LEF}}\left(\alpha,\beta\right) - C_n\left(\alpha,\beta,\delta_e = 0^\circ\right) \\ \delta C_{n_{\delta_a}} = C_{n_{\delta_a}}\left(\alpha,\beta\right) - C_n\left(\alpha,\beta,\delta_e = 0^\circ\right) \\ \delta C_{n_{\delta_{aLEF}}} = C_{n_{\delta_{aLEF}}}\left(\alpha,\beta\right) - C_{n_{LEF}}\left(\alpha,\beta\right) - \delta C_{n_{\delta_a}} \\ \delta C_{n_{\delta_r}} = C_{n_{\delta_r}}\left(\alpha,\beta\right) - C_n\left(\alpha,\beta,\delta_e = 0^\circ\right) \end{cases}.$$

控制系统模型介绍（续）

角速度的方程（机体系）

气动力矩系数

$$I\begin{bmatrix}\dot{p}\\ \dot{q}\\ \dot{r}\end{bmatrix}+\begin{bmatrix}p\\ q\\ r\end{bmatrix}\times I\begin{bmatrix}p\\ q\\ r\end{bmatrix}=\begin{bmatrix}0\\ -h_E r\\ h_E q\end{bmatrix}+\begin{bmatrix}\bar{L}\\ \bar{M}\\ \bar{N}\end{bmatrix}\qquad \begin{bmatrix}\bar{L}\\ \bar{M}\\ \bar{N}\end{bmatrix}=\frac{1}{2}\rho V_T^2 S\begin{bmatrix}bC_{l_T}(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\ldots)\\ \bar{c}C_{m_T}(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\ldots)\\ bC_{n_T}(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\ldots)\end{bmatrix}.$$

现有的飞行器的控制回路

- 基本都是基于PID控制
- 定点线性化后，通过频域分析调参
- 控制器参数选取的工作量很大
- 难以本质上克服强非线性

内容

二、全通道自抗扰控制设计

基于自抗扰控制的模型建立

角速度方程

$$I\begin{bmatrix}\dot{p}\\ \dot{q}\\ \dot{r}\end{bmatrix}+\begin{bmatrix}p\\ q\\ r\end{bmatrix}\times I\begin{bmatrix}p\\ q\\ r\end{bmatrix}=\begin{bmatrix}0\\ -h_E r\\ h_E q\end{bmatrix}+\frac{1}{2}\rho V_T^2 S\begin{bmatrix}bC_{l_T}(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\ldots)\\ \bar{c}C_{m_T}(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\ldots)\\ bC_{n_T}(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\ldots)\end{bmatrix}$$

关于输入 δ_e 的非线性函数

$$C_{l_T}=C_l(\alpha,\beta,\delta_e)+\left[\delta C_{l_{\delta_a}}+\delta C_{l_{\delta_{aLEF}}}\bullet\left(1-\frac{\delta_{LEF}}{25}\right)\right]\bullet\left(\frac{\delta_a}{20}\right)+\delta C_{l_\beta}(\alpha)\beta$$

$$+\delta C_{l_{\delta r}}\bullet\left(\frac{\delta_r}{30}\right)+\delta C_{l_\beta}(\alpha)+\frac{rb}{2V_T}\left[C_{l_r}(\alpha)+\delta C_{l_{rLEF}}(\alpha)\left(1-\frac{\delta_{LEF}}{25}\right)\right]+\frac{pb}{2V_T}\left[C_{l_p}(\alpha)+\delta C_{l_{pLEF}}(\alpha)\left(1-\frac{\delta_{LEF}}{25}\right)\right]+\delta C_{l_{LEF}}\bullet\left(1-\frac{\delta_{LEF}}{25}\right),$$

基于自抗扰控制的模型建立

角速度方程

$$I\begin{bmatrix}\dot p\\ \dot q\\ \dot r\end{bmatrix}+\begin{bmatrix}p\\ q\\ r\end{bmatrix}\times I\begin{bmatrix}p\\ q\\ r\end{bmatrix}=\begin{bmatrix}0\\ -h_E r\\ h_E q\end{bmatrix}+\frac{1}{2}\rho V_T^2 S\begin{bmatrix}bC_{l_T}\left(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\dots\right)\\ \bar{c}C_{m_T}\left(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\dots\right)\\ bC_{n_T}\left(\alpha,\beta,p,q,r,\delta_e,\delta_a,\delta_r,\dots\right)\end{bmatrix}$$

关于舵偏输入 δ_e 的非线性函数

$$C_{l_T}=C_l\left(\alpha,\beta,\delta_e\right)+\left[\delta C_{l_{\delta_a}}+\delta C_{l_{\delta_{aLEF}}}\bullet\left(1-\frac{\delta_{LEF}}{25}\right)\right]\bullet\left(\frac{\delta_a}{20}\right)+\delta C_{l_\beta}\left(\alpha\right)\beta$$

$$+\delta C_{l_{\delta r}}\bullet\left(\frac{\delta_r}{30}\right)+\delta C_{l_\beta}\left(\alpha\right)+\frac{rb}{2V_T}\left[C_{l_r}\left(\alpha\right)+\delta C_{l_{rLEF}}\left(\alpha\right)\left(1-\frac{\delta_{LEF}}{25}\right)\right]+\frac{pb}{2V_T}\left[C_{l_p}\left(\alpha\right)+\delta C_{l_{pLEF}}\left(\alpha\right)\left(1-\frac{\delta_{LEF}}{25}\right)\right]+\delta C_{l_{LEF}}\bullet\left(1-\frac{\delta_{LEF}}{25}\right),$$

利用拉格朗日中值定理等价转化为 δ_e 的线性形式

$$C_l\left(\alpha,\beta,\delta_e\right)=C_l\left(\alpha,\beta,\delta_e=\bar{\delta}_e\right)+\left.\frac{\partial C_l\left(\alpha,\beta,\xi\right)}{\partial\delta_e}\right|_{\xi\in\left(\bar{\delta}_e,\delta_e\right]}\left(\delta_e-\bar{\delta}_e\right)$$

$\bar{\delta}_e$:上一步的舵偏值

$$=C_l\left(\alpha,\beta,\delta_e=\bar{\delta}_e\right)-\left.\frac{\partial C_l\left(\alpha,\beta,\xi\right)}{\partial\delta_e}\right|_{\xi\in\left(\bar{\delta}_e,\delta_e\right]}\bar{\delta}_e+\left.\frac{\partial C_l\left(\alpha,\beta,\xi\right)}{\partial\delta_e}\right|_{\xi\in\left(\bar{\delta}_e,\delta_e\right]}\delta_e$$

基于控制器设计的模型建立（续）

角速度控制方程

$$\begin{bmatrix}\dot p\\ \dot q\\ \dot r\end{bmatrix}=I^{-1}\left\{-\begin{bmatrix}p\\ q\\ r\end{bmatrix}\times I\begin{bmatrix}p\\ q\\ r\end{bmatrix}+\begin{bmatrix}0\\ -h_E r\\ h_E q\end{bmatrix}\right\}+I^{-1}\begin{bmatrix}\tilde{\bar{L}}\\ \tilde{\bar{M}}\\ \tilde{\bar{N}}\end{bmatrix}+I^{-1}B\begin{bmatrix}\delta_e\\ \delta_a\\ \delta_r\end{bmatrix}$$

控制输入的线性部分

5个时变参数

$$B=\begin{bmatrix}b_{11} & b_{12} & b_{13}\\ b_{21} & 0 & 0\\ b_{31} & b_{32} & b_{33}\end{bmatrix}$$

$$\begin{cases}
b_{11}=\dfrac{1}{2}\rho V_T^2 Sb\left.\dfrac{\partial C_l(\alpha,\beta,\xi_l)}{\partial\delta_e}\right|_{\xi_l\in(\bar\delta_e,\delta_e]}, \quad b_{12}=\dfrac{1}{40}\rho V_T^2 Sb\left[\delta C_{l_{\delta_a}}+\delta C_{l_{\delta_{aLEF}}}\left(1-\dfrac{\delta_{LEF}}{25}\right)\right], \quad b_{13}=\dfrac{1}{60}\rho V_T^2 Sb\,\delta C_{l_{\delta r}}\\
b_{21}=\dfrac{1}{2}\rho V_T^2 S\bar c\left\{\left.\dfrac{\partial C_m(\alpha,\beta,\xi_1)}{\partial\delta_e}\right|_{\xi_1\in(\bar\delta_e,\delta_e]}+\left.\dfrac{\partial\delta C_{m_{ds}}(\alpha,\beta,\xi_2)}{\partial\delta_e}\right|_{\xi_2\in}+\left.\dfrac{\partial C_Z(\alpha,\beta,\xi_3)}{\partial\delta_e}\right|_{\xi_3\in(\bar\delta_e,\delta_e]}\left[x_{cgr}-x_{cg}\right]\right\}\\
b_{31}=\dfrac{1}{2}\rho V_T^2 Sb\left.\dfrac{\partial C_n(\alpha,\beta,\xi_n)}{\partial\delta_e}\right|_{\xi_n\in(\delta_e^*,\delta_e]}, \quad b_{32}=\dfrac{1}{40}\rho V_T^2 Sb\left\{\left[\delta C_{n_{\delta_a}}+\delta C_{n_{\delta_{aLEF}}}\left(1-\dfrac{\delta_{LEF}}{25}\right)\right]-\left[\delta C_{Y_{\delta_a}}+\delta C_{Y_{\delta_{aLEF}}}\left(1-\dfrac{\delta_{LEF}}{25}\right)\right]\left[x_{cgr}-x_{cg}\right]\dfrac{\bar c}{b}\right\}\\
b_{35}=b_{24}=x_T
\end{cases}$$

基于控制器设计的模型建立（续）

角速度控制模型

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = I^{-1}\left\{ -\begin{bmatrix} p \\ q \\ r \end{bmatrix} \times I \begin{bmatrix} p \\ q \\ r \end{bmatrix} + \begin{bmatrix} 0 \\ -h_E r \\ h_E q \end{bmatrix} \right\} + I^{-1}\begin{bmatrix} \tilde{\bar{L}} \\ \tilde{\bar{M}} \\ \tilde{\bar{N}} \end{bmatrix} + I^{-1}B\begin{bmatrix} \delta_e \\ \delta_a \\ \delta_r \end{bmatrix}$$

角速率的函数

气动力矩中与控制输入无关的项

$$\begin{aligned} \tilde{\bar{N}} = {} & \frac{1}{2}\rho V_T^2 Sb\left\{ C_n(\alpha,\beta,\delta_e=0) + \delta C_{n_{LEF}}\left(1-\frac{\delta_{LEF}}{25}\right) + \delta C_{n_\beta}(\alpha)\beta + \frac{rb}{2V_T}\left[C_{n_r}(\alpha) + \delta C_{n_{r_{LEF}}}(\alpha)\left(1-\frac{\delta_{LEF}}{25}\right)\right] + \frac{pb}{2V_T}\left[C_{n_p}(\alpha) + \delta C_{n_{p_{LEF}}}(\alpha)\left(1-\frac{\delta_{LEF}}{25}\right)\right]\right\} \\ & -\frac{1}{2}\rho V_T^2 Sb\left\{ C_Y(\alpha,\beta) + \delta C_{Y_{LEF}}\left(1-\frac{\delta_{LEF}}{25}\right) + \frac{rb}{2V_T}\left[C_{Y_r}(\alpha) + \delta C_{Y_{r_{LEF}}}(\alpha)\left(1-\frac{\delta_{LEF}}{25}\right)\right] + \frac{pb}{2V_T}\left[C_{Y_p}(\alpha) + \delta C_{Y_{pLEF}}(\alpha)\left(1-\frac{\delta_{LEF}}{25}\right)\right]\right\}\left[x_{cgr} - x_{cg}\right]\frac{\bar{c}}{b}, \end{aligned}$$

基于控制器设计的模型建立（续）

横向：滚转角速度跟踪

航向：侧滑角稳定

纵向：俯仰角速率稳定

需要利用侧滑角方程及角速度方程建立侧滑角 β 的控制模型

侧滑角方程

$$\dot{\beta}=p\sin\alpha-r\cos\alpha+\frac{1}{mV_T}\left(Y+F_z+mg\right)$$

角速度控制方程

$$\begin{bmatrix}\dot{p}\\ \dot{q}\\ \dot{r}\end{bmatrix}=I^{-1}\left\{-\begin{bmatrix}p\\ q\\ r\end{bmatrix}\times I\begin{bmatrix}p\\ q\\ r\end{bmatrix}+\begin{bmatrix}0\\ -h_E r\\ h_E q\end{bmatrix}\right\}+I^{-1}\begin{bmatrix}\tilde{\tilde{L}}\\ \tilde{\tilde{M}}\\ \tilde{\tilde{N}}\end{bmatrix}+I^{-1}B\begin{bmatrix}\delta_e\\ \delta_a\\ \delta_r\end{bmatrix}$$

基于控制器设计的模型建立（续）

侧滑角 β 方程

$$\begin{cases} \dot{\beta} = \omega_\beta, \\ \dot{\omega}_\beta = F_{\omega_\beta} + B_\beta \begin{bmatrix} \delta_e \\ \delta_a \\ \delta_r \end{bmatrix}, \end{cases}$$

$$\omega_\beta \triangleq p\sin\alpha - r\cos\alpha + \frac{1}{mV_T}\left(Y + F_z + mg_2\right)$$

$$F_{\omega_\beta} \triangleq \frac{d\left[\frac{1}{mV_T}\left(Y + F_z + mg_2\right)\right]}{dt} + \dot{\alpha}\left(p\cos\alpha + r\sin\alpha\right)$$

$$+\left[\sin\alpha \quad 0 \quad -\cos\alpha\right]\left[I^{-1}\left\{-\omega \times I\omega + \begin{bmatrix} 0 \\ -h_E r \\ h_E q \end{bmatrix}\right\} + I^{-1}\begin{bmatrix} \tilde{\bar{L}} \\ \tilde{\bar{M}} \\ \tilde{\bar{N}} \end{bmatrix}\right].$$

$$B_\beta \triangleq \left[\sin\alpha \quad 0 \quad -\cos\alpha\right] I^{-1} B$$

三维角速率、侧滑角 ⟵ **等价的舵面偏角**

$$\dot{p} = F_p + B_p \begin{bmatrix} \delta_e \\ \delta_a \\ \delta_r \end{bmatrix}, \quad \dot{q} = F_q + B_q \begin{bmatrix} \delta_e \\ \delta_a \\ \delta_r \end{bmatrix}, \quad \dot{r} = F_r + B_r \begin{bmatrix} \delta_e \\ \delta_a \\ \delta_r \end{bmatrix}$$

3个
一阶系统

$$\begin{cases} \dot{\beta} = \omega_\beta, \\ \dot{\omega}_\beta = F_{\omega_\beta} + B_\beta \begin{bmatrix} \delta_e \\ \delta_a \\ \delta_r \end{bmatrix}, \end{cases}$$

1个
二阶系统

$$\begin{bmatrix} B_p \\ B_q \\ B_r \end{bmatrix} \triangleq B_\omega, \quad \begin{bmatrix} F_p \\ F_q \\ F_r \end{bmatrix} \triangleq I^{-1} \left\{ - \begin{bmatrix} p \\ q \\ r \end{bmatrix} \times I \begin{bmatrix} p \\ q \\ r \end{bmatrix} + \begin{bmatrix} 0 \\ -h_E r \\ h_E q \end{bmatrix} \right\} + I^{-1} \begin{bmatrix} \tilde{\bar{L}} \\ \tilde{\bar{M}} \\ \tilde{\bar{N}} \end{bmatrix}$$

全通道自抗扰控制设计

横向：滚转角速度跟踪

航向：侧滑角稳定

纵向：俯仰角速率稳定

$$\begin{bmatrix} \dot{\beta} \\ \dot{\omega}_\beta \\ \dot{p} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} \omega_\alpha \\ \bar{F}_{\omega_\beta} \\ \bar{F}_p \\ \bar{F}_q \end{bmatrix} + \begin{bmatrix} 0 \\ \bar{B}_\beta \\ \bar{B}_p \\ \bar{B}_r \end{bmatrix} \begin{bmatrix} \delta_e \\ \delta_a \\ \delta_r \end{bmatrix}$$

理想的控制输入满足

$$\begin{bmatrix} \bar{B}_\beta \\ \bar{B}_p \\ \bar{B}_r \end{bmatrix}_{3X3} \begin{bmatrix} \delta_e \\ \delta_a \\ \delta_r \end{bmatrix} = \underbrace{\begin{bmatrix} -\bar{F}_{\omega_\beta} \\ -\bar{F}_p \\ -\bar{F}_q \end{bmatrix}}_{\text{补偿扰动}} + \underbrace{\begin{bmatrix} -k_{\beta p}\left(\beta-\beta^*\right)-k_{\beta d}\left(\dot{\beta}-\dot{\beta}^*\right) \\ -k_{pp}\left(p-p^*\right) \\ -k_{qp}\left(q-q^*\right) \end{bmatrix}}_{\text{PD控制}}.$$

步骤1：设计扩张状态观测器(ESO)实时估计“总扰动”

横向：滚转角速度跟踪

航向：侧滑角稳定

纵向：俯仰角速率稳定

$$\begin{bmatrix}\dot{\beta}\\ \dot{\omega}_\beta\\ \dot{p}\\ \dot{q}\end{bmatrix}=\begin{bmatrix}\omega_\alpha\\ \bar{F}_{\omega_\beta}\\ \bar{F}_p\\ \bar{F}_q\end{bmatrix}+\begin{bmatrix}0\\ \bar{B}_\beta\\ \bar{B}_p\\ \bar{B}_r\end{bmatrix}\begin{bmatrix}\delta_e\\ \delta_a\\ \delta_r\end{bmatrix}$$

ESO-1

$$\begin{cases}\dot{z}_{\beta 1,1}=z_{\beta 2,1}-\beta_{\beta 1,1}\left(z_{\beta 1,1}-\beta\right)\\ \dot{z}_{\beta 2,1}=z_{\beta 3,1}+u_{1,1r}-\beta_{\beta 2,1}\left(z_{\beta 1,1}-\beta\right).\\ \dot{z}_{\beta 3,1}=-\beta_{\beta 2,1}\left(z_{\beta 1,1}-\beta\right)\end{cases}$$

$$\begin{cases}z_{\beta 1,2}\text{估计}\beta\\ z_{\beta 2,2}\text{估计}\dot{\beta}\\ z_{\beta 3,2}\text{估计}\bar{F}_{\omega_\beta}\end{cases}$$

$\beta_{\alpha 1,2}=3\omega_{\alpha o,2},\ \beta_{\alpha 2,2}=3\omega_{\alpha o,2}^2,$

$\beta_{\alpha 3,2}=\omega_{\alpha o,2}^3,$

$\omega_{\alpha o,2}$：ESO-1的带宽

ESO-2

$$\begin{cases}\dot{z}_{p1,1}=z_{p2,1}+u_{2,1r}-\beta_{p1,1}\left(z_{p1,1}-p\right)\\ \dot{z}_{p2,1}=-\beta_{p2,1}\left(z_{p1,1}-p\right)\end{cases}.$$

$$\begin{cases}z_{p1,2}\text{估计}p\\ z_{p2,2}\text{估计}\bar{F}_p\end{cases}$$

$\beta_{p1,2}=2\omega_{po,2},\ \beta_{p2,2}=\omega_{po,2}^2,$

$\omega_{po,2}$：ESO-2的带宽

ESO-3

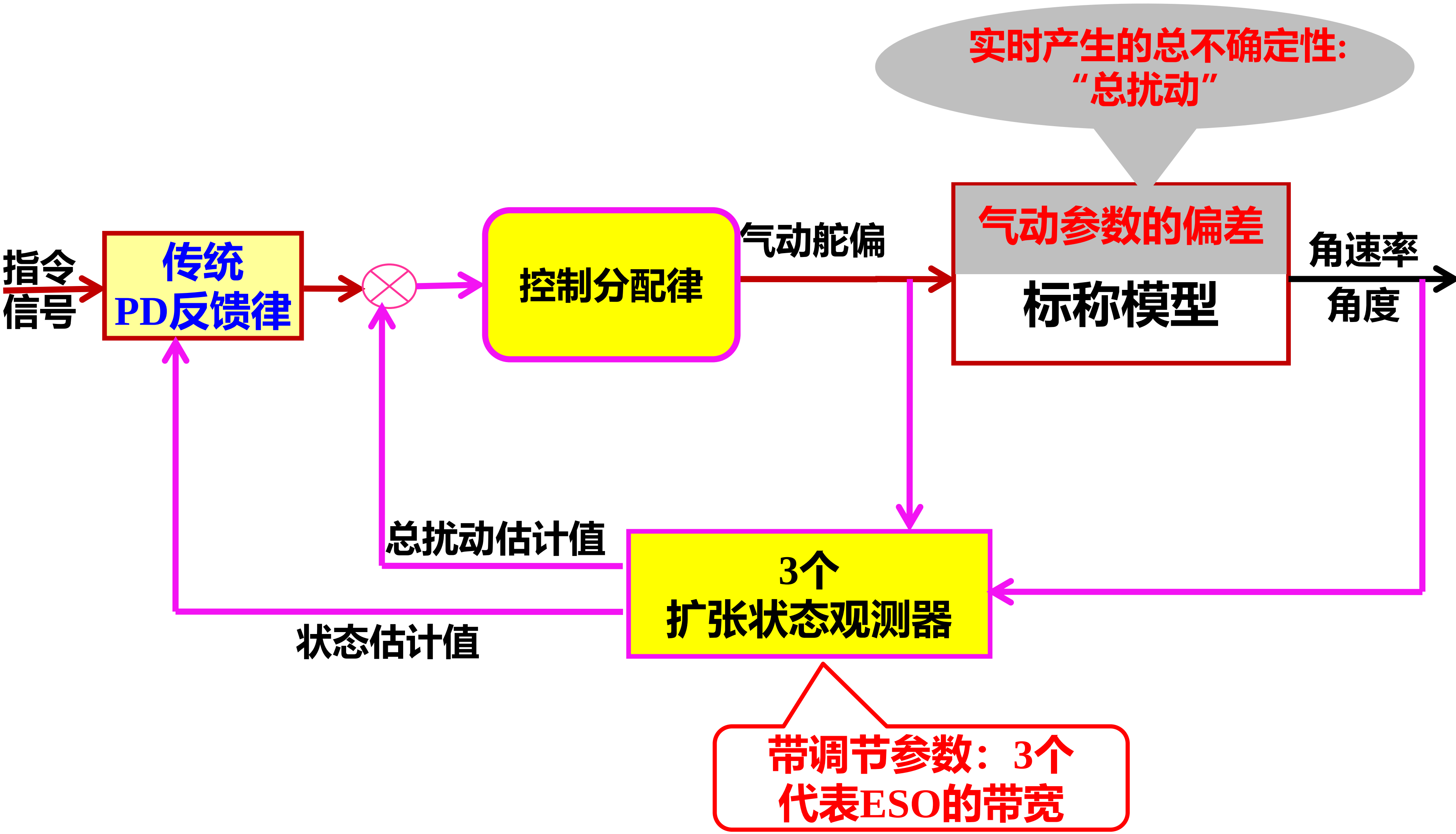
$$\begin{cases}\dot{z}_{q1,1}=z_{q2}+u_{3,1r}-\beta_{q1,1}\left(z_{q1,1}-q\right)\\ \dot{z}_{q2,1}=-\beta_{q2,1}\left(z_{q1,1}-q\right)\end{cases}.$$

$$\begin{cases}z_{q1,2}\text{估计q}\\ z_{q2,2}\text{估计}\bar{F}_q\end{cases}$$

$\beta_{r1,2}=2\omega_{ro,2},\ \beta_{r2,2}=\omega_{ro,2}^2,$

$\omega_{ro,2}$：ESO-3的带宽

全通道自抗扰控制设计（续）



全通道自抗扰控制设计（续）

横向：滚转角速度跟踪

航向：侧滑角稳定

纵向：俯仰角速率稳定

$$\begin{bmatrix} \dot{\beta} \\ \dot{\omega}_\beta \\ \dot{p} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} \omega_\alpha \\ \bar{F}_{\omega_\beta} \\ \bar{F}_p \\ \bar{F}_q \end{bmatrix} + \begin{bmatrix} 0 \\ \bar{B}_\beta \\ \bar{B}_p \\ \bar{B}_r \end{bmatrix} \begin{bmatrix} \delta_e \\ \delta_a \\ \delta_r \end{bmatrix}$$

控制输入

$$\begin{bmatrix} \delta_e \\ \delta_a \\ \delta_r \end{bmatrix} = \left(\begin{bmatrix} \bar{B}_\beta \\ \bar{B}_p \\ \bar{B}_r \end{bmatrix}_{3X3} \right)^{-1} \left(\begin{bmatrix} -z_{\beta 3,1} \\ -z_{p2,1} \\ -z_{q2} \end{bmatrix} + \begin{bmatrix} -k_{\beta p}\left(\beta-\beta^*\right)-k_{\beta d}\left(\dot{\beta}-\dot{\beta}^*\right) \\ -k_{pp}\left(p-p^*\right) \\ -k_{qp}\left(q-q^*\right) \end{bmatrix} \right)$$

补偿扰动

PD控制

$$\begin{cases} \dot{z}_{\beta 1,1} = z_{\beta 2,1} - \beta_{\beta 1,1}\left(z_{\beta 1,1}-\beta\right) \\ \dot{z}_{\beta 2,1} = z_{\beta 3,1} + u_{1,1r} - \beta_{\beta 2,1}\left(z_{\beta 1,1}-\beta\right). \\ \dot{z}_{\beta 3,1} = -\beta_{\beta 2,1}\left(z_{\beta 1,1}-\beta\right) \end{cases}$$

$$\begin{cases} \dot{z}_{p1,1} = z_{p2,1} + u_{2,1r} - \beta_{p1,1}\left(z_{p1,1}-p\right) \\ \dot{z}_{p2,1} = -\beta_{p2,1}\left(z_{p1,1}-p\right) \end{cases}.$$

$$\begin{cases} \dot{z}_{q1,1} = z_{q2} + u_{3,1r} - \beta_{q1,1}\left(z_{q1,1}-q\right) \\ \dot{z}_{q2,1} = -\beta_{q2,1}\left(z_{q1,1}-q\right) \end{cases}.$$

内容

一、飞机控制模型介绍

二、全通道自抗扰控制设计

三、参数调节方法及仿真结果

三、控制系统性能分析及参数调节方法

角速度方程

$$\begin{bmatrix}\dot{p}\\ \dot{q}\\ \dot{r}\end{bmatrix} = I^{-1}\left\{-\begin{bmatrix}p\\ q\\ r\end{bmatrix}\times I\begin{bmatrix}p\\ q\\ r\end{bmatrix}+\begin{bmatrix}0\\ -h_E r\\ h_E q\end{bmatrix}\right\} + I^{-1}\begin{bmatrix}\tilde{\bar{L}}\\ \tilde{\bar{M}}\\ \tilde{\bar{N}}\end{bmatrix} + I^{-1}B\begin{bmatrix}\delta_e\\ \delta_a\\ \delta_r\end{bmatrix}$$

$$B_\omega \triangleq I^{-1}B \in R^{3\times 5}$$

$$B = \begin{bmatrix} b_{11} & b_{12} & b_{13}\\ b_{21} & 0 & 0\\ b_{31} & b_{32} & b_{33}\end{bmatrix}$$

首要问题：$\bar{b}_{ij}$ 如何选取

$$B = \begin{bmatrix} \bar{b}_{11} & \bar{b}_{12} & \bar{b}_{13}\\ \bar{b}_{21} & 0 & 0\\ \bar{b}_{31} & \bar{b}_{32} & \bar{b}_{33}\end{bmatrix}$$

$$\begin{cases} b_{11} = \dfrac{1}{2}\rho V_T^2 Sb \left.\dfrac{\partial C_l(\alpha,\beta,\xi_l)}{\partial \delta_e}\right|_{\xi_l\in\left(\bar{\delta}_e,\delta_e\right]}, \quad b_{12} = \dfrac{1}{40}\rho V_T^2 Sb\left[\delta C_{l_{\delta_a}} + \delta C_{l_{\delta_{aLEF}}}\left(1-\dfrac{\delta_{LEF}}{25}\right)\right], \quad b_{13} = \dfrac{1}{60}\rho V_T^2 Sb\,\delta C_{l_{\delta r}} \\ b_{21} = \dfrac{1}{2}\rho V_T^2 S\bar{c}\left\{\left.\dfrac{\partial C_m(\alpha,\beta,\xi_1)}{\partial \delta_e}\right|_{\xi_1\in\left(\bar{\delta}_e,\delta_e\right]} + \left.\dfrac{\partial \delta C_{m_{ds}}(\alpha,\beta,\xi_2)}{\partial \delta_e}\right|_{\xi_2\in} + \left.\dfrac{\partial C_Z(\alpha,\beta,\xi_3)}{\partial \delta_e}\right|_{\xi_3\in\left(\bar{\delta}_e,\delta_e\right]}\left[x_{cgr}-x_{cg}\right]\right\} \\ b_{31} = \dfrac{1}{2}\rho V_T^2 Sb \left.\dfrac{\partial C_n(\alpha,\beta,\xi_n)}{\partial \delta_e}\right|_{\xi_n\in\left(\delta_e^*,\delta_e\right]}, \quad b_{32} = \dfrac{1}{40}\rho V_T^2 Sb\left\{\left[\delta C_{n_{\delta_a}} + \delta C_{n_{\delta_{aLEF}}}\left(1-\dfrac{\delta_{LEF}}{25}\right)\right] - \left[\delta C_{Y_{\delta_a}} + \delta C_{Y_{\delta_{aLEF}}}\left(1-\dfrac{\delta_{LEF}}{25}\right)\right]\left[x_{cgr}-x_{cg}\right]\dfrac{\bar{c}}{b}\right\} \end{cases}$$

首要问题：$\bar{b}_{ij}$ 如何选取

方法一：根据标称气动表进行实时插值计算

$$\bar{b}_{11}=\frac{1}{2}\rho V_T^2 Sb\frac{\partial C_l(\alpha,\beta,\delta_{e,r})}{\partial\delta_{e,r}},\quad \bar{b}_{12}=\frac{1}{40}\rho V_T^2 Sb\left[\delta C_{l_{\delta_a}}+\delta C_{l_{\delta_{aLEF}}}\left(1-\frac{\delta_{LEF}}{25}\right)\right],\quad \bar{b}_{13}=\frac{1}{60}\rho V_T^2 Sb\delta C_{l_{\delta r}}$$

$$\bar{b}_{21}=\frac{1}{2}\rho V_T^2 S\bar{c}\left\{\frac{\partial C_m(\alpha,\beta,\delta_{e,r})}{\partial\delta_{e,r}}+\frac{\partial C_{m_{ds}}(\alpha,\beta,\delta_{e,r})}{\partial\delta_{e,r}}+\frac{\partial C_Z(\alpha,\beta,\delta_{e,r})}{\partial\delta_{e,r}}\left[x_{cgr}-x_{cg}\right]\right\}$$

$$\bar{b}_{31}=\frac{1}{2}\rho V_T^2 Sb\frac{\partial C_n(\alpha,\beta,\delta_{e,r})}{\partial\delta_{e,r}}$$

$$\bar{b}_{32}=\frac{1}{40}\rho V_T^2 Sb\left\{\left[\delta C_{n_{\delta_a}}+\delta C_{n_{\delta_{aLEF}}}\left(1-\frac{\delta_{LEF}}{25}\right)\right]-\left[\delta C_{Y_{\delta_a}}+\delta C_{Y_{\delta_{aLEF}}}\left(1-\frac{\delta_{LEF}}{25}\right)\right]\left[x_{cgr}-x_{cg}\right]\frac{\bar{c}}{b}\right\}$$

$$\bar{b}_{33}=\frac{1}{60}\rho V_T^2 Sb\left\{\delta C_{n_{\delta_r}}-\delta C_{Y_{\delta_r}}\left[x_{cgr}-x_{cg}\right]\frac{\bar{c}}{b}\right\}$$

其中 $\delta_{e,r}$ 为飞机的实时舵偏偏角

首要问题：$\bar{b}_{ij}$ 如何选取

方法二：根据拟合/先验信息取为常值

$$\bar{b}_{11}=\frac{1}{2}\rho^*\left(V_T^*\right)^2 Sb\frac{\partial C_l\left(\alpha^*,\beta^*,\delta_e^*\right)}{\partial\delta_e},\quad \bar{b}_{12}=\frac{1}{40}\rho^*\left(V_T^*\right)^2 Sb\left[\delta C_{l_{\delta_a}}+\delta C_{l_{\delta_{aLEF}}}\left(1-\frac{\delta_{LEF}}{25}\right)\right]$$

$$\bar{b}_{13}=\frac{1}{60}\rho^*\left(V_T^*\right)^2 Sb\delta C_{l_{\delta r}}$$

$$\bar{b}_{21}=\frac{1}{2}\rho^*\left(V_T^*\right)^2 S\bar{c}\left\{\frac{\partial C_m\left(\alpha^*,\beta^*,\delta_e^*\right)}{\partial\delta_e}+\frac{\partial C_{m_{ds}}\left(\alpha^*,\beta^*,\delta_e^*\right)}{\partial\delta_e}+\frac{\partial C_Z\left(\alpha^*,\beta^*,\delta_e^*\right)}{\partial\delta_e}\left[x_{cgr}-x_{cg}\right]\right\}$$

$$\bar{b}_{31}=\frac{1}{2}\rho^*\left(V_T^*\right)^2 Sb\frac{\partial C_n\left(\alpha^*,\beta^*,\delta_e^*\right)}{\partial\delta_e}$$

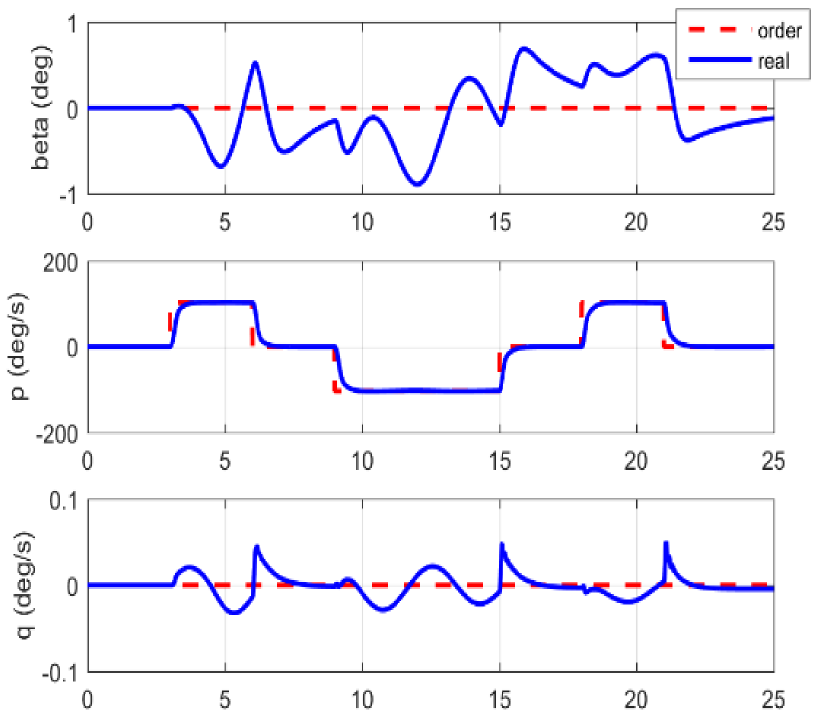
$$\bar{b}_{32}=\frac{1}{40}\rho^*\left(V_T^*\right)^2 Sb\left\{\left[\delta C_{n_{\delta_a}}+\delta C_{n_{\delta_{aLEF}}}\left(1-\frac{\delta_{LEF}}{25}\right)\right]-\left[\delta C_{Y_{\delta_a}}+\delta C_{Y_{\delta_{aLEF}}}\left(1-\frac{\delta_{LEF}}{25}\right)\right]\left[x_{cgr}-x_{cg}\right]\frac{\bar{c}}{b}\right\}$$

$$\bar{b}_{33}=\frac{1}{60}\rho^*\left(V_T^*\right)^2 Sb\left\{\delta C_{n_{\delta_r}}-\delta C_{Y_{\delta_r}}\left[x_{cgr}-x_{cg}\right]\frac{\bar{c}}{b}\right\}$$

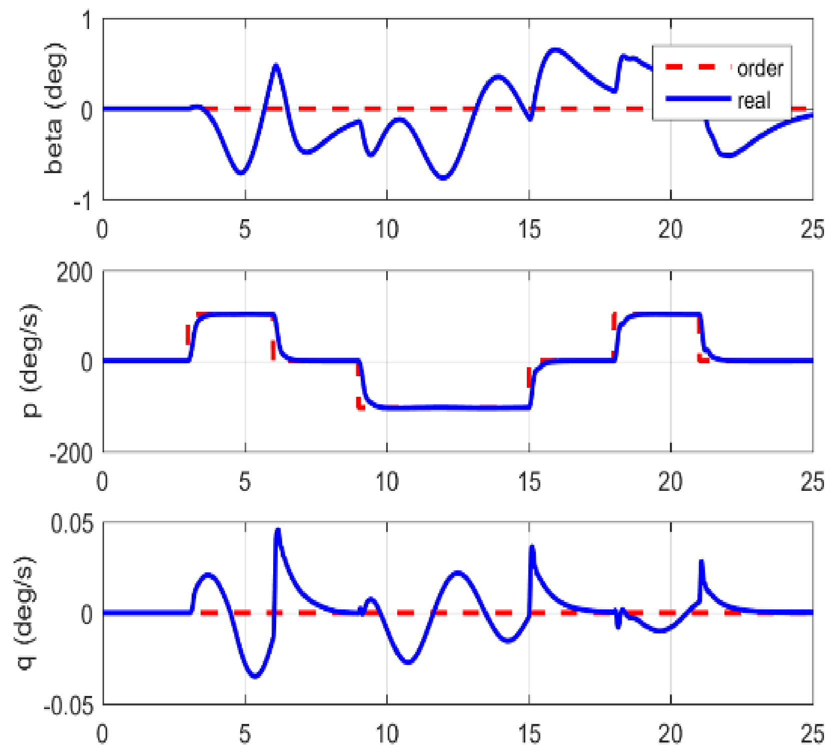
其中 $\left(\alpha^*,\beta^*,\delta_e^*\right)$ 为飞机平稳飞行时的取值

首要问题：$\bar{b}_{ij}$ 如何选取

方案一与方案二的闭环响应基本一致



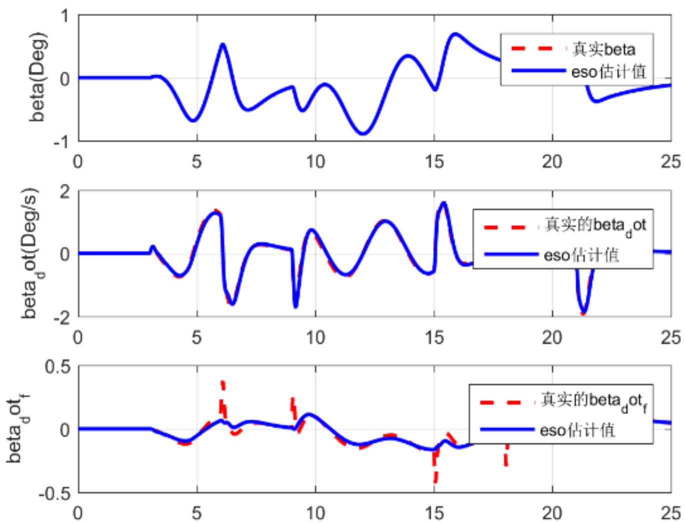
方案一



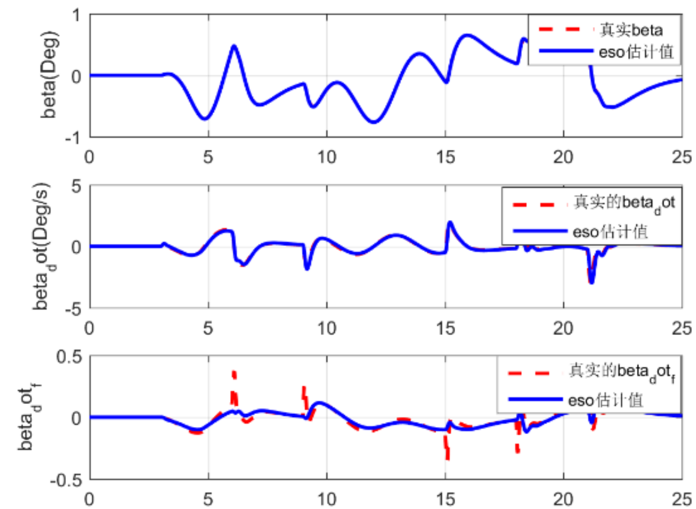
方案二

首要问题：$\bar{b}_{ij}$ 如何选取

ESO对“总扰动”的估计效果



方案一



方案二

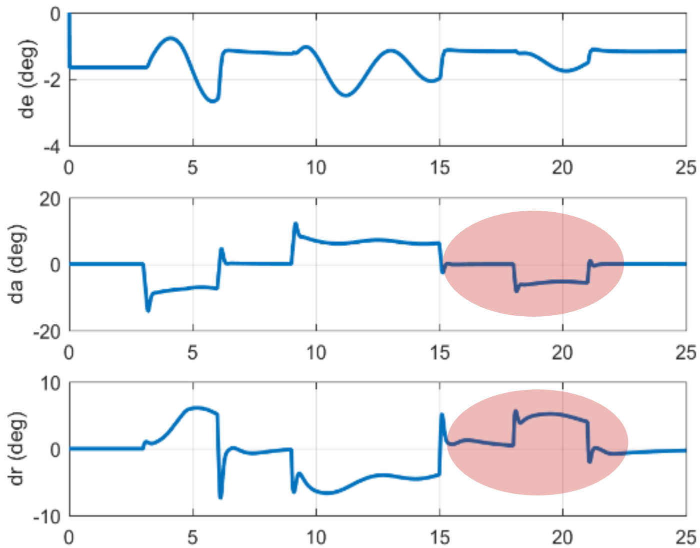
“总扰动”： $\bar{F}_{\omega_\alpha} = F_{\omega_\alpha} + \left(B_i - \bar{B}_i\right)\left[\delta_e, \delta_a, \delta_r, F_T \sin\delta_z, F_T \cos\delta_z \sin\delta_y\right]^T$

由于“总扰动“包含了B 的偏差

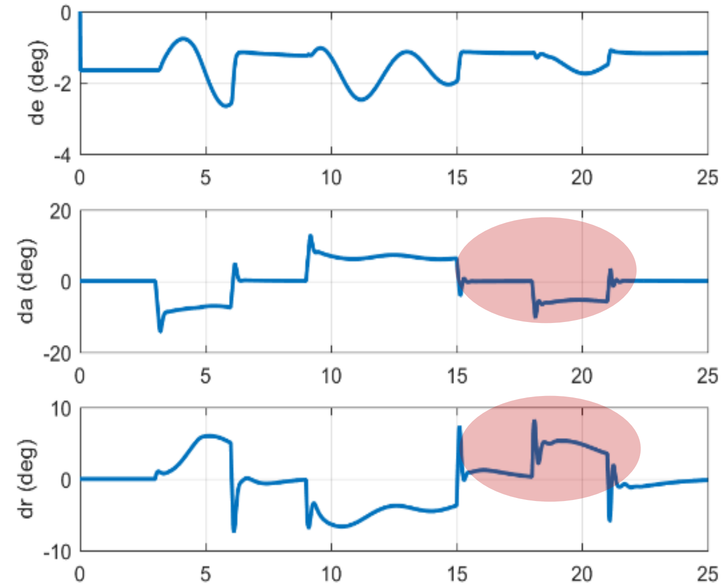
因此方案一和方案二都可对 B 的偏差进行补偿

三、控制系统性能分析及参数调节方法

➤方案一比方案二对应的控制输入更平滑



方案一



方案二

➤方案一比方案二的计算量大

方案一需要根据标称气动表数据进行实时插值计算

方案二取舵偏增益阵估计值为常值

ESO参数调节

ESO-1

$$\begin{cases}\dot{z}_{\beta 1,1}=z_{\beta 2,1}-\beta_{\beta 1,1}\left(z_{\beta 1,1}-\beta\right)\\ \dot{z}_{\beta 2,1}=z_{\beta 3,1}+u_{1,1r}-\beta_{\beta 2,1}\left(z_{\beta 1,1}-\beta\right).\\ \dot{z}_{\beta 3,1}=-\beta_{\beta 2,1}\left(z_{\beta 1,1}-\beta\right)\end{cases}$$

$$\begin{cases}z_{\beta 1,2}\text{估计}\beta\\ z_{\beta 2,2}\text{估计}\dot{\beta}\\ z_{\beta 3,2}\text{估计}\bar{F}_{\omega_\beta}\end{cases}$$

$\beta_{\alpha 1,2}=3\omega_{\alpha o,2},\ \beta_{\alpha 2,2}=3\omega_{\alpha o,2}^2,$

$\beta_{\alpha 3,2}=\omega_{\alpha o,2}^3,$

$\omega_{\alpha o,2}$: ESO-1的带宽

ESO-2

$$\begin{cases}\dot{z}_{p1,1}=z_{p2,1}+u_{2,1r}-\beta_{p1,1}\left(z_{p1,1}-p\right)\\ \dot{z}_{p2,1}=-\beta_{p2,1}\left(z_{p1,1}-p\right)\end{cases}.$$

$$\begin{cases}z_{p1,2}\text{估计}p\\ z_{p2,2}\text{估计}\bar{F}_p\end{cases}$$

$\beta_{p1,2}=2\omega_{po,2},\ \beta_{p2,2}=\omega_{po,2}^2,$

$\omega_{po,2}$: ESO-2的带宽

ESO-3

$$\begin{cases}\dot{z}_{q1,1}=z_{q2}+u_{3,1r}-\beta_{q1,1}\left(z_{q1,1}-q\right)\\ \dot{z}_{q2,1}=-\beta_{q2,1}\left(z_{q1,1}-q\right)\end{cases}.$$

$$\begin{cases}z_{q1,2}\text{估计q}\\ z_{q2,2}\text{估计}\bar{F}_q\end{cases}$$

$\beta_{r1,2}=2\omega_{ro,2},\ \beta_{r2,2}=\omega_{ro,2}^2,$

$\omega_{ro,2}$: ESO-3的带宽

观测器带宽
大于扰动主要成分的频率

$\omega_{po,1}=15;$

$\omega_{qo,1}=5;$

$\omega_{\beta o,1}=15;$

PD反馈的参数设计

横向：滚转角速度稳定

航向：偏航角速度稳定

纵向：迎角跟踪

$$\begin{bmatrix}\dot{\beta}\\ \dot{\omega}_{\beta}\\ \dot{p}\\ \dot{q}\end{bmatrix}=\begin{bmatrix}\omega_{\alpha}\\ \bar{F}_{\omega_{\beta}}\\ \bar{F}_{p}\\ \bar{F}_{q}\end{bmatrix}+\begin{bmatrix}0\\ \bar{B}_{\beta}\\ \bar{B}_{p}\\ \bar{B}_{r}\end{bmatrix}\begin{bmatrix}\delta_{e}\\ \delta_{a}\\ \delta_{r}\end{bmatrix}$$

控制输入

$$\begin{bmatrix}\delta_{e}\\ \delta_{a}\\ \delta_{r}\end{bmatrix}=\left(\begin{bmatrix}\bar{B}_{\beta}\\ \bar{B}_{p}\\ \bar{B}_{r}\end{bmatrix}_{3X3}\right)^{-1}\left(\begin{bmatrix}-z_{\beta 3,1}\\ -z_{p2,1}\\ -z_{q2}\end{bmatrix}+\begin{bmatrix}-k_{\beta p}\left(\beta-\beta^{*}\right)-k_{\beta d}\left(\dot{\beta}-\dot{\beta}^{*}\right)\\ -k_{pp}\left(p-p^{*}\right)\\ -k_{qp}\left(q-q^{*}\right)\end{bmatrix}\right)$$

PD反馈的参数设计：系统预期响应

$$\begin{cases}k_{\beta p,1}=2k_{\beta 1},k_{\beta d,1}=k_{\beta 1}^{2},k_{\beta 1}=1,\\ k_{pp,1}=10,\\ k_{qp,1}=3.\end{cases}$$

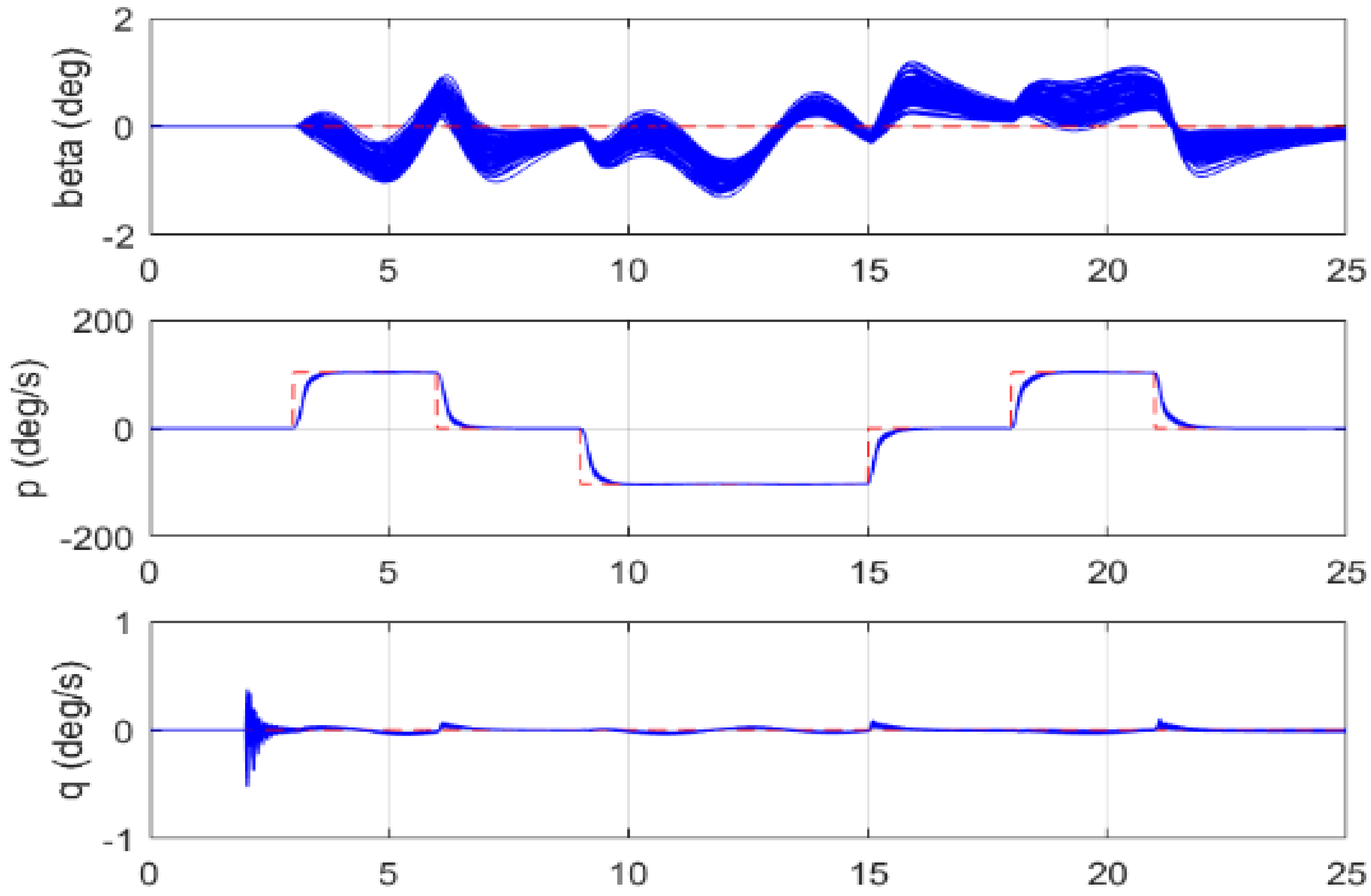
正常飞行状态：气动参数拉偏

43个气动参数的随机拉偏30%（均匀分布）

气动参数					
C_X	$\delta C_{X_{LEF}}$	C_{X_q}	$\delta C_{X_{q_{LEF}}}$	C_Y	$\delta C_{Y_{LEF}}$
$\delta C_{Y_{\delta_a}}$	$\delta C_{Y_{\delta_{a_{LEF}}}}$	$\delta C_{Y_{\delta_r}}$	$\delta C_{Y_{p_{LEF}}}$	C_{Y_p}	C_{Y_r}
$\delta C_{Y_{r_{LEF}}}$	C_Z	$\delta C_{Z_{LEF}}$	C_{Z_q}	$\delta C_{Z_{q_{LEF}}}$	C_l
$\delta C_{l_{LEF}}$	$\delta C_{l_{\delta_a}}$	$\delta C_{l_{\delta_{a_{LEF}}}}$	$\delta C_{l_{\delta_r}}$	C_{l_r}	$\delta C_{l_{r_{LEF}}}$
C_{l_p}	$\delta C_{l_{p_{LEF}}}$	δC_{l_β}	δC_{n_β}	C_m	$\delta C_{m_{LEF}}$
C_{m_q}	$\delta C_{m_{q_{LEF}}}$	δC_m	$\delta C_{m_{ds}}$	C_n	$\delta C_{n_{LEF}}$
$\delta C_{n_{\delta_a}}$	$\delta C_{n_{\delta_{a_{LEF}}}}$	$\delta C_{n_{\delta_r}}$	C_{n_r}	$\delta C_{n_{r_{LEF}}}$	C_{n_p}
$\delta C_{n_{p_{LEF}}}$					

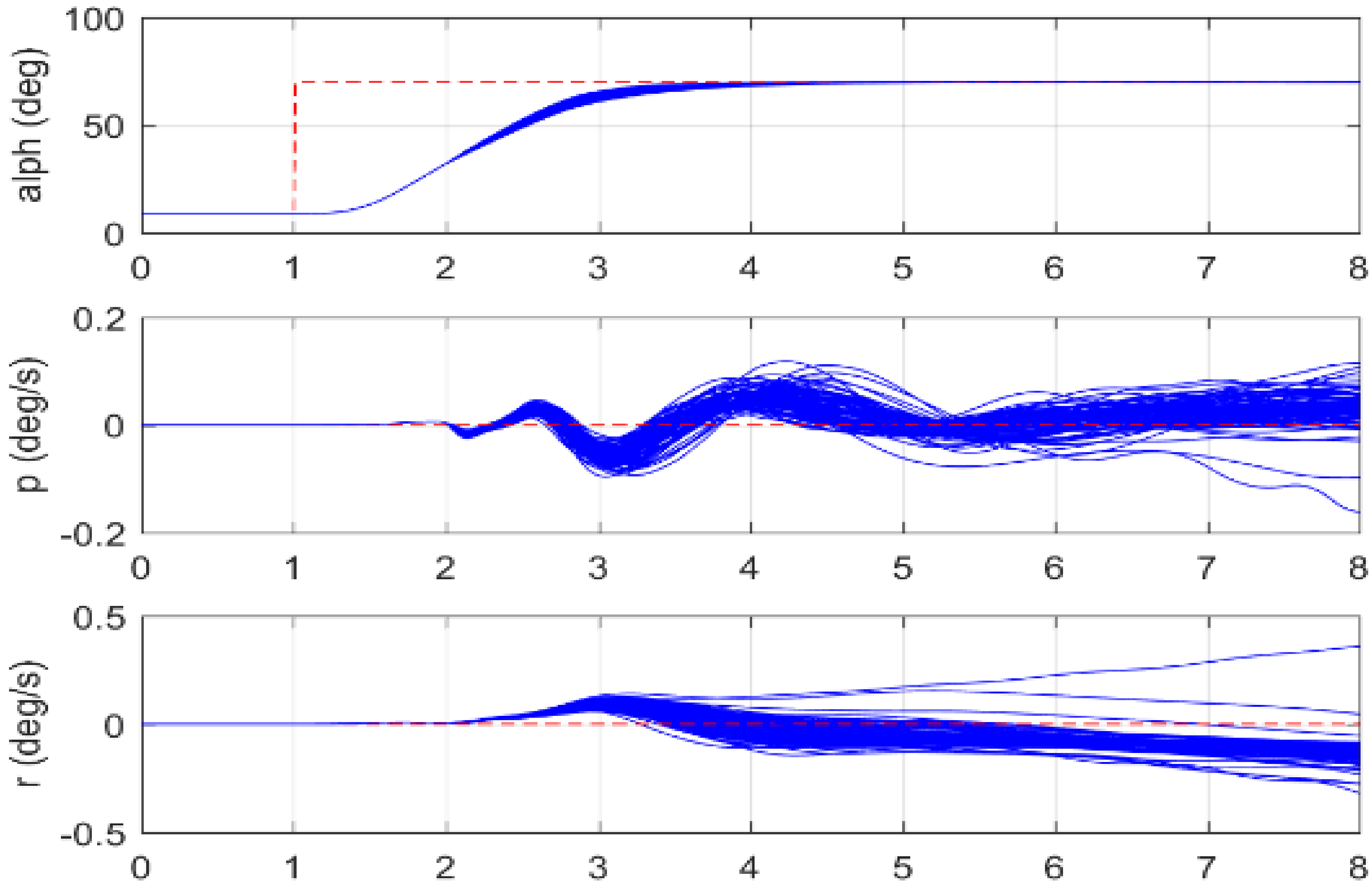
控制系统的动态性能分析(续)

正常飞行状态下闭环响应: 100种随机偏差参数组合



控制系统的动态性能分析(续)

控制迎角下的闭环响应: 100种随机偏差参数组合



控制系统的动态性能分析

$$x \triangleq \begin{bmatrix} \beta \\ \omega_\beta \\ p \\ r \end{bmatrix}，设\bar{x}^* \triangleq \begin{bmatrix} \bar{\beta}^* \\ \bar{\omega}_\beta^* \\ \bar{p}^* \\ \bar{r}^* \end{bmatrix}为x的理想轨迹：\begin{bmatrix} \dot{\bar{\beta}}^* \\ \dot{\bar{\omega}}_\beta^* \\ \dot{\bar{p}}^* \\ \dot{\bar{q}}^* \end{bmatrix} = \begin{bmatrix} \bar{\omega}_\beta^* \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -k_{\alpha p}\left(\bar{\beta}^*\right) - k_{\alpha d}\left(\bar{\omega}_\beta^*\right) \\ -k_p \bar{p}^* \\ -k_r \bar{q}^* \end{bmatrix}, \bar{x}^*(t_0) = x(t_0)$$

定理1. **在一定的条件下，自抗扰控制的闭环系统满足，存在 ω_0^* 使得 $\forall \omega_0 \geq \omega_0^*$**

$$\left\| x(t) - \bar{x}^*(t) \right\| \leq \gamma_1^* \frac{\max\{\ln \omega_e, 1\}}{\omega_e}, \quad \forall t \geq t_0$$

其中 γ_1^* 为一个常值.

- **基于自抗扰控制的闭环系统良好的瞬态性能；**
- **控制器参数的主要物理意义**

PD参数：调节理想轨迹的性能

ESO带宽：调节实际轨迹与理想轨迹的误差大小

Wenchao Xue*, and Yi Huang, Performance Analysis of 2-DOF Tracking Control for a Class of nonlinear Uncertain Systems with Discontinuous Disturbances, International Journal of Robust and nonlinear Control, 2018, 28:1456–1473.

自抗扰控制系统稳定性的基本理论结果

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自抗扰控制思想：运动控制系统的例子

- **对象** $\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2, w(t), t) + bu \end{cases}$

  $y = x_1$

- **标准型** $\ddot{y} = bu$

- **总扰动估计值** $\hat{f}(t) \approx f(x_1, x_2, w(t), t)$

总扰动的估计：扩张状态观测器

$$x_3 = f(y, \dot{y}, w, t)$$

$$\ddot{y} = f(x_1, x_2, w(t), t) + bu \quad \Longrightarrow \quad \begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 + bu, \\ \dot{x}_3 = \dot{f} \\ y = x_1 \end{cases}$$

Extended State Observer (ESO)

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - \beta_1 G_1(\hat{X}_1 - y) \\ \dot{\hat{X}}_2 = \hat{X}_3 - \beta_2 G_2(\hat{X}_1 - y) + bu \\ \dot{\hat{X}}_3 = -\beta_3 G_3(\hat{X}_1 - y) \end{cases}$$

$$\Downarrow$$

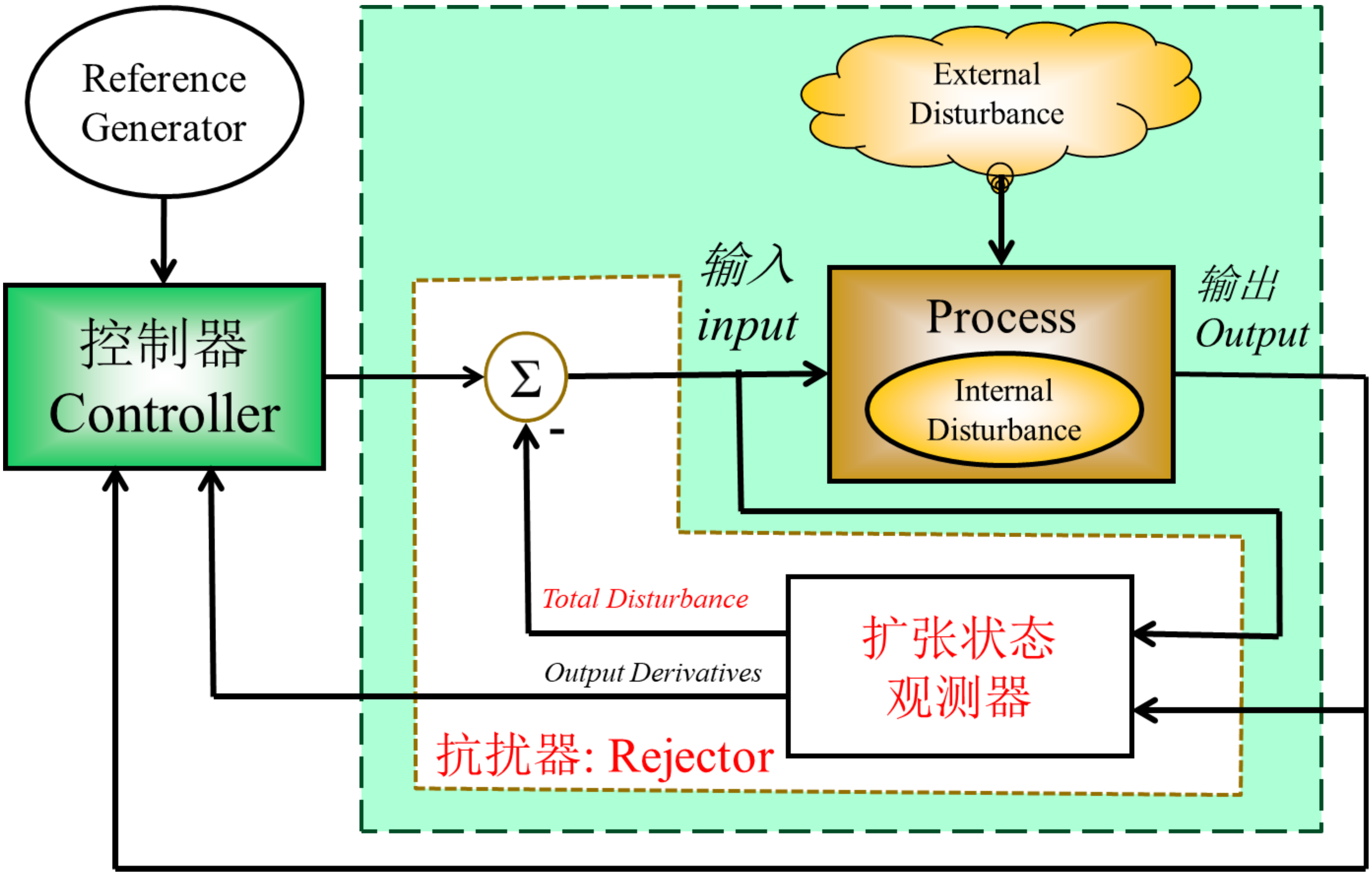
$$\hat{X}_1 \approx x_1 \quad \hat{X}_2 \approx x_2 \quad \hat{X}_3 \approx x_3 = f$$

例子：运动控制系统

自抗扰设计

$$\ddot{y} = f(x_1, x_2, w(t), t) + bu \xrightarrow{u = u_0 - \frac{\hat{f}}{b},\ \hat{f} \approx f} \ddot{y} \approx bu_0$$

自抗扰控制的核心部分：扩张状态观测器



线性ADRC设计

A parameterized linear ADRC (LADRC) was proposed in Gao Z. (2003, CDC)

使得控制器的参数具有物理意义，更加易于调节

设有受到未知外扰作用的非线不确定对象

$$\begin{cases} \dot{X}_1 = X_2 \\ \cdots \\ \dot{X}_{n-1} = X_n \\ \dot{X}_n = X_{n+1} + \bar{B}(t)U(t) \end{cases}$$

$$X_{n+1} = F(X(t), w(t), t) + \left(B(X,t) - \bar{B}(t)\right)U(t)$$

线性ADRC

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - \beta_1(\hat{X}_1 - Y) \\ \cdots \\ \dot{\hat{X}}_n = \hat{X}_{n+1} - \beta_n(\hat{X}_1 - Y) + \bar{B}(t)U(t) \\ \dot{\hat{X}}_{n+1} = -\beta_{n+1}(\hat{X}_1 - Y) \end{cases}$$

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} - \bar{B}^{-1}(t)\sum_{i=1}^{n} k_i \hat{X}_i$$

闭环动态设计：极点配置

增益：极点配置

预期系统的响应

带宽参数化的线性ADRC

设有受到未知外扰作用的非线不确定对象

$$\begin{cases} \dot{X}_1 = X_2 \\ \cdots \\ \dot{X}_{n-1} = X_n \\ \dot{X}_n = X_{n+1} + \bar{B}(t)U(t) \end{cases}$$

$$X_{n+1} = F(X,t) + d(t) + \left(B(X,t) - \bar{B}(t)\right)U(t)$$

LADRC

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - \beta_1(\hat{X}_1 - Y) \\ \cdots \\ \dot{\hat{X}}_n = \hat{X}_{n+1} - \beta_n(\hat{X}_1 - Y) + \bar{B}(t)U(t) \\ \dot{\hat{X}}_{n+1} = -\beta_{n+1}(\hat{X}_1 - Y) \end{cases}$$

$$s^{n+1} + \sum_{i=1}^{n+1} \beta_i s^{n-i+1} = (s + \omega_e)^{n+1}, \quad \omega_e > 0$$

LESO带宽

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} - \bar{B}^{-1}(t)\sum_{i=1}^{n} k_i \hat{X}_i$$

闭环动态设计：极点配置

$$s^n + \sum_{i=1}^{n} k_i s^{i-1} = (s + \omega_c)^n, \quad \omega_c > 0$$

预期动态的带宽

$$\begin{cases}\dot{X}_1 = X_2 \\ \cdots \\ \dot{X}_{n-1} = X_n \\ \dot{X}_n = X_{n+1} + \bar{B}(t)U(t)\end{cases}$$

标准型 + 总扰动

ESO (1995)

$$\begin{array}{l}\hat{X}_1 \to Y \\ \cdots \\ \hat{X}_n \to X_n (= Y^{(n-1)}) \\ \hat{X}_{n+1} \to X_{n+1}\end{array}$$

ADRC (Han,1998)

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} + U_0(Y, \hat{X}_2, \cdots \hat{X}_n)$$

抵消总扰动的效应

对标称系统的控制

$$\begin{cases}\dot{X}_1 = X_2 \\ \cdots \\ \dot{X}_{n-1} = X_n \\ \dot{X}_n = \bar{B}(t)U_0(\bullet)\end{cases}$$

鲁棒性

预期的性能

双自由度设计

自抗扰控制系统的稳定性分析

- 积分串联型结构的系统
  - 线性ESO的估计误差分析
  - 闭环系统分析
- 一般结构下的系统
  - 可观性分析
  - ESO设计

自抗扰控制系统的稳定性分析

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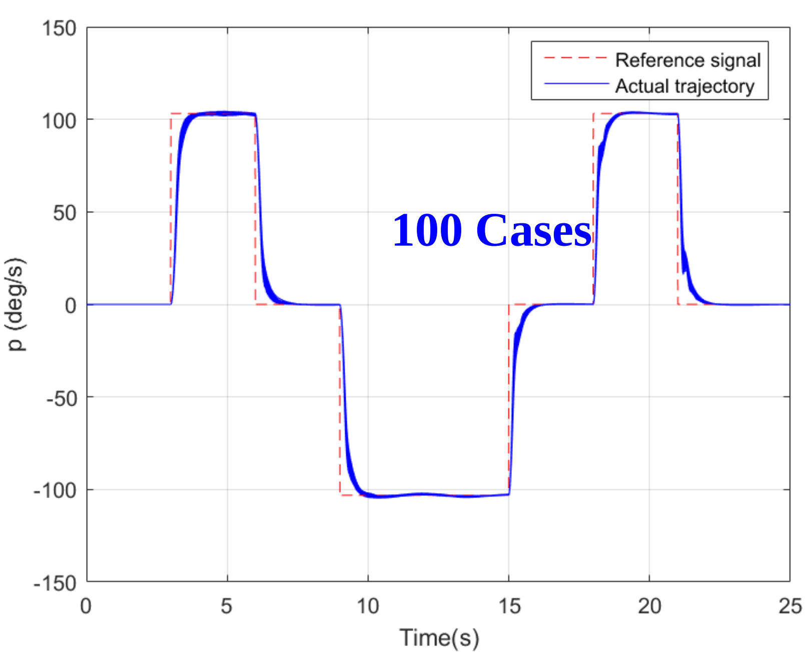
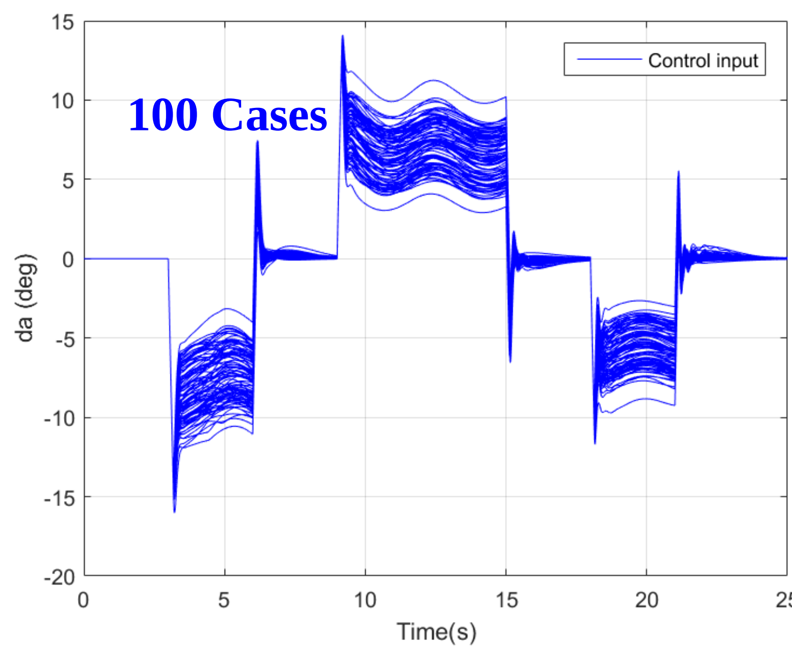
自抗扰控制系统的特点：
在不确定性保证了闭环系统输出达到期望的瞬态

Simulation results of flight control: roll rate tracking under ±40% random derivations of parameters

闭环系统的角速率响应



控制输入



$$\dot{p} = b(t)u + F(p,u,t)$$

p ：角速率

$b(t)$: 控制输入增益

F : 总扰动

u : 控制输入

线性ESO的估计误差分析

不确定系统：

$$\begin{cases} \dot{X}_1 = X_2 \\ \cdots \\ \dot{X}_{n-1} = X_n \\ \dot{X}_n = X_{n+1} + \bar{B}(t)U(t) \end{cases}$$

线性ESO：

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 + l_1(\hat{X}_1 - Y) \\ \cdots \\ \dot{\hat{X}}_n = \hat{X}_{n+1} + l_n(\hat{X}_1 - Y) + \bar{B}(t)U(t) \\ \dot{\hat{X}}_{n+1} = l_{n+1}(\hat{X}_1 - Y) \end{cases}$$

总扰动： $X_{n+1} = F(X,t) + d(t) + \left(B(X,t) - \bar{B}(t)\right)U(t)$

ESO的带宽： ω_e $\quad s^{n+1} + \sum_{i=1}^{n+1} l_i s^{n+1-i} = (s+\omega_e)^{n+1}$

经典的理论分析结果：

[Q Zheng, L Gao, Z Gao, 2007CDC] Assuming that $\dot{X}_{n+1}(.)$ is a bounded function, then there exists a finite $T_1 > 0$ and a positive integer k such that

$$\left|X_i - \hat{X}_i\right| \le O\left(\frac{1}{\omega_e^k}\right) \qquad \forall t \ge T_1 > 0, \omega_e > 0, \ i = 1, 2, \ldots, n+1$$

线性ESO的估计误差分析：定理1

定理1：假设总扰动的导数 $\dot{X}_{n+1}$ 有界，则

$$\lim_{t\to\infty}\left|X_i(\mathrm{t})-\hat{X}_i(\mathrm{t})\right|\le O\left(\frac{1}{\omega_e^{n+2-i}}\right),\qquad \omega_e>0,\qquad i=1,2,\dots,n+1.$$

证明：首先定义ESO的估计误差 $\hat{E}\triangleq\begin{bmatrix}\hat{E}_1\\ \hat{E}_2\\ \vdots\\ \hat{E}_{n+1}\end{bmatrix}=\begin{bmatrix}X_1-\hat{X}_1\\ X_2-\hat{X}_2\\ \vdots\\ X_{n+1}-\hat{X}_{n+1}\end{bmatrix}$

进而得到ESO估计误差的动态方程

$$\begin{bmatrix}\dot{\hat{E}}_1\\ \dot{\hat{E}}_2\\ \vdots\\ \dot{\hat{E}}_{n+1}\end{bmatrix}=\begin{bmatrix}-l_1 & 1 & \cdots & 0\\ -l_2 & 0 & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ -l_{n+1} & 0 & \cdots & 0\end{bmatrix}\begin{bmatrix}\hat{E}_1\\ \hat{E}_2\\ \vdots\\ \hat{E}_{n+1}\end{bmatrix}+\begin{bmatrix}0\\ 0\\ \vdots\\ \dot{X}_{n+1}\end{bmatrix}$$

其中： $s^{n+1}+\sum_{i=1}^{n+1}l_i s^{n+1-i}=(s+\omega_e)^{n+1}$ 或者 $l_i=C_{n+1}^i\omega_e{}^i=\dfrac{(n+1)!}{i!(n+1-i)!}\omega_e{}^i$

线性ESO的估计误差分析：定理1（续）

再引入新状态及等价变换： $$\xi \triangleq \begin{bmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_{n+1} \end{bmatrix} = \begin{bmatrix} \omega_e^n & 0 & \cdots & 0 \\ 0 & \omega_e^{n-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} \hat{E}_1 \\ \hat{E}_2 \\ \vdots \\ \hat{E}_{n+1} \end{bmatrix}$$

奇异摄动理论中的典型变换，

使得变换后的状态在

同一个时间尺度上

即有： $\xi_i = \omega_e^{n+1-i} \hat{E}_i, \dot{\xi} = \omega_e^{n+1-i} \dot{\hat{E}},$

进而得到新状态的动态方程： $$\dot{\xi} = \begin{bmatrix} \omega_e^n & 0 & \cdots & 0 \\ 0 & \omega_e^{n-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \dot{\hat{E}}$$

$$= \begin{bmatrix} \omega_e^n & 0 & \cdots & 0 \\ 0 & \omega_e^{n-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \left(\begin{bmatrix} -l_1 & 1 & \cdots & 0 \\ -l_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -l_{n+1} & 0 & \cdots & 0 \end{bmatrix} \hat{E} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \dot{X}_{n+1} \end{bmatrix} \right)$$

$$= \omega_e \begin{bmatrix} -C_{n+1}^1 & 1 & \cdots & 0 \\ -C_{n+1}^2 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -C_{n+1}^{n+1} & 0 & \cdots & 0 \end{bmatrix} \xi + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \dot{X}_{n+1} \end{bmatrix}$$

系统矩阵的元素都为ESO

带宽的比例函数

线性ESO的估计误差分析：定理1（续）

$$\dot{\xi} = \omega_e \begin{bmatrix} -C_{n+1}^1 & 1 & 0 & \cdots & 0 \\ -C_{n+2}^2 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -C_{n+1}^n & 0 & 0 & \cdots & 1 \\ -C_{n+1}^{n+1} & 0 & 0 & \cdots & 0 \end{bmatrix} \xi + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \dot{X}_{n+1} \end{bmatrix}$$

因为矩阵 $\begin{bmatrix} -C_{n+1}^1 & 1 & 0 & \cdots & 0 \\ -C_{n+2}^2 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -C_{n+1}^n & 0 & 0 & \cdots & 1 \\ -C_{n+1}^{n+1} & 0 & 0 & \cdots & 0 \end{bmatrix}$ **为赫尔维兹矩阵，则存在一个正定矩阵** P **使得：**

$$\begin{bmatrix} -C_{n+1}^1 & 1 & 0 & \cdots & 0 \\ -C_{n+2}^2 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -C_{n+1}^n & 0 & 0 & \cdots & 1 \\ -C_{n+1}^{n+1} & 0 & 0 & \cdots & 0 \end{bmatrix}^T P + P \begin{bmatrix} -C_{n+1}^1 & 1 & 0 & \cdots & 0 \\ -C_{n+2}^2 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -C_{n+1}^n & 0 & 0 & \cdots & 1 \\ -C_{n+1}^{n+1} & 0 & 0 & \cdots & 0 \end{bmatrix} = -I$$

定义矩阵 $V(\xi) \triangleq \xi^T P \xi$ **，则有**

$$\frac{dV(\xi)}{dt} = -\omega_e \xi^T \xi + 2\xi^T P \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \dot{X}_{n+1} \end{bmatrix}$$

线性ESO的估计误差分析：定理1（续）

定义矩阵 $V(\xi)\triangleq\xi^T P\xi$ ，令 c_1, c_2 分别为矩阵 P 的最小和最大特征根，则有

$$\frac{dV(\xi)}{dt}=-\omega_e\xi^T\xi+2\xi^T P\begin{bmatrix}0\\0\\\vdots\\\dot{X}_{n+1}\end{bmatrix}$$

$$\Downarrow\quad c_1\|\xi\|^2\le V(\xi)\triangleq\xi^T P\xi\le c_2\|\xi\|^2$$

$$\frac{dV(\xi)}{dt}\le-\omega_e\frac{V(\xi)}{c_2}+2\|\xi\|c_2\left\|\dot{X}_{n+1}\right\|\le-\omega_e\frac{V(\xi)}{c_2}+2\frac{\sqrt{V(\xi)}}{\sqrt{c_1}}c_2\left\|\dot{X}_{n+1}\right\|$$

$$\Downarrow\quad\frac{d\sqrt{V(\xi)}}{dt}=\frac{1}{2\sqrt{V(\xi)}}\frac{dV(\xi)}{dt}$$

$$\frac{d\sqrt{V(\xi)}}{dt}\le-\omega_e\frac{\sqrt{V(\xi)}}{2c_2}+\frac{c_2}{\sqrt{c_1}}\left\|\dot{X}_{n+1}\right\|$$

线性ESO的估计误差分析：定理1（续）

$$\frac{d\sqrt{V(\xi)}}{dt} \le -\omega_e \frac{\sqrt{V(\xi)}}{2c_2} + \frac{c_2}{\sqrt{c_1}}\left\|\dot{X}_{n+1}\right\|$$

又由于 $\dot{X}_{n+1}$ 有界，则存在一个常数 M，使得

$$\left\|\dot{X}_{n+1}\right\| \le M, \qquad \forall t \ge t_0.$$

有

$$\frac{d\sqrt{V(\xi)}}{dt} \le -\omega_e \frac{\sqrt{V(\xi)}}{2c_2} + \frac{c_2}{\sqrt{c_1}} M, \qquad \forall t \ge t_0$$

由Gronwall–Bellman不等式有

$$\sqrt{V(\xi)} \le e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{V(\xi(t_0))} + \frac{c_2}{\sqrt{c_1}} M \int_{t_0}^{t} e^{-\frac{\omega_e}{2c_2}(t-s)} ds, \qquad \forall t \ge t_0$$

线性ESO的估计误差分析：定理1（续）

根据 $V(\xi)\triangleq\xi^T P\xi$ **和** $c_1\|\xi\|^2\le V(\xi)\triangleq\xi^T P\xi\le c_2\|\xi\|^2$ **进一步得：**

$$\|\xi\|\le\sqrt{\frac{V(\xi)}{c_1}}\le e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{\frac{V(\xi(t_0))}{c_1}}+\frac{c_2}{c_1}M\int_{t_0}^{t}e^{-\frac{\omega_e}{2c_2}(t-s)}ds,\qquad\forall t\ge t_0$$

又由于$\xi_i=\omega_e^{n+1-i}\hat{E}_i$和$\hat{E}_i=\frac{1}{\omega_e^{n+1-i}}\xi_i$，**则有**

$$\begin{aligned}\left|\hat{E}_i(t)\right|&=\frac{1}{\omega_e^{n+1-i}}\left|\xi_i(t)\right|\\&\le\frac{1}{\omega_e^{n+1-i}}\|\xi(t)\|\\&\le\frac{1}{\omega_e^{n+1-i}}\left[e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{\frac{V(\xi(t_0))}{c_1}}+\frac{c_2}{c_1}M\int_{t_0}^{t}e^{-\frac{\omega_e}{2c_2}(t-s)}ds\right].\end{aligned}$$

线性ESO的估计误差分析：定理1（续）

再取极限可得到：

$$
\begin{aligned}
\lim_{t\to\infty}\left|\hat{E}_i(t)\right| &\le \lim_{t\to\infty}\frac{1}{\omega_e^{n+1-i}}\left[e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{\frac{V(\xi(t_0))}{c_1}}+\frac{c_2}{c_1}M\int_{t_0}^{t}e^{-\frac{\omega_e}{2c_2}(t-s)}ds\right]\\
&\le \lim_{t\to\infty}\frac{1}{\omega_e^{n+1-i}}\left[e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{\frac{V(\xi(t_0))}{c_1}}+\frac{c_2}{c_1}M\int_{t_0}^{t}e^{-\frac{\omega_e}{2c_2}(t-s)}ds\right]\\
&\le \lim_{t\to\infty}\frac{1}{\omega_e^{n+1-i}}\left[e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{\frac{V(\xi(t_0))}{c_1}}+\frac{c_2}{c_1}M\frac{2c_2}{\omega_e}\left(1-e^{-\frac{\omega_e}{2c_2}(t-t_0)}\right)\right]\\
&\le \frac{1}{\omega_e^{n+1-i}}\frac{c_2}{c_1}M\frac{2c_2}{\omega_e}.
\end{aligned}
$$

所以， $\left|X_i-\hat{X}_i\right|\le O\left(\frac{1}{\omega_e^{n+2-i}}\right),\ \forall t\ge T_1>t_0,\ \omega_e>0,\ i=1,2,\ldots,n+1$ **证毕。**

线性ESO的估计误差分析：定理2

定理2：假设总扰动的导数有界，则

$$\left|X_i(\mathrm{t})-\hat{X}_i(\mathrm{t})\right|\le O\left(\frac{1}{\omega_e^{n+2-i}}\right),\quad \forall t\ge T\triangleq t_0+2(n+1)c_2\max\left\{\frac{\ln\omega_e}{\omega_e},0\right\},\quad i=1,2,\dots,n+1$$

证明：同上，定义矩阵$V(\xi)\triangleq\xi^T P\xi$**，有**

$$\frac{dV(\xi)}{dt}=-\omega_e\xi^T\xi+2\xi^T P\begin{bmatrix}0\\0\\\vdots\\\dot{X}_{n+1}\end{bmatrix}$$

$$\frac{dV(\xi)}{dt}\le-\omega_e\frac{V(\xi)}{2c_2}+2\|\xi\|c_2\left\|\dot{X}_{n+1}\right\|\le-\omega_e\frac{V(\xi)}{2c_2}+2\frac{\sqrt{V(\xi)}}{\sqrt{c_1}}c_2\left\|\dot{X}_{n+1}\right\|$$

$$\frac{d\sqrt{V(\xi)}}{dt}\le-\omega_e\frac{\sqrt{V(\xi)}}{2c_2}+\frac{c_2}{\sqrt{c_1}}\left\|\dot{X}_{n+1}\right\|$$

线性ESO的估计误差分析：定理2（续）

又由于 $\dot{X}_{n+1}$ 有界，则存在一个常数 M，使得

$$\left\|\dot{X}_{n+1}\right\| \le M, \qquad \forall t \ge t_0.$$

因此有

$$\frac{d\sqrt{V(\xi)}}{dt} \le -\omega_e \frac{\sqrt{V(\xi)}}{2c_2} + \frac{c_2}{\sqrt{c_1}} M$$

由Gronwall–Bellman不等式有

$$\sqrt{V(\xi)} \le e^{-\frac{\omega_e}{2c_2}(t-t_0)} \sqrt{V(\xi(t_0))} + \frac{c_2}{\sqrt{c_1}} M \int_{t_0}^{t} e^{-\frac{\omega_e}{c_2}(t-s)} ds$$

设初始估计误差满足：

$$\left|\hat{E}(t_0)\right| < \rho_0$$

则有

$$\left\|\xi(t_0)\right\| \le \bar{\mu}_e(\omega_e)\rho_0 \qquad \text{其中}\, \bar{\mu}_e(\omega_e) \triangleq \max\left\{1, \omega_e^n\right\}$$

线性ESO的估计误差分析：定理2（续）

设初始估计误差满足：

$$\left|\hat{E}(t_0)\right| < \rho_0$$

则有

$$\left\|\xi(t_0)\right\| \le \bar{\mu}_e(\omega_e)\rho_0 \qquad \bar{\mu}_e(\omega_e) \triangleq \max\left\{1, \omega_e^n\right\}$$

则对 $\sqrt{V(\xi)} \le e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{V(\xi(t_0))} + \frac{c_2}{\sqrt{c_1}} M \int_{t_0}^{t} e^{-\frac{\omega_e}{c_2}(t-s)} ds$ **的右端第1项得如下分析结果**

当 $\forall t \ge T \triangleq t_0 + 2(n+1)c_2 \max\left\{\frac{\ln \omega_e}{\omega_e}, 0\right\}, \ \omega_e > 0, \ i = 1, 2, \ldots, n+1$ **时**

$$e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{V(\xi(t_0))} \le \frac{1}{\omega_e^{n+1}}\sqrt{c_2}\,\rho_0 \bar{\mu}_e(\omega_e) \le \max\left\{\frac{1}{\omega_e^{n+1}}, \frac{1}{\omega_e}\right\}\sqrt{c_2}\,\rho_0.$$

线性ESO的估计误差分析：定理2（续）

又由于 $\xi_i = \omega_e^{n+1-i}\hat{E}_i$ 和 $\hat{E}_i = \dfrac{1}{\omega_e^{n+1-i}}\xi_i$，**则**

当 $\forall t \ge T \triangleq t_0 + 2(n+1)c_2 \max\left\{\dfrac{\ln\omega_e}{\omega_e}, 0\right\}, \omega_e > 0,\ i = 1,2,\ldots,n+1$ **时**

$$\left|\hat{E}_i(t)\right| = \frac{1}{\omega_e^{n+1-i}}\left\|\xi_i(t)\right\|$$

$$\le \frac{1}{\omega_e^{n+1-i}}\left[e^{-\frac{\omega_e}{2c_2}t}\sqrt{\frac{V(\xi(t_0))}{c_1}} + \frac{c_2}{c_1}M\int_{t_0}^{t} e^{-\frac{\omega_e}{2c_2}(t-s)}ds\right]$$

$$\le \frac{1}{\omega_e^{n+1-i}}\left[\max\left\{\frac{1}{\omega_e^{n+1}}, \frac{1}{\omega_e}\right\}\sqrt{c_2}\rho_0 + \frac{c_2}{c_1}M\frac{2c_2}{\omega_e}\right]$$

因此 $\left|X_i - \hat{X}_i\right| \le O\left(\dfrac{1}{\omega_e^{n+2-i}}\right), \quad \forall t \ge T_1 > t_0, \omega_e > 0,\ i = 1,2,\ldots,n+1$ **证毕.**

线性ESO的估计误差分析：定理3

定理3：假设总扰动的导数，即 $\dot{X}_{n+1}(.)$，满足 $\lim\limits_{t\to\infty}\dot{X}_{n+1}(.)=0$，则：

$$\lim_{t\to\infty}\left|X_i-\hat{X}_i\right|=0,\quad i=1,2,\ldots,n+1$$

证明：同上定义矩阵 $V(\xi)\triangleq\xi^T P\xi$，有

$$\frac{dV(\xi)}{dt}=-\omega_e\xi^T\xi+2\xi^T P\begin{bmatrix}0\\0\\\vdots\\\dot{X}_{n+1}\end{bmatrix}$$

$$\frac{dV(\xi)}{dt}\le-\omega_e\frac{V(\xi)}{2c_2}+2\|\xi\|c_2\left\|\dot{X}_{n+1}\right\|\le-\omega_e\frac{V(\xi)}{2c_2}+2\frac{\sqrt{V(\xi)}}{\sqrt{c_1}}c_2\left\|\dot{X}_{n+1}\right\|$$

$$\frac{d\sqrt{V(\xi)}}{dt}\le-\omega_e\frac{\sqrt{V(\xi)}}{2c_2}+\frac{c_2}{\sqrt{c_1}}\left\|\dot{X}_{n+1}\right\|$$

由Gronwall–Bellman不等式有

$$\sqrt{V(\xi)}\le e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{V(\xi(t_0))}+\frac{c_2}{\sqrt{c_1}}\int_{t_0}^{t}e^{-\frac{\omega_e}{c_2}(t-s)}\left\|\dot{X}_{n+1}(s)\right\|ds$$

线性ESO的估计误差分析：定理3（续）

又由于 $\lim\limits_{t\to\infty}\dot{X}_{n+1}=0$, 则有 $\lim\limits_{t\to\infty}\varepsilon(t)=0$ 其中 $\varepsilon(t)\triangleq\sup\limits_{s\in\left[\frac{t}{2},\infty\right)}\left\|\dot{X}_{n+1}(s)\right\|$

因此有

$$\sqrt{V(\xi)}\le e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{V(\xi(t_0))}+\frac{c_2}{\sqrt{c_1}}\left(\int_{t_0}^{t/2}+\int_{t/2}^{t}\right)e^{-\frac{\omega_e}{2c_2}(t-s)}\left\|\dot{X}_{n+1}(s)\right\|ds$$

$$\le e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{V(\xi(t_0))}+\frac{c_2}{\sqrt{c_1}}\int_{t_0}^{t/2}e^{-\frac{\omega_e}{2c_2}(t-s)}\left\|\dot{X}_{n+1}(s)\right\|ds+\varepsilon(t)\frac{c_2}{\sqrt{c_1}}\int_{t/2}^{t}e^{-\frac{\omega_e}{2c_2}(t-s)}ds$$

$$\le e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{V(\xi(t_0))}+\frac{c_2}{\sqrt{c_1}}M\int_{t_0}^{t/2}e^{-\frac{\omega_e}{2c_2}(t-s)}ds+\varepsilon(t)\frac{c_2}{\sqrt{c_1}}\int_{t/2}^{t}e^{-\frac{\omega_e}{2c_2}(t-s)}ds$$

$$\le e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{V(\xi(t_0))}+\frac{c_2}{\sqrt{c_1}}M\frac{2c_2}{\omega_e}\left(e^{-\frac{\omega_e}{2c_2}\frac{t}{2}}-e^{-\frac{\omega_e}{2c_2}(t-t_0)}\right)+\varepsilon(t)\frac{c_2}{\sqrt{c_1}}\frac{2c_2}{\omega_e}\left(1-e^{-\frac{\omega_e}{2c_2}\frac{t}{2}}\right).$$

线性ESO的估计误差分析：定理3（续）

进一步可以得到：

$$
\begin{aligned}
\lim_{t\to\infty}\sqrt{V(\xi)} \le{}& \lim_{t\to\infty} e^{-\frac{\omega_e}{2c_2}(t-t_0)}\sqrt{V(\xi(t_0))} \\
&+\lim_{t\to\infty}\frac{c_2}{\sqrt{c_1}}M\frac{2c_2}{\omega_e}\left(e^{-\frac{\omega_e}{2c_2}\frac{t}{2}}-e^{-\frac{\omega_e}{2c_2}(t-t_0)}\right) \\
&+\lim_{t\to\infty}\varepsilon(t)\frac{c_2}{\sqrt{c_1}}\frac{2c_2}{\omega_e}\lim_{t\to\infty}\left(1-e^{-\frac{\omega_e}{2c_2}\frac{t}{2}}\right) \\
={}& 0
\end{aligned}
$$

线性ESO的估计误差分析：峰值现象(peaking phenomenon)

引理1： 若存在 $i \in \{1,2\ldots,n\}$ 使得 $\left|X_i(t_0)-\hat{X}_i(t_0)\right| \neq 0$ ，则存在某个总扰动及初值使得ESO估计误差满足：

$$\lim_{\omega_e\to\infty}\sup_t\left|X_i(\mathrm{t})-\hat{X}_i(\mathrm{t})\right|=\infty$$

证明：同定理1-3，有

$$\xi(t)=\exp\left\{\omega_e\begin{bmatrix}-C_{n+1}^1 & 1 & \cdots & 0\\ -C_{n+1}^2 & 0 & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ -C_{n+1}^{n+1} & 0 & \cdots & 0\end{bmatrix}(t-t_0)\right\}\xi(t_0) \quad \xi \triangleq \begin{bmatrix}\xi_1\\ \xi_2\\ \vdots\\ \xi_{n+1}\end{bmatrix}=\begin{bmatrix}\omega_e^n & 0 & \cdots & 0\\ 0 & \omega_e^{n-1} & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \cdots & 1\end{bmatrix}\begin{bmatrix}\hat{E}_1\\ \hat{E}_2\\ \vdots\\ \hat{E}_{n+1}\end{bmatrix}$$

$$\begin{bmatrix}\hat{E}_1(t)\\ \hat{E}_2(t)\\ \vdots\\ \hat{E}_{n+1}(t)\end{bmatrix}=\begin{bmatrix}\omega_e^{-n} & 0 & \cdots & 0\\ 0 & \omega_e^{-(n-1)} & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \cdots & 1\end{bmatrix}G\begin{bmatrix}\omega_e^n & 0 & \cdots & 0\\ 0 & \omega_e^{n-1} & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \cdots & 1\end{bmatrix}\begin{bmatrix}\hat{E}_1(t_0)\\ \hat{E}_2(t_0)\\ \vdots\\ \hat{E}_{n+1}(t_0)\end{bmatrix}$$

$$=\begin{bmatrix}G_{11} & G_{12}\omega_e^{-1} & \cdots & G_{1,n+1}\omega_e^{-n}\\ G_{21}\omega_e^{1} & G_{22} & \cdots & G_{2,n+1}\omega_e^{-(n-1)}\\ \vdots & \vdots & \ddots & \vdots\\ G_{n+1,1}\omega_e^{n} & G_{n+1,2}\omega_e^{n-1} & \cdots & G_{n+1,n+1}\end{bmatrix}\begin{bmatrix}\hat{E}_1(t_0)\\ \hat{E}_2(t_0)\\ \vdots\\ \hat{E}_{n+1}(t_0)\end{bmatrix}$$

非对角矩阵

其中 $G \triangleq \exp\left\{\omega_e\begin{bmatrix}-C_{n+1}^1 & 1 & \cdots & 0\\ -C_{n+1}^2 & 0 & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ -C_{n+1}^{n+1} & 0 & \cdots & 0\end{bmatrix}(t-t_0)\right\}$

线性ESO的估计误差分析：引理1（续）

不妨设总扰动的导数为0，并且除第1个状态外，其它状态初始估计误差为0，则有：

$$\begin{bmatrix} \hat{E}_1(t) \\ \hat{E}_2(t) \\ \vdots \\ \hat{E}_{n+1}(t) \end{bmatrix} = \begin{bmatrix} G_{11}(\mathrm{t}) \\ G_{21}(\mathrm{t})\omega_e^1 \\ \vdots \\ G_{n+1,1}(\mathrm{t})\omega_e^n \end{bmatrix} \hat{E}_1(t_0)$$

由G的表达式可知，其各分量满足：

$$G_{ij}(\mathrm{t}) = \sum_{k=0}^{n} a_k \left(\omega_e\left(t-t_0\right)\right)^k e^{-\omega_e(t-t_0)} \hat{E}_1(t_0)$$

则第2个状态的估计值满足：

$$\hat{E}_2\left(t_0 + \frac{c}{\omega_e}\right) = \omega_e^1 \left(\sum_{k=0}^{n} a_k c^k\right) e^{-c} \hat{E}_1(t_0)$$

取 c 使得 $\lim\limits_{\omega_e\to\infty} \hat{E}_2\left(t_0 + \frac{c}{\omega_e}\right) = \left(\sum_{k=0}^{n} a_k c^k\right) e^{-c} \hat{E}_1(t_0) \lim\limits_{\omega_e\to\infty} \omega_e^1 = \infty.$ 即证。

线性自抗扰控制的跟踪误差分析：定理4

Uncertain Systems:
$$\begin{cases}\dot{X}_1 = X_2\\ \cdots\\ \dot{X}_{n-1} = X_n\\ \dot{X}_n = X_{n+1} + \bar{B}(t)U(t)\end{cases}$$

总扰动: $X_{n+1} = F(X,t) + d(t) + \left(B(X,t) - \bar{B}(t)\right)U(t)$

Linear ESO:
$$\begin{cases}\dot{\hat{X}}_1 = \hat{X}_2 - l_1(\hat{X}_1 - Y)\\ \cdots\\ \dot{\hat{X}}_n = -l_n(\hat{X}_1 - Y) - \hat{X}_{n+1} + \bar{B}(t)U(t)\\ \dot{\hat{X}}_{n+1} = -l_{n+1}(\hat{X}_1 - Y)\end{cases}$$

ESO的带宽: ω_e $\quad s^{n+1} + \sum_{i=1}^{n+1} l_i s^{n+1-i} = (s+\omega_e)^{n+1}$

控制器设计: $U(\mathrm{t}) = \bar{B}^{-1}(-k_1\hat{X}_1 - k_2\hat{X}_2 \cdots - k_n\hat{X}_n - \hat{X}_{n+1})$

定理4：在一定的条件下，存在ω_e^*，使得 $\forall \omega_e \geq \omega_e^*$：

$$\left\| y(t) - y^*(t) \right\| \leq \gamma_1^* \frac{\max\{\ln(\rho_2\omega_e), \ln\omega_e, 1\}}{\omega_e}, \quad \forall t \geq t_0$$

瞬态性能：接近预期轨迹

其中 γ_1^* 为正常数.

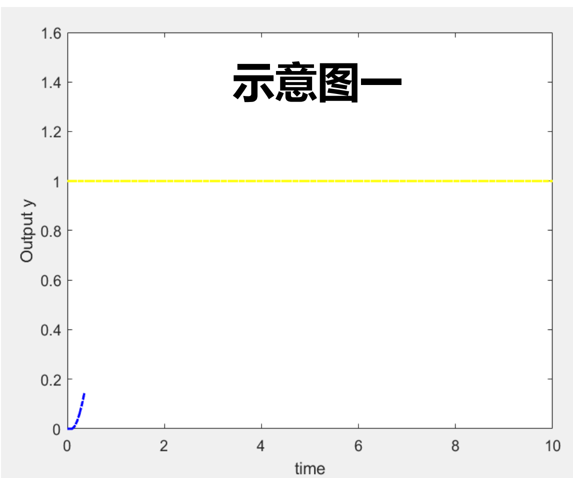
Wenchao Xue*, and Yi Huang, Performance Analysis of 2-DOF Tracking Control for a Class of nonlinear Uncertain Systems with Discontinuous Disturbances, International Journal of Robust and nonlinear Control, 2018, 28:1456–1473.

定理4：考虑上述闭环系统，存在 ω_e^*，使得 $\forall \omega_e \geq \omega_e^*$：

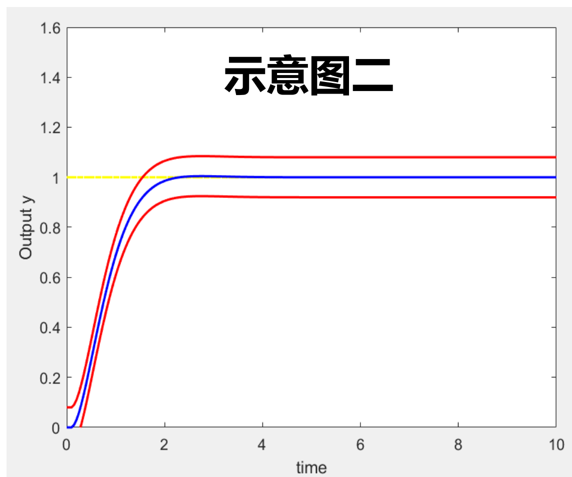
$$\|y(t)-y^*(t)\| \leq \gamma_1^* \frac{\max\{\ln(\rho_2\omega_e), \ln\omega_e, 1\}}{\omega_e}, \quad \forall t \geq t_0$$

→ **瞬态性能：接近预期轨迹**

其中 γ_1^* 为正常数.



$y \to r \ (t \to \infty)$，
但超调、震荡…



输出曲线在理想轨线(蓝线)的小范围内(红线之间)波动

$$y^* = x_1^*$$

$$\begin{bmatrix} \dot{x}_1^* \\ \dot{x}_2^* \\ \vdots \\ \dot{x}_n^* \end{bmatrix} = A_k \left(\begin{bmatrix} x_1^* \\ x_2^* \\ \vdots \\ x_n^* \end{bmatrix} - \begin{bmatrix} r \\ r^{(1)} \\ \vdots \\ r^{(n-1)} \end{bmatrix} \right) + \begin{bmatrix} 0 \\ \vdots \\ 0 \\ r^{(n)} \end{bmatrix}$$

$$A_k = \begin{bmatrix} 0 & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 \\ k_{1,c} & k_{2,c} & \cdots & k_{n,c} \end{bmatrix}$$

自抗扰控制系统的稳定性分析

- 积分串联型结构的系统
  - 线性ESO的估计误差分析
  - 闭环系统分析
- 一般结构下的系统
  - 可观性分析
  - ESO设计

我国最早从事现代控制理论的单位：
中国科学院系统控制重点实验室

关肇直院士，1962-1982年任控制理论研究室主任

宋健院士，1962-1979年任控制理论研究室副主任

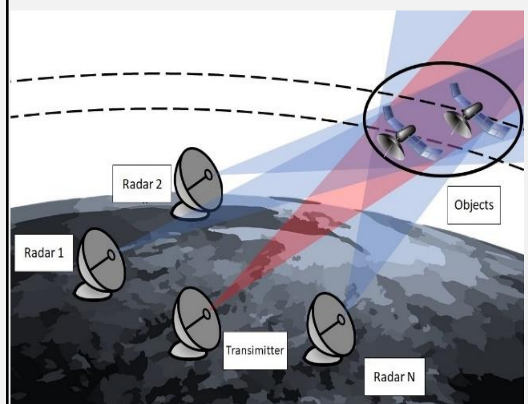
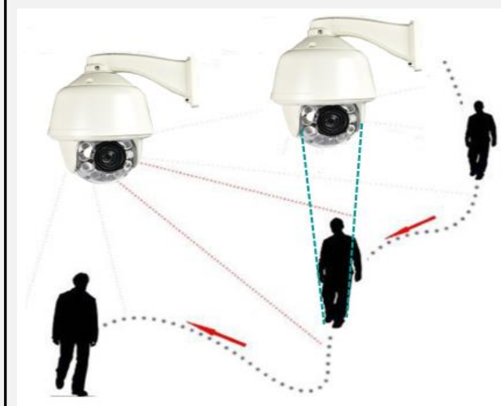
1960年前后，现代控制理论开始建立，以适应导弹和人造卫星等尖端控制技术的高度复杂性和高精度需求。

钱学森和关肇直等国内一批优秀的科学家，以他们敏锐的洞察力，意识到现代控制理论在我国工业和国防现代化中的重要作用。

在钱学森倡导下，经中国科学院和国防部第五研究院商定，于1962年3月在中国科学院数学研究所成立控制理论研究室。

1970-1980年代控制室编写的现代控制系统理论小丛书系列

系统的状态估计问题

Practical Systems	Volatility Estimation 	Orbit Determination 	Camera Tracking 
Observation data	Disturbed stock price	Radar measurements	Inaccurate angles
System state	Volatility	Position and velocity	Position and velocity

能观性(observable)
估计状态的前提（1960s, Kalman）

$\mathbf{x}(\mathbf{t}) \in \mathbf{R}^{\mathbf{n}}$为状态；　$\mathbf{y}(\mathbf{t}) \in \mathbf{R}^{\mathbf{m}}$为量测输出；

线性定常系统
$$\begin{cases} \dot{x}(t) = Ax(t) \\ y(t) = Cx(t), \end{cases} \qquad t \in [0, +\infty).$$

定义1. 对于系统(1), 我们称其为**能观**的，如果

$$y(t), \forall t \in [0, +\infty) \text{ 唯一确定 } x(t), \forall t \in [0, +\infty).$$

引理1. 系统(1)能观当且仅当

$$\text{rank}\left(\begin{bmatrix} A - \lambda I_n \\ C \end{bmatrix}\right) = n, \ \forall \lambda \in \mathbb{C}.$$

线性定常系统：能观性和观测器设计

$$\begin{cases} \dot{x}(t) = Ax(t) \\ y(t) = Cx(t), \end{cases} \qquad t \in [0, +\infty).$$

引理2. **如果系统(1)能观的，则如下观测器**

$$\dot{\hat{x}}(t) = A\hat{x}(t) + L(y - C\hat{x}(t)) \quad t \in [0, +\infty).$$

满足 $(A - LC)$ **可任意配置特征根，且特征根具有负实部时：**

$$\lim_{t\to\infty} \hat{x}(t) - x(t) = 0.$$

Chi-Tsong Chen, Linear System Theory and Design, Oxford University Press, USA, 1998, pp.231-266.

不确定系统：能观性的定义

$x(t) \in R^n$，$\xi(t) \in R^q$，$y(t) \in R^m$，(A,B,C)：一般矩阵

$$\begin{cases} \dot{x}(t) = Ax(t) + \mathrm{B}u(t) + B\xi(t), \\ y(t) = Cx(t), \end{cases} \quad t \in [0, +\infty). \qquad \mathbf{(3.1)}$$

$\xi(t)$ 作为已知输入时的能观性

定义1. 对于系统（3.1)，我们称其为能观的，如果$\xi(t)$=0，u(t)=0，$y(t)$=0，$\forall t \in [0, +\infty)$ 意味着 $x(t) = 0, \forall t \in [0, +\infty)$.

$\xi(t)$ 作为未知输入时的强能观性

定义2. 对于系统（3.1)，我们称其为强能观的，如果对所考虑范围内的任意$\xi(t)$，u(t)=0，$y(t)$=0，$\forall t \in [0, +\infty)$ 意味着 $x(t) = 0$，$\forall t \in [0, +\infty)$.

不确定系统：能观性的判据

$$x(t) \in R^n \quad \xi(t) \in R^q \quad y(t) \in R^m$$

$$\begin{cases} \dot{x}(t) = Ax(t) + \mathrm{B}u(t) + B\xi(t), \\ y(t) = Cx(t), \end{cases} \quad t \in [0, +\infty). \qquad \textbf{(3.1)}$$

推论1. **系统 (3.1) 强能观当且仅当**

$$\mathrm{rank}\left(\begin{bmatrix} A - \lambda I_n & B \\ C & 0_{m\times q} \end{bmatrix}\right) = n + q, \ \forall \lambda \in \mathbb{C}.$$

对于系统(3.1)，如下情况一定不能观：

扰动维数大于量测维数

X. Zhang; W. Xue; S. Chen; H. Fang; Generalized observability analysis and observer design for a class of systems with multiple uncertain dynamics and measurement biases, Systems & Control Letters, 2021*

不确定系统：能观的组合状态

推论1. 系统 (3.1) 强能观当且仅当

$$\operatorname{rank}\left(\begin{bmatrix} A-\lambda \mathrm{I}_n & B \\ C & 0_{m\times q}\end{bmatrix}\right)=n+q,\ \forall\lambda\in\mathbb{C}.$$

组合状态的能观性

定义3： 令 $w \triangleq [x^T,\ \xi^T]^T$, 对于任意一个具有合适维数的矩阵 F, 称 $v \triangleq Fw$ 是系统状态 x, 不确定动态 ξ 的一个（线性）**组合状态**。

定义4. 对于系统 (3.1)，称**组合状态** v **能观**，如果 $y(t)=0,\ \forall t\in[0,+\infty)$ 意味着 $v(t)=0,\ \forall t\in[0,+\infty)$.

X. Zhang; W. Xue; S. Chen; H. Fang; Generalized observability analysis and observer design for a class of systems with multiple uncertain dynamics and measurement biases, Systems & Control Letters, 2021*

不确定系统：如何获得能观的组合状态？

$$\begin{cases}\dot{x}(t) = Ax(t) + \mathrm{Bu}(t) + B\xi(t), \\ y(t) = Cx(t),\end{cases}$$

引入如下变化矩阵：

$$M_r \triangleq \begin{bmatrix} M_{1,r_1} \\ M_{2,r_2} \\ \vdots \\ M_{m,r_m} \end{bmatrix}, \quad 其中 M_{i,r_i} \triangleq \begin{bmatrix} c_i & \mathbf{0}_{1\times q} \\ c_i A & c_i B \\ \vdots & \vdots \\ c_i A^{r_i} & c_i A^{r_i - 1} B \end{bmatrix},$$

产生过程：
对y(t)的第i个变量依次求1阶, 2阶, …, 导数，直到出现扰动$\xi(t)$

$$r_i \triangleq \begin{cases} \min\{h \in \mathbb{Z}^+ \mid c_i A^{h-1} B \neq \mathbf{0}\}, & 当 h < \infty, \\ \mathrm{rank}([c_i^T \quad (c_i A)^T \quad \cdots \quad (c_i A^{n-q-1})^T]^T), & 其他情形 \end{cases}$$

不确定系统：如何获得能观的组合状态？

$$\begin{cases} \dot{x}(t) = Ax(t) + \mathrm{Bu}(t) + B\xi(t), \\ y(t) = Cx(t), \end{cases}$$

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产生过程：
对y(t)的第i个变量依次求1阶, 2阶, …, 导数，直到出现扰动$\xi(t)$

$$r_i \triangleq \begin{cases} \min\{h \in \mathbb{Z}^+ | c_iA^{h-1}B \neq \mathbf{0}\}, & \text{当} h < \infty, \\ \mathrm{rank}([c_i^T \quad (c_iA)^T \quad \cdots \quad (c_iA^{n-q-1})^T]^T), & \text{其他情形} \end{cases}$$

考虑特殊情况（最简单的积分串联型）：

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} \Longrightarrow M_r = I_{n+1}$$

不确定系统：能观的组合状态

$$\begin{cases}\dot{x}(t) = Ax(t) + \mathrm{Bu}(t) + B\xi(t),\\ y(t) = Cx(t),\end{cases}$$

设$M_r \triangleq \begin{bmatrix} M_{1,r_1} \\ M_{2,r_2} \\ \vdots \\ M_{m,r_m} \end{bmatrix}$, $M_{i,r_i} \triangleq \begin{bmatrix} c_i & \mathbf{0}_{1\times q} \\ c_i A & c_i B \\ \vdots & \vdots \\ c_i A^{r_i} & c_i A^{r_i-1}B \end{bmatrix}$,

$$r_i \triangleq \begin{cases}\min\{h \in \mathbb{Z}^+ | c_i A^{h-1}B \neq \mathbf{0}\}, & \text{当}\exists h \in \mathbb{Z}^+, \mathrm{s.t.}\, c_i A^{h-1}B \neq \mathbf{0},\\ \mathrm{rank}([c_i^T \quad (c_iA)^T \quad \cdots \quad (c_iA^{n-q-1})^T]^T), & \text{其他情形。}\end{cases}$$

定理 2. 对于系统 (3.1), 如果

$$\mathbf{rank}(\boldsymbol{M_r}) = \mathbf{rank}\left(\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \mathbf{0}_{\boldsymbol{m\times q}} \end{bmatrix}\right),$$

扰动维数=量测维数，

扰动到量测的相对阶向量之和=n

则所有能观的组合状态构成的集合为

积分串联型系统时$M_r = I_{n+1}$

$$S_o \triangleq \{\tilde{v} | \tilde{v} = GM_r w\}, \quad G \in \mathbb{R}^{*\times l}, w \triangleq [x^T, \xi^T]^T$$

X. Zhang; W. Xue; S. Chen; H. Fang; Generalized observability analysis and observer design for a class of systems with multiple uncertain dynamics and measurement biases, Systems & Control Letters, 2021*

不确定系统：能观的组合状态

$x(t) \in R^n \quad \xi(t) \in R^q \quad y(t) \in R^m$

$$\begin{cases} \dot{x}(t) = Ax(t) + \mathrm{Bu}(t) + B\xi(t), \\ y(t) = Cx(t), \end{cases} \quad t \in [0, +\infty). \qquad \textbf{(3.1)}$$

在定理2的条件下，可以找到等价坐标变换 $\tilde{v} = M_r w$, 使得

$$\begin{cases} \dot{\tilde{v}}(t) = \tilde{A}\tilde{v}(t) + \tilde{B}\dot{\xi}(t), \\ y(t) = \tilde{C}\tilde{v}(t), \end{cases} \quad t \in [0, +\infty),$$

由m个“积分串联型+总扰动”的形式组成

其中, $\tilde{A} \triangleq M_r \begin{bmatrix} A & B \\ 0_{(q+p)\times n} & 0_{q\times q} \end{bmatrix} M_r^{-1}$, $\quad \tilde{C} \triangleq [C \quad 0_{m\times q}] M_r^{-1}$, $\tilde{B} \triangleq M_r [0_{q\times n} \quad \mathrm{I}_q]^T$.

X. Zhang; W. Xue; S. Chen; H. Fang; Generalized observability analysis and observer design for a class of systems with multiple uncertain dynamics and measurement biases, Systems & Control Letters, 2021*

不确定系统：扩张状态观测器设计

$$\begin{cases}\dot{\hat{\tilde{v}}}(t)=\tilde{A}\hat{\tilde{v}}(t)+\boldsymbol{L}\left(y(t)-\tilde{C}\hat{\tilde{v}}(t)\right),\\ \hat{\tilde{x}}(t)=[\mathbf{I}_n\quad \mathbf{0}_{n\times(q+\tilde{p})}]\hat{\tilde{v}}(t),\end{cases}\quad t\in[0,+\infty).\qquad (3.2)$$

观测器 (3.2) 中的增益计算公式如下：

$$\boldsymbol{L}=\left(\tilde{A}-\frac{1}{\epsilon}\left(\boldsymbol{P}^{(\epsilon)}\right)^{-1}\boldsymbol{\Lambda}\boldsymbol{P}^{(\epsilon)}\right)\tilde{C}^T,$$

其中, $\boldsymbol{\Lambda}\triangleq\mathrm{diag}([\Lambda_1\quad \Lambda_2\quad\cdots\quad\Lambda_m]),$

$$\Lambda_i\triangleq[\lambda_{i,1}\quad\lambda_{i,2}\quad\cdots\quad\lambda_{i,\tilde{r}_i+1}],\ i,j=1,2,\cdots m,$$

$$\boldsymbol{P}^{(\epsilon)}\triangleq\begin{bmatrix}P_{1,1}^{(\epsilon)}&\cdots&P_{1,m}^{(\epsilon)}\\ \vdots&\ddots&\vdots\\ P_{m,1}^{(\epsilon)}&\cdots&P_{m,m}^{(\epsilon)}\end{bmatrix},\ P_{i,j}^{(\epsilon)}\triangleq\begin{bmatrix}p_{i,j}^{(\epsilon)1,1}&\cdots&p_{i,j}^{(\epsilon)1,s_j}&\delta_{i,j}\\ \vdots&\ddots&\vdots&\vdots\\ p_{i,j}^{(\epsilon)\tilde{r}_i+1,1}&\cdots&p_{i,j}^{(\epsilon)\tilde{r}_i+1,\tilde{r}_j}&\delta_{i,j}\end{bmatrix},$$

$$p_{i,j}^{(\epsilon)h,k}\triangleq\begin{cases}\delta_{i,j}, & \text{if } k=\tilde{r}_j+1\ ,\\ \frac{\lambda_{i,j}}{\epsilon}p_{i,j}^{(\epsilon)h,k+1}-a_{i,j}^{(k+1)}, & \text{otherwise},\end{cases}\quad \epsilon\in(0,1].$$

不确定系统：扩张状态观测器设计

$$\begin{cases} \dot{\hat{\tilde{v}}}(t) = \tilde{A}\hat{\tilde{v}}(t) + \boldsymbol{L}\left(y(t) - \tilde{C}\hat{\tilde{v}}(t)\right), \\ \hat{\tilde{x}}(t) = [\mathbf{I}_n \quad \mathbf{0}_{n\times(q+\tilde{p})}]\hat{\tilde{v}}(t), \end{cases} \quad t \in [0, +\infty). \qquad \textbf{(3.2)}$$

考虑特殊情况（最简单的积分串联型）：

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} \Longrightarrow M_r = I_{\mathrm{n+1}} \Longrightarrow \tilde{v} = [x^T, \xi^T]^T$$

观测器 (3.2) 中的增益形式可退化为：

$$L = \begin{bmatrix} \frac{1}{\varepsilon} l_1 & \frac{1}{\varepsilon^2} l_2 & \dots & \frac{1}{\varepsilon^{n+1}} l_{n+1} \end{bmatrix}^T, \qquad (s+1)^{n+1} = s^{n+1} + l_1 s^n + \dots + s^{n+1}$$

ESO带宽

扩张状态观测器的估计误差分析

定理3. 在满足定理2的条件及增益L的算法下，如果 $\dot{\xi}(t)$ 是有界的，则存在独立于 ϵ 的正常数 γ_1 和 γ_2 使得

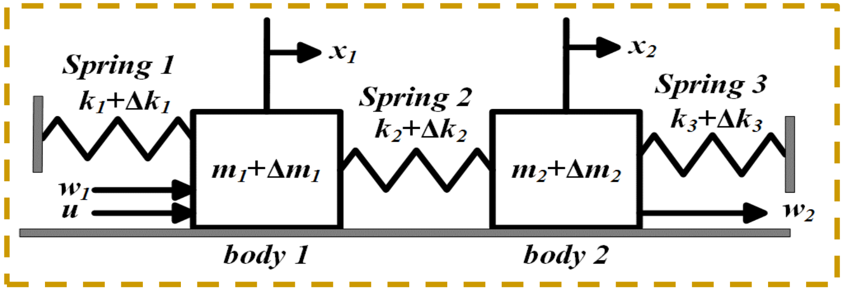
$$\|\widehat{v}(t)-\widetilde{v}(t)\| \le \gamma_1\left(\exp\left\{-\frac{\gamma_2}{\epsilon}t\right\}+\epsilon\right), \quad \forall t \in [0,\infty).$$

- 系统能观：可等价转换为状态、扰动
- 系统不能观：分析是否等于/近似为部分状态

刻画了整个时间域上的估计误差范围

X. Zhang; W. Xue; S. Chen; H. Fang; Generalized observability analysis and observer design for a class of systems with multiple uncertain dynamics and measurement biases, Systems & Control Letters, 2021*

实验结果: 弹簧+质量块耦合系统



细长飞行器

柔性机器人

吊缆(细长绳、杆)

刚性+弹性耦合系统

$$\begin{cases} \dot{x}(t) = Ax(t) + b_u u(t) + B_f \, f(x,u,t) \\ y(t) = c^T x(t) = x_2(t) \end{cases} \qquad x = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}^T$$

只能量测得到第2个质量块的位置

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\dfrac{k_1+k_2}{m_1} & \dfrac{k_2}{m_1} & 0 & 0 \\ \dfrac{k_2}{m_2} & -\dfrac{k_2+k_3}{m_2} & 0 & 0 \end{bmatrix}, \quad b_u = \begin{bmatrix} 0 \\ 0 \\ \dfrac{1}{m_1} \\ 0 \end{bmatrix}, B_f = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, c = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad f(x,u,t) = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} \dfrac{m_1 w_1 - \Delta m_1 u + \left((k_1+k_2)\Delta m_1 - (\Delta k_1 + \Delta k_2) m_1\right) x_1 + (\Delta k_2 m_1 - k_2 \Delta m_1) x_2}{m_1 (m_1 + \Delta m_1)} \\ \dfrac{m_2 w_2 + (\Delta k_2 m_2 - k_2 \Delta m_2) x_1 + \left((k_2+k_3)\Delta m_2 - (\Delta k_2 + \Delta k_3) m_2\right) x_2}{m_2 (m_2 + \Delta m_2)} \end{bmatrix}$$

已知模型部分

两个质量块可能收到的干扰不相同

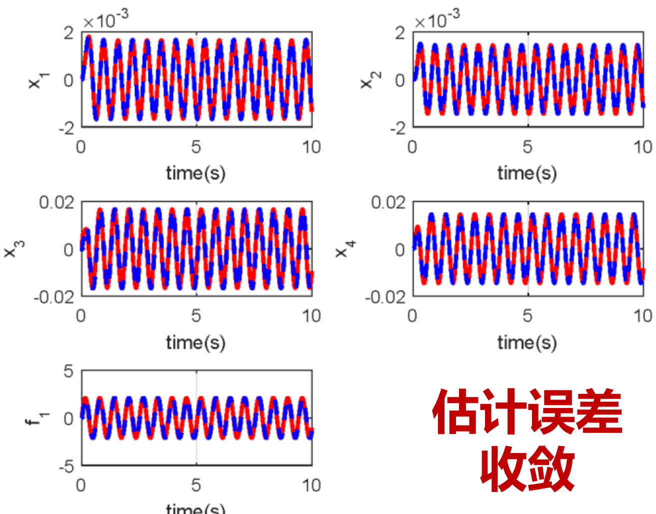
实验结果: 弹簧+质量块耦合系统

- 情况1：**仅有扰动f_1, 不存在f_2**

能观性条件成立

$$\begin{cases} \Delta m_i = 0\,(kg), \Delta k_j = 0\,(N/m), \\ w_1 = 5\sin(10t)/m_1\,(N),\ w_2 = 0\,(N) \end{cases}$$

$$\begin{cases} f_1 = 5\sin(10t)/m_1\,(N) \\ f_2 = 0\,(N) \end{cases}$$



估计误差收敛

- 情况2：**扰动f_1和f_2同时存在**

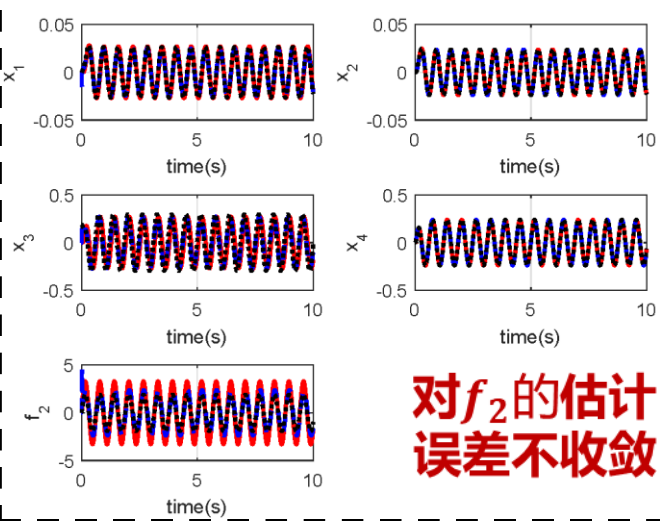
能观性条件不成立

$$\begin{cases} \Delta m_i = 0\,(kg), \Delta k_j = 0\,(N/m), \\ w_1 = 0\,(N),\ w_2 = 5\sin(10t)/m_1\,(N) \end{cases}$$

$$\begin{cases} f_1 = 0\,(N) \\ f_2 = 5\sin(10t)/m_1\,(N) \end{cases}$$

估计偏差

$$e_b(t) = \begin{bmatrix} 0 & 1.04\times10^{-5} & 0 \\ 0 & 0 & 0 \\ -0.0038 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -0.028 & 0 \end{bmatrix} \begin{bmatrix} f_2^{(1)} \\ f_2^{(2)} \\ f_2^{(3)} \end{bmatrix}$$



对f_2的估计误差不收敛

S. Chen, W. Bai, Y. Hu, Y. Huang, Z. Gao. On the conceptualization of total disturbance and its profound implications.

实验结果：弹簧+质量块耦合系统

刚性+弹性耦合系统：

$$\begin{cases} \dot{x}(t) = Ax(t) + b_u u(t) + B_f f(x,u,t) \\ y(t) = c^T x(t) = x_2(t) \end{cases}$$

扰动

$$f(x,u,t) = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} \dfrac{m_1 w_1 - \Delta m_1 u + \left((k_1+k_2)\Delta m_1 - (\Delta k_1 + \Delta k_2)m_1\right)x_1 + (\Delta k_2 m_1 - k_2 \Delta m_1)x_2}{m_1(m_1+\Delta m_1)} \\ \dfrac{m_2 w_2 + (\Delta k_2 m_2 - k_2\Delta m_2)x_1 + \left((k_2+k_3)\Delta m_2 - (\Delta k_2 + \Delta k_3)m_2\right)x_2}{m_2(m_2+\Delta m_2)} \end{bmatrix}$$

非积分器串联型

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\dfrac{k_1+k_2}{m_1} & \dfrac{k_2}{m_1} & 0 & 0 \\ \dfrac{k_2}{m_2} & -\dfrac{k_2+k_3}{m_2} & 0 & 0 \end{bmatrix}, \quad b_u = \begin{bmatrix} 0 \\ 0 \\ \dfrac{1}{m_1} \\ 0 \end{bmatrix}, B_f = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, c = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

对系统进行变换，使得变换后的状态和扰动（原扰动及导数组合）可观

- 计算输出到输入的相对阶：4
- 将系统写成积分串联型系统
- 得到系统中总扰动

积分器串联型内核

$$\begin{cases} \dot{\tilde{x}}_1 = \tilde{x}_2, \\ \dot{\tilde{x}}_2 = \tilde{x}_3, \\ \dot{\tilde{x}}_3 = \tilde{x}_4, \\ \dot{\tilde{x}}_4 = bu + f_{total}, \end{cases}$$

总扰动

$$y = \tilde{x}_1 = x_2, b = \frac{k_2}{m_1 m_2}.$$

$$\begin{aligned} f_{total} = & -\frac{(k_1+\Delta k_1)(k_2+\Delta k_2)+(k_1+\Delta k_1)(k_3+\Delta k_3)+(k_2+\Delta k_2)(k_3+\Delta k_3)}{(m_1+\Delta m_1)(m_2+\Delta m_2)}\tilde{x}_1 \\ & -\frac{(m_2+\Delta m_2)(k_1+\Delta k_1+k_2+\Delta k_2)+(m_1+\Delta m_1)(k_2+\Delta k_2+k_3+\Delta k_3)}{(m_1+\Delta m_1)(m_2+\Delta m_2)}\tilde{x}_2 \\ & +\left(\frac{k_2+\Delta k_2}{(m_1+\Delta m_1)(m_2+\Delta m_2)} - \frac{k_2}{m_1 m_2}\right)u + \frac{k_2+\Delta k_2}{m_1(m_2+\Delta m_2)}w_1 + \frac{1}{m_2+\Delta m_2}w_2^{(2)} \\ & -\frac{(k_1+\Delta k_1+k_2+\Delta k_2)(m_2+\Delta m_2)+2(k_2+\Delta k_2+k_3+\Delta k_3)(m_1+\Delta m_1)}{(m_1+\Delta m_1)(m_2+\Delta m_2)^2}w_2 \end{aligned}$$

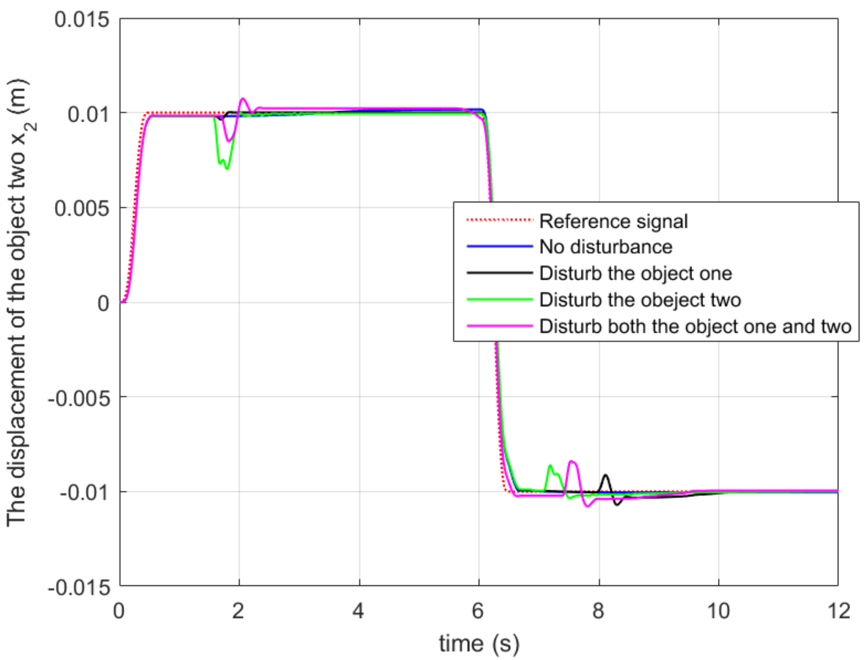
- 5阶扩张状态观测器
- ADRC控制

S. Chen, W. Bai, Y. Hu, Y. Huang, Z. Gao. On the conceptualization of total disturbance and its profound implications.

实验结果: 弹簧+质量块耦合系统

实验1 （外界扰动）

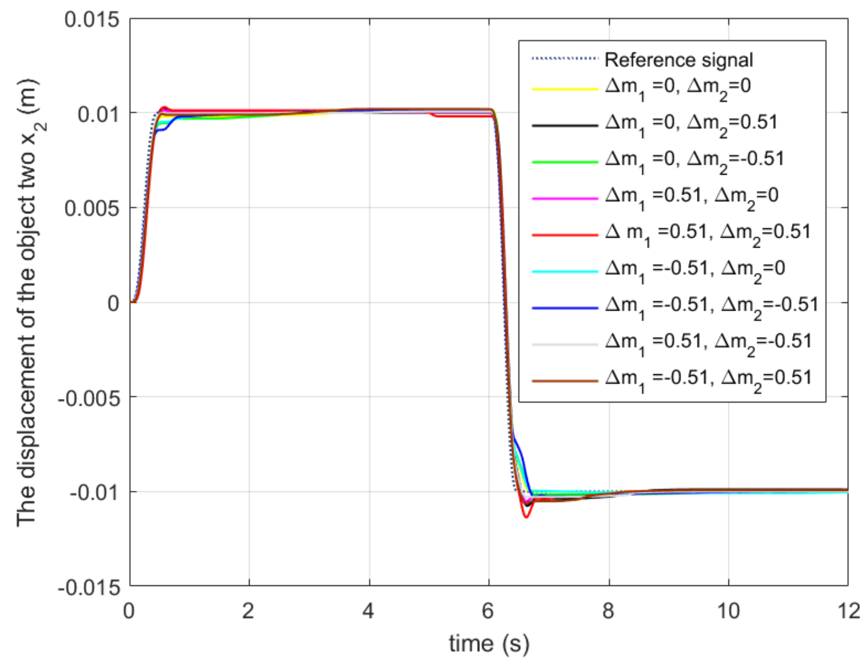
- ●情况1：无扰动
- ●情况2：扰动物体1（2秒与8秒附近）
- ●情况3：扰动物体2（2秒与8秒附近）
- ●情况4：扰动物体1与物体2（2秒与8秒附近）



实验2 （质量偏差）

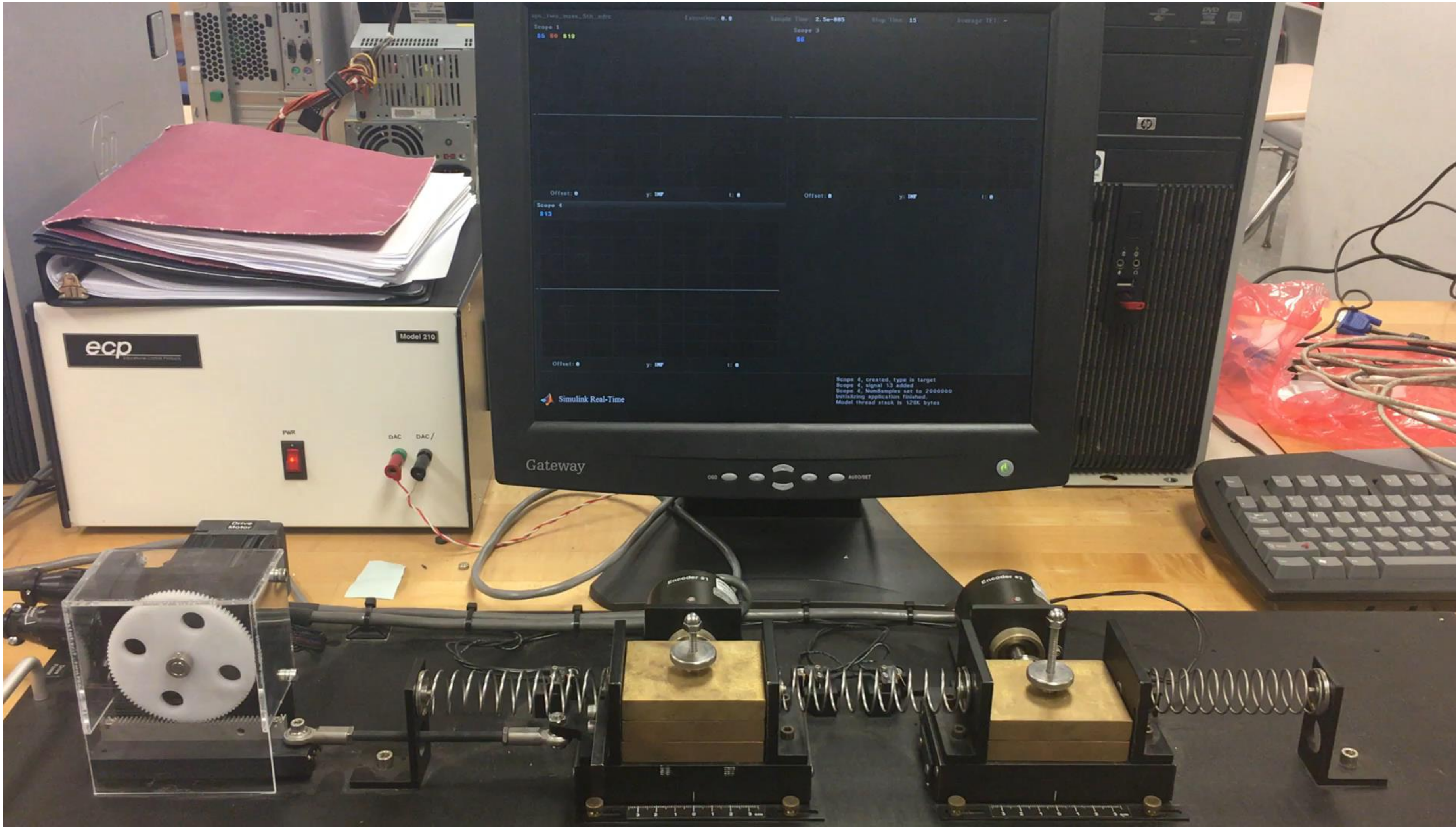
$$\left\{\left(\Delta m_1, \Delta m_2\right) \mid \Delta m_1 \in \Delta_m, \Delta m_2 \in \Delta_m\right\},$$

$$\Delta_m = \left\{0(kg), -0.51(kg), 0.51(kg)\right\}$$



S. Chen, W. Bai, Y. Hu, Y. Huang, Z. Gao. On the conceptualization of total disturbance and its profound implications.

实验结果: 弹簧+质量块耦合系统



https://space.bilibili.com/408884199

更一般的系统：扰动、量测零偏

含不确定动态及量测零偏的系统

$$x(t) \in R^n \quad \xi(t) \in R^q \quad y(t) \in R^m \quad b \in R^p$$

$$\begin{cases} \dot{x}(t) = Ax(t) + B\xi(t), \\ y(t) = Cx(t) + Db, \end{cases} \quad t \in [0, +\infty). \qquad (2.1)$$

$\xi(t)$ 及 b 作为未知输入时的强能观性

定义2. 对于系统（2.1），我们称其为强能观的，如果对所考虑范围内的任意$\xi(t)$和b, $y(t), \forall t \in [0, +\infty)$唯一确定 $x(t), \forall t \in [0, +\infty)$.

$\xi(t)$ 及 b 作为已知输入时的能观性

定义1'. 对于系统（2.1），我们称其为能观的，如果u(t)和b已知，$y(t)$, $\forall t \in [0, +\infty)$ 唯一确定 $x(t), \forall t \in [0, +\infty)$.

$x(t) \in R^n \quad \xi(t) \in R^q \quad y(t) \in R^m \quad b \in R^p$

含有量测未知零偏
$$\begin{cases} \dot{x}(t) = Ax(t) + B\xi(t), \\ y(t) = Cx(t) + Db, \end{cases} \quad t \in [0, +\infty). \quad \textbf{(3)}$$

推论 2. 系统 (3) 强能观当且仅当

$$\mathrm{rank}\left(\begin{bmatrix} A - \lambda \mathrm{I}_n & \mathbf{0}_{n\times p} & B \\ \mathbf{0}_{p\times n} & -\lambda \mathrm{I}_p & \mathbf{0}_{p\times q} \\ C & D & \mathbf{0}_{m\times q} \end{bmatrix}\right) = n + q + p, \ \forall \lambda \in \mathbb{C}.$$

对于系统(3)，如下情况一定不能观：

i) 扰动维数大于量测维数

ii) 零偏维数等于量测维数且存在扰动

X. Zhang; W. Xue; S. Chen; H. Fang; Generalized observability analysis and observer design for a class of systems with multiple uncertain dynamics and measurement biases, Systems & Control Letters, 2021*

含有量测未知零偏 $\begin{cases} \dot{x}(t) = Ax(t) + B\xi(t), \\ y(t) = Cx(t) + Db, \end{cases} \quad t \in [0, +\infty).$ **(3)**

推论 2. 系统 (3) 强能观当且仅当

$$\operatorname{rank}\left(\begin{bmatrix} A-\lambda \mathrm{I}_n & 0_{n\times p} & B \\ 0_{p\times n} & -\lambda \mathrm{I}_p & 0_{p\times q} \\ C & D & 0_{m\times q} \end{bmatrix}\right) = n + q + p, \forall \lambda \in \mathbb{C}.$$

组合状态的能观性

定义2： 令 $w \triangleq [x^T \quad \xi^T \quad b^T]^T$，对于任意一个具有合适维数的矩阵 F，称 $v \triangleq Fw$ 是系统状态 x，不确定动态 ξ 以及量测零偏 b的一个（线性）**组合状态**。

定义3. 对于系统 (2.1)，称**组合状态 v 能观**，如果 $y(t), \forall t \in [0, +\infty)$ 唯一确定 $v(t), \forall t \in [0, +\infty)$.

X. Zhang; W. Xue; S. Chen; H. Fang; Generalized observability analysis and observer design for a class of systems with multiple uncertain dynamics and measurement biases, Systems & Control Letters, 2021*

谢谢！

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自抗扰控制设计方法的新进展

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自抗扰控制设计方法的新进展

- 降阶ESO的设计方法
- 基于自抗扰控制的反步法设计
- 预测自抗扰控制设计方法

正常阶的ESO：(n+1)-th order

非线性不确定系统

$$\begin{cases} \dot{x} = Ax + B\left(f(\bullet) + \bar{b}u \right) \\ y = C^T x + n \end{cases}$$

$x: state \quad y: output \quad u: control\ input$

$n: measurement\ noise$

$f(\bullet) \triangleq a^T x + d(t) + \left(b - \bar{b}\right)u: total\ disturbance$

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}$$

线性ESO

$$\begin{cases} \dot{\hat{x}}_1 = \hat{x}_2 - \beta_1(\hat{x}_1 - y) \\ \vdots \\ \dot{\hat{x}}_n = \hat{x}_{n+1} - \beta_n(\hat{x}_1 - y) + bu \\ \dot{\hat{x}}_{n+1} = -\beta_{n+1}(\hat{x}_1 - y) \end{cases},$$

ESO带宽

$$s^{n+1} + \sum_{i=1}^{n+1} \beta_i s^{n+1-i} = (s + \omega_e)^{n+1},$$

传递函数：从扰动到 $\hat{x}_{n+1}$

$$Q_N = \frac{\hat{x}_{n+1}(s)}{f(s)} = \frac{\omega_e^{\,n+1}}{(s+\omega_e)^{n+1}},$$

传递函数：从噪声到 $\hat{x}_{n+1}$

$$G_N = \frac{\hat{x}_{n+1}(s)}{n(s)} = \frac{\omega_e^{\,n+1}}{(s+\omega_e)^{n+1}} s^n$$

降阶ESO (RESO)：n-th order

非线性不确定系统

$$\begin{cases} \dot{x} = Ax + B\left(f(\bullet) + \bar{b}u\right) \\ y = C^T x + n \end{cases}$$

$x: state \quad y: output \quad u: control\ input$

$n: measurement\ noise$

$f(\bullet) \triangleq a^T x + d(t) + \left(b - \bar{b}\right)u : total\ disturbance$

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}$$

RESO

$$\begin{cases} \dot{\eta}_1 = \begin{cases} -\beta_1\eta_1 - \beta_1^2 x_1 - \beta_1\bar{b}(t)u(t), \text{ if } n = 1. \\ -\beta_1\eta_1 + \eta_2 + (\beta_2 - \beta_1^2)y, \text{ if } n > 1. \end{cases} \\ \dot{\eta}_2 = -\beta_2\eta_2 + \eta_3 + (\beta_2 - \beta_1\beta_2)y \\ \dots \\ \dot{\eta}_{n-1} = -\beta_{n-1}\eta_1 + \eta_n + (\beta_n - \beta_1\beta_{n-1})y + \bar{b}(t)u(t) \\ \dot{\eta}_n = -\beta_n\eta_1 - \beta_1\beta_n y \end{cases}$$

$$\hat{x}_{i+1} = \eta_i + \beta_{i-1}y, \quad i = 1, 2, \dots, n,$$

RESO带宽

$$s^n + \sum_{i=1}^{n} \beta_i s^{n-i} = (s + \omega_e)^n$$

传递函数： 从扰动到 $\hat{x}_{n+1}$ $\quad Q_R = \dfrac{\hat{x}_{n+1}(s)}{f(s)} = \dfrac{\omega_e{}^n}{(s+\omega_e)^n},$

传递函数：从噪声到 $\hat{x}_{n+1}$ $\quad G_R = \dfrac{\hat{x}_{n+1}(s)}{n(s)} = \dfrac{\omega_e{}^n}{(s+\omega_e)^n} s^n$

RESO V.S ESO

	ESO	RESO
传递函数：从扰动到 $\hat{x}_{n+1}$ ：	$Q_N=\dfrac{\omega_e^{\,n+1}}{(s+\omega_e)^{n+1}}$,	$Q_R=\dfrac{\omega_e^{\,n}}{(s+\omega_e)^{n}}$,
传递函数：从噪声到 $\hat{x}_{n+1}$ ：	$G_N=\dfrac{\omega_e^{\,n+1}}{(s+\omega_e)^{n+1}}s^n$,	$G_R=\dfrac{\omega_e^{\,n}}{(s+\omega_e)^{n}}s^n$

$$\angle\frac{Q_R(j\omega)}{Q_N(j\omega)}>0$$

总扰动的跟踪

RESO > ESO

$$|G_R(j\omega)|>|G_N(j\omega)|$$

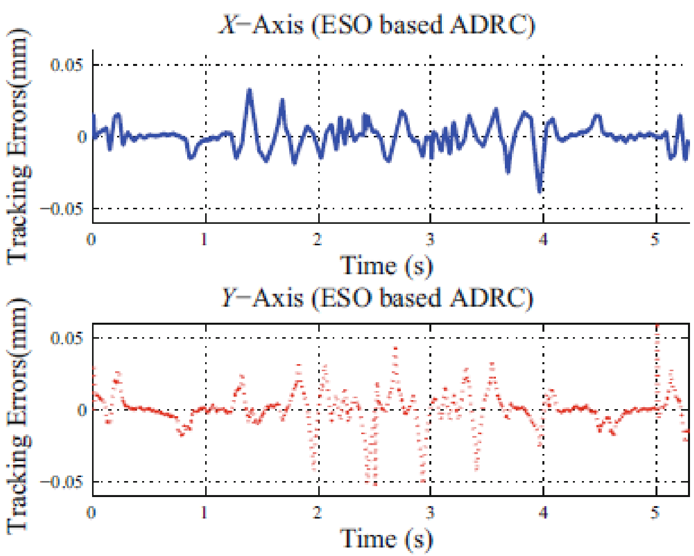
对量测噪声的滤波

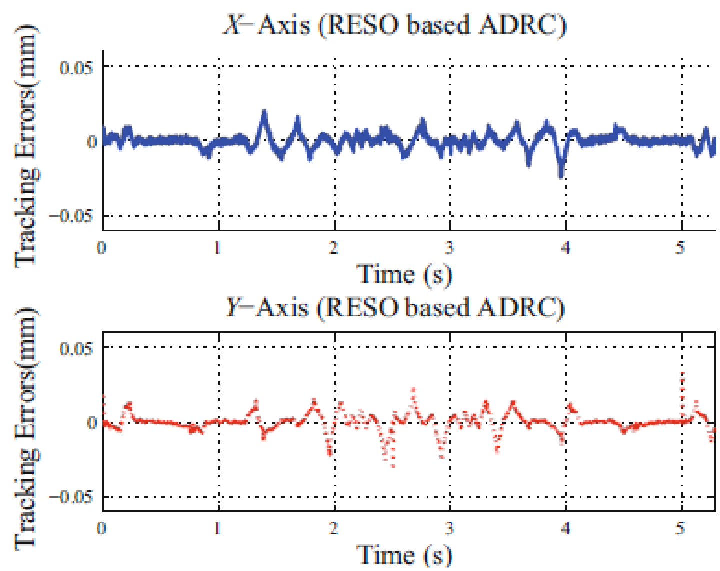
RESO < ESO

仿真结果: 伺服系统

$$
\begin{cases}
\dot{x}_1 = x_2, \\
\dot{x}_2 = x_3 + b_0 u, \\
y = x_1,
\end{cases}
\qquad x_3 = -a_1 x_2 - f_d.
$$

RESO使得跟踪误差更小；ESO使得滤波误差更小；





J. Guo, W. Xue*, and T. Hu, Active Disturbance Rejection Control for PMLM Servo System in CNC Machining, Journal of Systems Science & Complexity, vol. 29, issue 1, pp. 74–98, 2016

自抗扰控制设计方法的新进展

- 降阶ESO的设计方法
- 基于自抗扰控制的反步法设计
- 预测自抗扰控制设计方法

飞机：
攻角控制系统

X_1 : **angle of attack**
X_2 : **angular rate**

无人碾压机：
路径跟踪控制

X_1 : **cross-track error**
X_2 : **error course angle**

$$\dot{X}_1 = b_1(X_1,t)X_2 + f_1(X_1,t)$$
$$\dot{X}_2 = b_2(X_1,X_2,t)U + f_2(X_1,X_2,t)$$

- **下三角系统: 不匹配的扰动**
- **状态: 所有都可以被量测**

W. Xue, and Y. Huang, On Performance Analysis of ADRC for a Class of MIMO Lower-triangular Nonlinear Uncertain Systems, ISA Transactions, vol. 53, pp. 955-962, 2014.

S.Chen, W. Xue, Y. Lin and Y. Huang, On Active Disturbance Rejection Control for Path Following of Automated Guided Vehicle with Uncertain Velocities, ACC 2019, 2019.

对于下三角系统的多个ESO设计

$$\dot{X}_1 = b_1(X_1,t)X_2 + f_1(X_1,t)$$
$$\dot{X}_2 = b_2(X_1,X_2,t)U + f_2(X_1,X_2,t)$$

$$\dot{X}_1 = b_1(X_1,t)X_2 + f_1(X_1,t)$$

虚拟输入

第1个子系统的总扰动

$$\begin{cases} \dot{\hat{X}}_1 = b_1(X_1,t)X_2 - \beta_{1,1}(\hat{X}_1 - X_1) + \hat{\Delta}_1 \\ \dot{\hat{\Delta}}_1 = -\beta_{1,2}(\hat{X}_1 - X_1) \end{cases}$$

第1个子系统的RESO

$$\dot{X}_2 = b_2(X_1,X_2,t)U + f_2(X_1,X_2,t)$$

第2个子系统的总扰动

$$\begin{cases} \dot{\hat{X}}_2 = \hat{b}_2(t)U - \beta_{2,1}(\hat{X}_2 - X_2) + \hat{\Delta}_2 \\ \dot{\hat{\Delta}}_2 = -\beta_{2,2}(\hat{X}_2 - X_2) \end{cases}$$

第2个子系统的RESO

$$\bar{X}_2 = b_1(\cdot)^{-1}(-K_1X_1 - \hat{\Delta}_1(\cdot))$$

虚拟输入的参考信号

$$U = \hat{b}_2(\cdot)^{-1}\left(-K_2(X_2 - \bar{X}_2) - \hat{\Delta}_2(\cdot) + \dot{\bar{X}}_2\right)$$

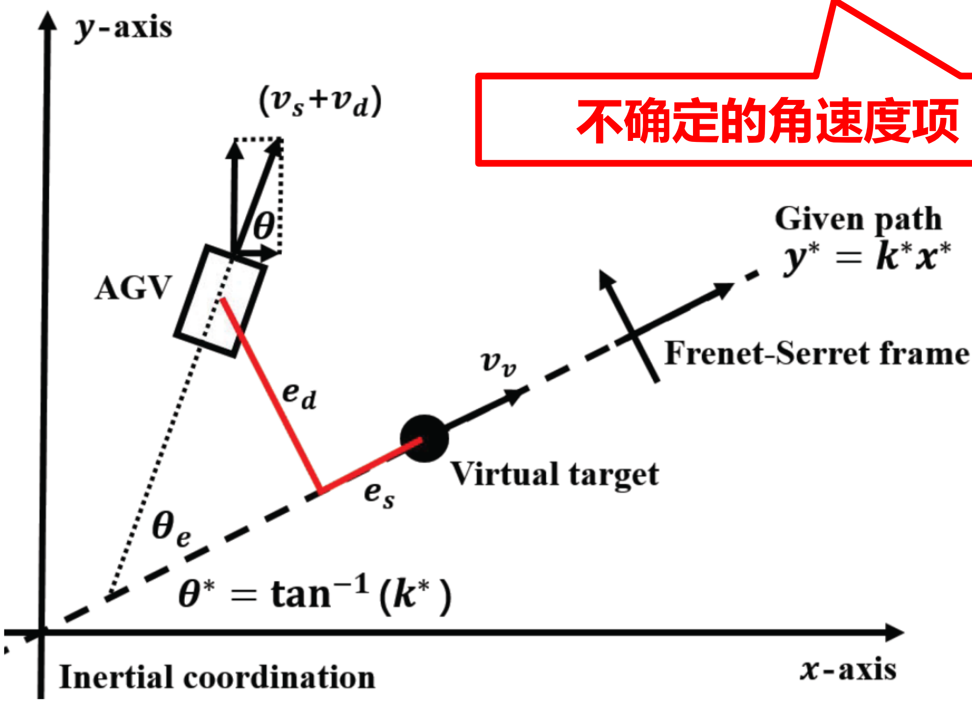
控制输入设计

无人碾压机：路径跟踪控制

$$\begin{cases}\dot{x}(t) = (v_s + v_d(t))\cos(\theta(t)), \\ \dot{y}(t) = (v_s + v_d(t))\sin(\theta(t)), & t \geq t_0, \\ \dot{\theta}(t) = \omega_s(t) + \omega_d(t), \end{cases}$$

不确定的角速度项

不确定的线速度项



下三角系统

$$\boxed{\begin{cases}\dot{e}_d(t) = v_s \sin(\theta_e(t)) + f_1, \\ \dot{\theta}_e(t) = \omega_s(t) + f_2, \end{cases}}$$

S. Chen, W. Xue, Y. Lin and Y. Huang, On Active Disturbance Rejection Control for Path, 2018. Following of Automated Guided Vehicle with Uncertain Velocities, ACC 2019.

设计两个RESO

$$
\begin{cases}
\dot{e}_d(t) = v_s \sin(\theta_e(t)) + f_1, \\
\dot{\theta}_e(t) = \omega_s(t) + f_2,
\end{cases}
$$

$$
\begin{cases}
\dot{\xi}_1(t) = -\beta_1 \xi_1(t) - \beta_1^2 e_d(t) - \beta_1 v_s \sin(\theta_e(t)), \\
\hat{f}_1(t) = \beta_1 e_d(t) + \xi_1(t),
\end{cases}
$$

$$
\begin{cases}
\dot{\xi}_2(t) = -\beta_2 \xi_2(t) - \beta_2^2 \theta_e(t) - \beta_2 \omega_s(t), \\
\hat{f}_2(t) = \beta_2 \theta_e(t) + \xi_2(t),
\end{cases}
$$

虚拟输入的参考值

$$
\theta_{ed}(t) = \begin{cases}
\arcsin\left(\dfrac{-\hat{f}_1(t) - k_d e_d(t)}{v_s} \right), & \text{if } \left| \dfrac{-\hat{f}_1(t) - k_d e_d(t)}{v_s} \right| < 1, \\
\pi/2, & \text{if } \dfrac{-\hat{f}_1(t) - k_d e_d(t)}{v_s} > 1, \\
-\pi/2, & \text{if } \dfrac{-\hat{f}_1(t) - k_d e_d(t)}{v_s} < -1.
\end{cases}
$$

控制输入加入饱和

$$
\omega_s(t) = -\hat{f}_2(t) - k_\theta(\theta_e(t) - \theta_{ed}(t)) + \dot{\theta}_{ed}(t),
$$

控制输入设计

$$\begin{cases}\dot{\xi}_1(t) = -\beta_1\xi_1(t) - \beta_1^2 e_d(t) - \beta_1 v_s \sin(\theta_e(t)),\\ \hat{f}_1(t) = \beta_1 e_d(t) + \xi_1(t),\end{cases}$$

$$\begin{cases}\dot{\xi}_2(t) = -\beta_2\xi_2(t) - \beta_2^2\theta_e(t) - \beta_2\omega_s(t),\\ \hat{f}_2(t) = \beta_2\theta_e(t) + \xi_2(t),\end{cases}$$

$$\theta_{ed}(t) = \begin{cases}\arcsin\left(\frac{-\hat{f}_1(t) - k_d e_d(t)}{v_s}\right), & \text{if } \left|\frac{-\hat{f}_1(t) - k_d e_d(t)}{v_s}\right| < 1,\\ \pi/2, & \text{if } \frac{-\hat{f}_1(t) - k_d e_d(t)}{v_s} > 1,\\ -\pi/2, & \text{if } \frac{-\hat{f}_1(t) - k_d e_d(t)}{v_s} < -1.\end{cases}$$

$$\omega_s(t) = -\hat{f}_2(t) - k_\theta(\theta_e(t) - \theta_{ed}(t)) + \dot{\theta}_{ed}(t),$$

Theorem: Under certain conditions we have that for any given $\varepsilon > 0$ and $t_\varepsilon > 0$, there exist $(k_\theta^*, \beta_1^*, \beta_2^*)$ such that

$$\sup_{t\in[t_0,\infty)} |e_d(t) - e_d^*(t)| \leq \varepsilon, \tag{2}$$

where

真实误差与理想轨迹之间的误差可以调节的任意小

$$\dot{e}_d^*(t) = -k_d e_d^*(t), \quad t \geq t_0, \quad e_d^*(t_0) = e_d(t_0), \tag{3}$$

for $\forall k_\theta \geq k_\theta^*$, $\forall \beta_1 \geq \beta_1^*$ and $\forall \beta_2 \geq \beta_2^*$.

误差的理想轨迹

无人碾压机：路径跟踪控制的仿真结果

$$
\begin{cases}
\dot{x}(t) = (v_s + v_d(t))\cos(\theta(t)), \\
\dot{y}(t) = (v_s + v_d(t))\sin(\theta(t)), & t \geq t_0, \\
\dot{\theta}(t) = \omega_s(t) + \omega_d(t),
\end{cases} \qquad (1)
$$

4种不确定性

$$
\begin{cases}
C1: v_d = \omega_d = 0, \\
C2: v_d = 0.1(1 + 2\sin(2t))v_s, \quad \omega_d = 2 + 3\sin(2t), \\
C3: v_d = 0.3v_s\sin(2t), \quad \omega_d = \begin{cases} -5, & 0 \leq t < 1, \\ 5, & 1 \leq t < 3, \\ -5, & 3 \leq t < 5, \\ 5, & t \geq 5, \end{cases} \\
C4: v_d = \begin{cases} -0.3v_s, & 0 \leq t < 1, \\ 0.3v_s, & 1 \leq t < 3, \\ -0.3v_s, & 3 \leq t < 5, \\ 0.3v_s, & t \geq 5, \end{cases} \quad \omega_d = 5\sin(2t),
\end{cases}
$$

自抗扰控制的参数固定

$$
k_d = 2, \quad k_\theta = 100, \quad \beta_1 = 10, \quad \beta_2 = 20.
$$

轨迹跟踪：仿真结果

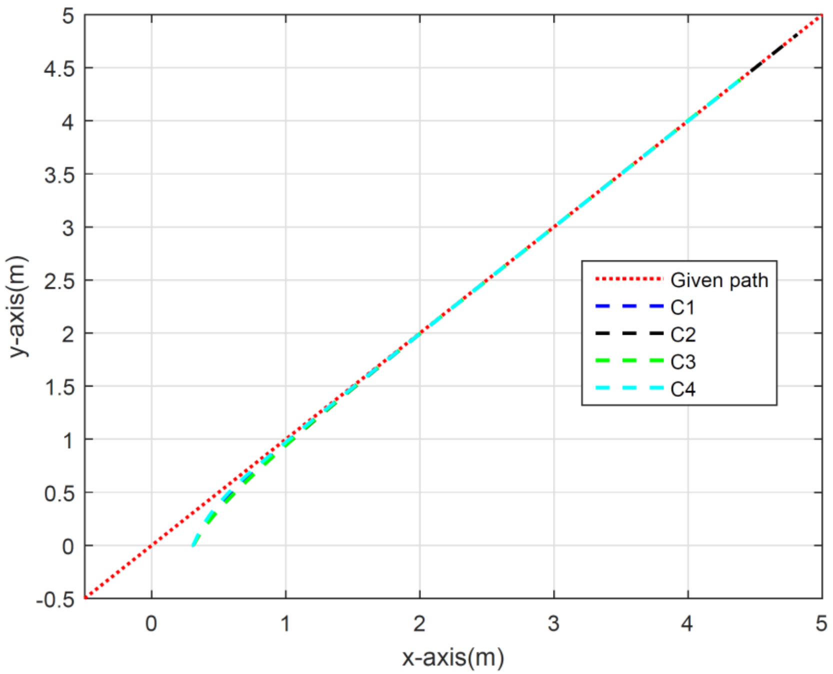


Fig. 2. Tracking results of the AGV with uncertainties (70).

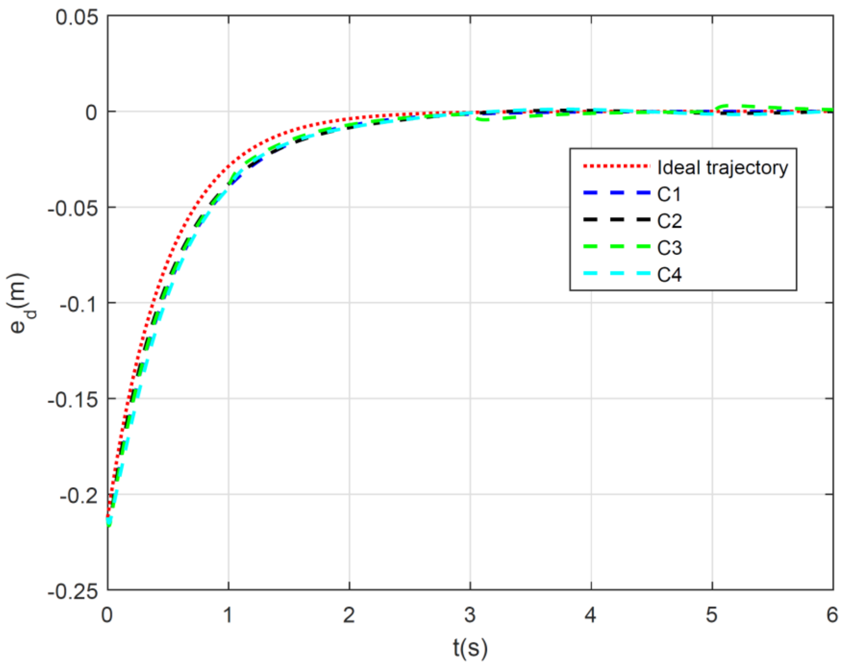


Fig. 3. Cross-track errors for four cases of uncertainties (70).

$$\dot{X}_1 = b_1(X_1,t)X_2 + f_1(X_1,t)$$

虚拟输入

第1个子系统的总扰动

$$\begin{cases} \dot{\hat{X}}_1 = b_1(X_1,t)X_2 - \beta_{1,1}(\hat{X}_1 - X_1) + \hat{\Delta}_1 \\ \dot{\hat{\Delta}}_1 = -\beta_{1,2}(\hat{X}_1 - X_1) \end{cases}$$

第1个子系统的RESO

$$\bar{X}_2 = b_1(\cdot)^{-1}(-K_1X_1 - \hat{\Delta}_1(\cdot))$$

虚拟输入的参考信号

$$\dot{X}_2 = b_2(X_1,X_2,t)U + f_2(X_1,X_2,t)$$

第2个子系统的总扰动

$$\begin{cases} \dot{\hat{X}}_2 = \hat{b}_2(t)U - \beta_{2,1}(\hat{X}_2 - X_2) + \hat{\Delta}_2 \\ \dot{\hat{\Delta}}_2 = -\beta_{2,2}(\hat{X}_2 - X_2) \end{cases}$$

第2个子系统的RESO

$$U = \hat{b}_2(\cdot)^{-1}\left(-K_2(X_2 - \bar{X}_2) - \hat{\Delta}_2(\cdot) + \dot{\bar{X}}_2\right)$$

控制输入设计

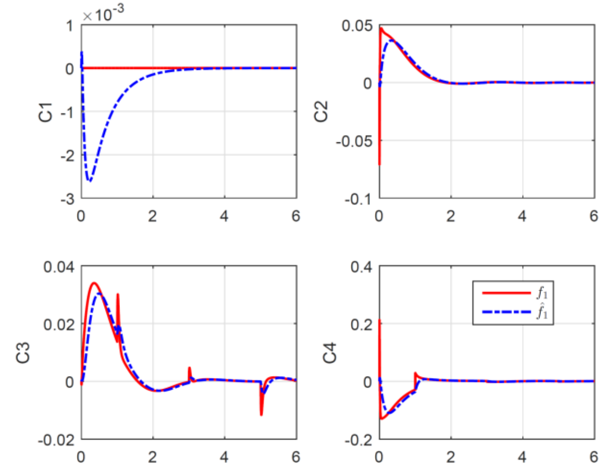


Fig. 4. The estimations of the “total disturbance” f_1 for four cases of uncertainties (70).

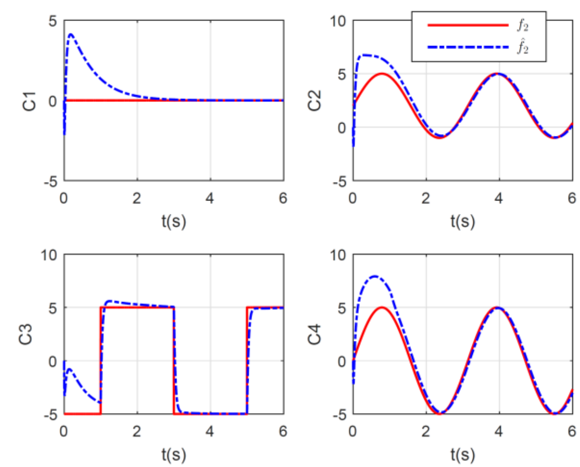


Fig. 5. The estimations of the “total disturbance” f_2 for four cases of uncertainties (70).

自抗扰控制设计方法的新进展

- 降阶ESO的设计方法
- 基于自抗扰控制的反步法设计
- 预测自抗扰控制设计方法

非线性不确定时滞系统

$$\begin{cases} \dot{x}(t) = Ax(t) + B\big(u(t-\tau) + \delta(x(t),t)\big) \\ y(t) = C^T x(t) \end{cases}$$

输入延迟　　总扰动

(注：输入输出延迟模型可等价变换为输入延迟模型)

总扰动：	$\delta(x,t) \in R$	控制输入：	$u(t) \in R$
时延：	$\tau \in R$	量测与被控输出：	$y(t) \in R$
系统状态：	$x(t) \in R^n$	已知模型部分：	A, B, C

理想轨线 $x^*(t)$

$$\begin{cases} \dot{x}^*(t) = Ax^*(t) - BK^T\big(x^*(t) - r(t)\big) \\ x^*(t_0) = x(t_0) \end{cases}$$

控制参数　　参考信号

$$A_K = A - BK^T$$

设计特征值

快速、无超调

控制目的：设计输入使状态跟踪理想轨线

理想控制输入

$$u^*(t) = -K^T\left(x(t+\tau) - r(t+\tau)\right) - \delta\left(x(t+\tau), t+\tau\right)$$

预估系统状态与扰动

针对时延系统改进的自抗扰控制算法

- **PO-ADRC: Predictor Observer based ADRC**
  (基于预测观测器的自抗扰)
- **DD-ADRC: Delayed designed ADRC**
  (匹配时延自抗扰)
- **PP-ADRC: Polynomial based predictive ADRC**
  (基于多项式预测的自抗扰)
- **SP-ADRC: Smith Predictor based ADRC**
  (基于斯密斯预估器的自抗扰)

Predictor Observer based ADRC (PO-ADRC)

无扰系统

$$\begin{cases} \dot{X}(t) = AX(t) + Bu(t-\tau) \\ Y(t) = C^T X(t) \end{cases}$$

预测观测器

$$\begin{cases} \dot{\hat{X}}(t+\tau) = A\hat{X}(t+\tau) + Bu(t) + e^{A\tau}L\left(Y(t) - \hat{Y}(t+\tau)\right) \\ \hat{Y}(t+\tau) = C^T\hat{X}(t) + C^T\int_0^\tau e^{A\theta}L\left(Y(t-\theta) - \hat{Y}(t+\tau-\theta)\right)d\theta \end{cases}$$

[13] M. Krstic. Delay Compensation for Nonlinear, Adaptive, and PDE Systems. Cambridge, MA: Birkhäuser Boston, 2009.

$$\begin{cases} \begin{bmatrix} \dot{\hat{x}}_{PO}(t+\tau) \\ \dot{\hat{\delta}}_{PO}(t+\tau) \end{bmatrix} = A_e \begin{bmatrix} \hat{x}_{PO}(t+\tau) \\ \hat{\delta}_{PO}(t+\tau) \end{bmatrix} + B_e u(t) + e^{A_e\tau}L_{e,PO}\left(y(t) - \hat{y}_{PO}(t+\tau)\right) \\ \hat{y}_{PO}(t+\tau) = C^T\hat{x}_{PO}(t) + C_e^T\int_0^\tau e^{A\theta}L_{e,PO}\left(y(t-\theta) - \hat{y}_{PO}(t+\tau-\theta)\right)d\theta \end{cases}$$

扩张状态预测观测器

$$A_e = \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}, B_e = \begin{bmatrix} B \\ 0 \end{bmatrix}, C_e = \begin{bmatrix} C \\ 0 \end{bmatrix}$$

Predictor Observer based ADRC (PO-ADRC)

扩张状态预测观测器

$$\begin{cases} \begin{bmatrix} \dot{\hat{x}}_{PO}(t+\tau) \\ \dot{\hat{\delta}}_{PO}(t+\tau) \end{bmatrix} = A_e \begin{bmatrix} \hat{x}_{PO}(t+\tau) \\ \hat{\delta}_{PO}(t+\tau) \end{bmatrix} + B_e u(t) + e^{A_e \tau} L_{e,PO}\left(y(t) - \hat{y}_{PO}(t+\tau)\right) \\ \hat{y}_{PO}(t+\tau) = C^T \hat{x}_{PO}(t) + C_e^T \int_0^\tau e^{A\theta} L_{e,PO}\left(y(t-\theta) - \hat{y}_{PO}(t+\tau-\theta)\right) d\theta \end{cases}$$

主动抗扰的控制设计

$$u(t) = -K^T\left(\hat{x}_{PO}(t+\tau) - r(t+\tau)\right) - \hat{\delta}_{PO}(t+\tau)$$

状态与扰动的预估值

[14] W. Xue, P. Liu, S. Chen, Y. Huang. On extended state predictor observer based active disturbance rejection control for uncertain systems with sensor delay. In 16th International Conference on Control, Automation and Systems, 2016.

Delayed designed ADRC (DD-ADRC)

匹配时延的扩张状态观测器

$$\begin{cases} \begin{bmatrix} \dot{\hat{x}}_{DD}(t) \\ \dot{\hat{\delta}}_{DD}(t) \end{bmatrix} = A_e \begin{bmatrix} \hat{x}_{DD}(t) \\ \hat{\delta}_{DD}(t) \end{bmatrix} + \boxed{B_e u(t-\tau)} + L_{e,DD}\left(y(t) - \hat{y}_{DD}(t)\right) \\ \hat{y}_{DD}(t) = C^T \hat{x}_{DD}(t) \end{cases}$$

时延匹配的控制输入

主动抗扰的控制设计

$$u(t) = -K^T\left(\hat{x}_{DD}(t) - r(t)\right) - \hat{\delta}_{DD}(t)$$

状态与扰动的预估值

[15] S. Zhao, Z. Gao. Modified active disturbance rejection control for time-delay systems. ISA Transactions, 2014.

Polynomial based predictive ADRC (PP-ADRC)

预测环节

泰勒展开

$$e^{\tau s}=1+\tau s+O\left(\tau^2 s^2\right)$$

量测预测

$$y_p\left(t\right)=\tau\dot{y}\left(t\right)+y\left(t\right)$$

扩张状态观测器

$$\begin{cases}\begin{bmatrix}\dot{\hat{x}}_{PP}\left(t+\tau\right)\\ \dot{\hat{\delta}}_{PP}\left(t+\tau\right)\end{bmatrix}=A_e\begin{bmatrix}\hat{x}_{PP}\left(t+\tau\right)\\ \hat{\delta}_{PP}\left(t+\tau\right)\end{bmatrix}+B_e u\left(t\right)+\boxed{L_{e,PP}\left(y_P\left(t\right)-\hat{y}_{PP}\left(t+\tau\right)\right)}\\ \hat{y}_{PP}\left(t+\tau\right)=C^T\hat{x}_{PP}\left(t+\tau\right)\end{cases}$$

量测预测值

主动抗扰的控制设计

$$u\left(t\right)=-K^T\left(\hat{x}_{PP}\left(t+\tau\right)-r\left(t+\tau\right)\right)-\hat{\delta}_{PP}\left(t+\tau\right)$$

状态与扰动的预估值

[16] J. Han. Auto-Disturbances Rejection Control for Time-Delay Systems. Control engineering of China, 2008.

Smith Predictor based ADRC (SP-ADRC)

Smith预估器(预测环节)

$$y_{P,SP}(t)=y(t)-\tilde{y}_{SP}(t)+\tilde{y}_{SP}(t+\tau),\quad \begin{cases}\dot{\tilde{x}}_{SP}(t)=A\tilde{x}_{SP}(t)+Bu(t-\tau)\\ \tilde{y}_{SP}(t)=C^T\tilde{x}_{SP}(t)\end{cases}$$

量测预测

扩张状态观测器

$$\begin{cases}\begin{bmatrix}\dot{\hat{x}}_{SP}(t+\tau)\\ \dot{\hat{\delta}}_{SP}(t+\tau)\end{bmatrix}=A_e\begin{bmatrix}\hat{x}_{SP}(t+\tau)\\ \hat{\delta}_{SP}(t+\tau)\end{bmatrix}+B_e u(t)+\boxed{L_{e,SP}\left(y_{P,SP}(t)-\hat{y}_{SP}(t+\tau)\right)}\\ \hat{y}_{SP}(t+\tau)=C^T\hat{x}_{SP}(t+\tau)\end{cases}$$

量测预测值

主动抗扰的控制设计

$$u(t)=-K^T\left(\hat{x}_{SP}(t+\tau)-r(t+\tau)\right)-\hat{\delta}_{SP}(t+\tau)$$

状态与扰动的预估值

[17] Q. Zheng, Z. Gao. Predictive active disturbance rejection control for processes with time delay. ISA Transactions, 2014.

● 可处理的给定时延大小

	可处理的给定时延大小 (无扰情况)
PO-ADRC	任意给定
DD-ADRC	有界
PP-ADRC	有界
SP-ADRC	任意给定

Sen Chen, Wenchao Xue*, Sheng Zhong, and Yi Huang, On comparison of modified ADRCs for nonlinear uncertain systems with time delay, SCIENCE CHINA Information Sciences, Vol. 61 070223:1–070223:15, 2018.

● 对开环稳定条件的要求

	开环稳定条件 (有扰情况)
PO-ADRC	非必要条件
DD-ADRC	非必要条件
PP-ADRC	非必要条件
SP-ADRC	必要条件

Sen Chen, Wenchao Xue*, Sheng Zhong, and Yi Huang, On comparison of modified ADRCs for nonlinear uncertain systems with time delay, SCIENCE CHINA Information Sciences, Vol. 61 070223:1–070223:15, 2018.

● 抗扰性能(低频段)

假设参考信号与初始值为零

	抗扰性能 (低频段)
PO-ADRC	$\lim_{\omega\to 0} G_{y\delta,a}(j\omega)=0$ $a=PO,DD,PP,SP$
DD-ADRC	
PP-ADRC	
SP-ADRC	

$$Y_a(s)=G_{y\delta,a}(s)\Delta_a(s),\quad a=PO,DD,PP,SP.$$

扰动到输出的传递函数

$Y_a(s)=L(y_a)$ 输出的拉普拉斯变换

$\Delta_a(s)=L(\delta_a)$ 扰动的拉普拉斯变换

Sen Chen, Wenchao Xue*, Sheng Zhong, and Yi Huang, On comparison of modified ADRCs for nonlinear uncertain systems with time delay, SCIENCE CHINA Information Sciences, Vol. 61 070223:1–070223:15, 2018.

● 抗扰性能(低频段)

	抗扰性能 (低频段，一阶系统)
$\lim\limits_{\omega\to 0}\dfrac{G_{y\delta,\mathrm{PP}}(j\omega)}{G_{y\delta,\mathrm{DD}}(j\omega)}$	$O\left(\dfrac{1}{\omega_o}\right)$
$\lim\limits_{\omega\to 0}\dfrac{G_{y\delta,\mathrm{PP}}(j\omega)}{G_{y\delta,\mathrm{SP}}(j\omega)}$	
$\lim\limits_{\omega\to 0}\dfrac{G_{y\delta,\mathrm{PP}}(j\omega)}{G_{y\delta,\mathrm{PO}}(j\omega)}$	

PP-ADRC: 更强的低频段抗扰性能(较大观测器带宽ω_o)

$$\det\left(A_e - L_{e,a}C_e^T\right) = \left(s+\omega_o\right)^2$$

相同的观测器带宽ω_o

Sen Chen, Wenchao Xue*, Sheng Zhong, and Yi Huang, On comparison of modified ADRCs for nonlinear uncertain systems with time delay, SCIENCE CHINA Information Sciences, Vol. 61 070223:1–070223:15, 2018.

仿真分析

一阶非线性不确定时延系统

$$\begin{cases} \dot{x}_1(t) = ax_1(t) + bu(t-\tau) + \delta(x_1, t) \\ y(t) = x_1(t) \end{cases}$$

延迟

总扰动

$$a = -6.9\times10^{-3},\ b = 4.35\times10^{-2},\ \tau = 60$$

[18] W. Tan, F. Fang, et al. Linear control of a boiler-turbine unit: analysis and design. ISA Transaction, 2008.

不确定性

$$Case\,1:\ \delta = \begin{cases} 0,\ 0\ (\mathrm{s}) \le t < 1000\ (\mathrm{s}) \\ 5,\ t \ge 1000\ (\mathrm{s}) \end{cases}$$

(突变外扰)

$$Case\,2:\ \delta = (ax_1)^2 + \sin\left(x_1 / 2\pi\right) + e^{a(x_1-300)}$$

(非线性未建模动态)

$$Case\,3:\ \delta = 0.2ax_1 / b$$

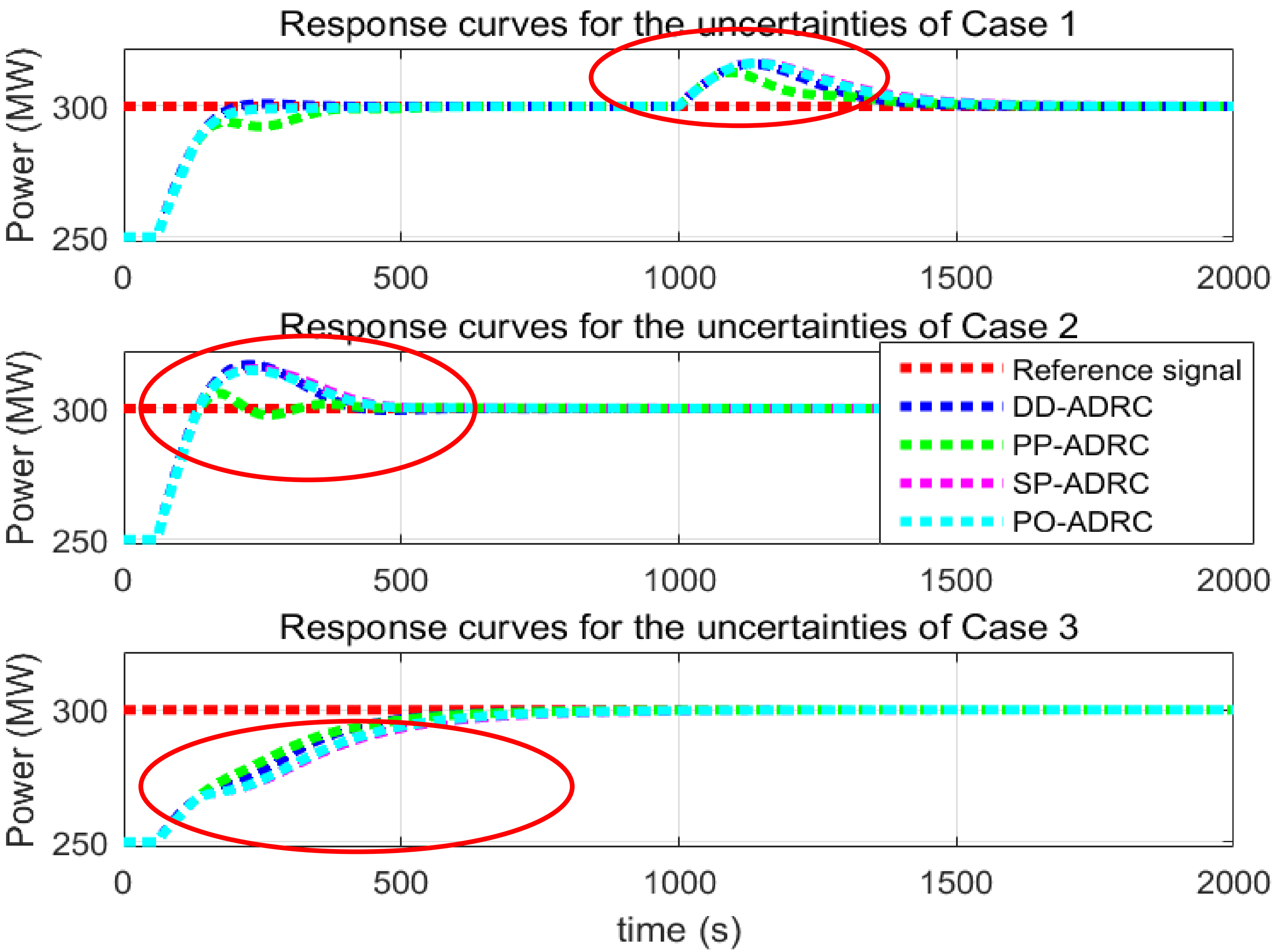
(参数不确定)

参考信号： $r = 300$

控制器参数： $K = 0.014,\ \omega_o = 0.015$

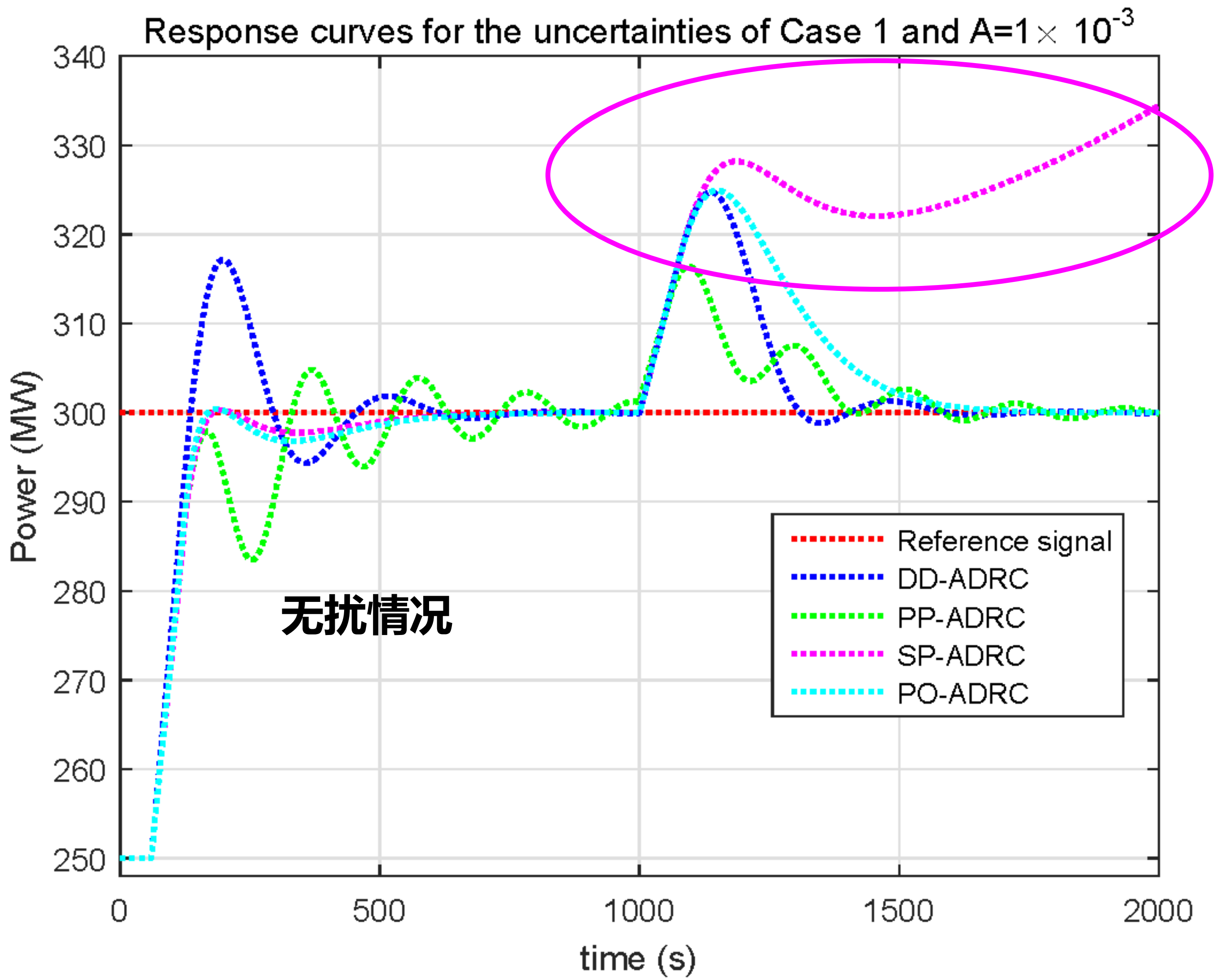
陈森，非线性不确定系统的自抗扰控制研究[D]，中国科学院大学, 2019.

四种针对时延系统改进ADRC的仿真结果(三种不确定性)



陈森，非线性不确定系统的自抗扰控制研究[D]，中国科学院大学, 2019.

四种针对时延系统改进ADRC的仿真结果(对象开环不稳定)



$\mathbf{a = 1 \times 10^{-3}}$**(对象开环不稳定)**

陈森，非线性不确定系统的自抗扰控制研究[D]，中国科学院大学, 2019.

谢谢！

薛文超

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卡尔曼滤波简介

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中国科学院数学与系统科学研究院

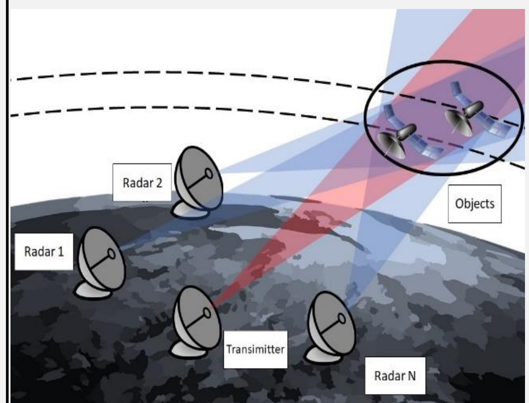
概要

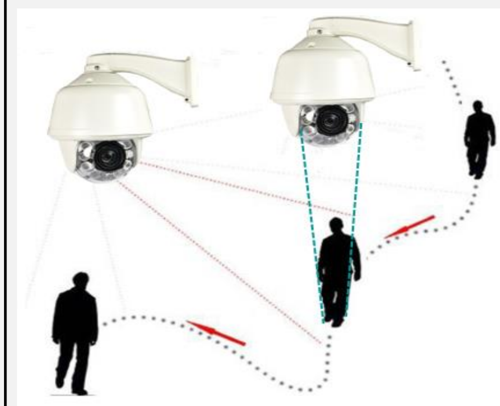
- 研究背景和问题描述
- 卡尔曼滤波算法及其主要特性
- 扩展卡尔曼滤波算法

研究背景

滤波：最优的状态估计

实际系统	Volatility Estimation	Orbit Determination	Camera Tracking
Observation data	Disturbed stock price	Radar measurements	Measurements of sensors in network
System state	Volatility	Position and velocity	Position and velocity





准备知识：随机变量

- **随机变量：随机试验各种结果的实值单值函数**

  X：将随机试验各种结果映射到实数

  （扔硬币出现正面记为1，扔硬币出现反面记为0）

- **随机变量X最基本的特性：概率分布函数**

  (probability distribution function, PDF) $F_X(x) = P(X \le x)$

- **PDF满足的特性：**

$$
\begin{aligned}
F_X(x) &\in [0,1] \\
F_X(-\infty) &= 0 \\
F_X(\infty) &= 1 \\
F_X(a) &\le F_X(b) \quad \text{if } a \le b \\
P(a < X \le b) &= F_X(b) - F_X(a)
\end{aligned}
$$

准备知识：概率分布函数(PDF)和概率密度函数(pdf)

- **概率密度函数(Probability density function, pdf)：PDF的导数**

$$f_X(x) = \frac{dF_X(x)}{dx}$$

- **pdf满足的特性：**

$$\begin{aligned}
F_X(x) &= \int_{-\infty}^{x} f_X(z)\,dz \\
f_X(x) &\geq 0 \\
\int_{-\infty}^{\infty} f_X(x)\,dx &= 1 \\
P(a < x \leq b) &= \int_a^b f_X(x)\,dx
\end{aligned}$$

准备知识：条件概率分布函数和条件概率密度函数

- 给定事件A条件下的概率分布

$$
\begin{aligned}
F_X(x|A) &= P(X \leq x|A) \\
&= \frac{P(X \leq x, A)}{P(A)}
\end{aligned}
$$

- 给定事件A条件下的概率密度

$$
f_X(x|A) = \frac{dF_X(x|A)}{dx}
$$

准备知识：期望和方差

- **g(X)为随机变量的一个函数（将随机变量映射为随机变量），则g(X)的期望为** $E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$

- **X的期望为：** $\overline{x} \triangleq E(X) = \int_{-\infty}^{\infty} x f_X(x)\, dx$

- **X的方差：**

$$\begin{aligned} \sigma_X^2 &= E[(X-\bar{x})^2] \\ &= \int_{-\infty}^{\infty} (x-\bar{x})^2 f_X(x)\, dx \end{aligned}$$

- **常用标记：** $X \sim (\bar{x}, \sigma^2)$

- **条件期望：** $E(X|A) = \int_{-\infty}^{\infty} x f_X(x|A)\, dx$

准备知识：高维随机变量

$$
\begin{aligned}
F_X(x) &= P(X \leq x) \\
F_Y(y) &= P(Y \leq y)
\end{aligned}
$$

- **联合分布函数：** $F_{XY}(x, y) = P(X \leq x, Y \leq y)$

(扔两个硬币，出现正面+1，出现反面+0)

$$
\begin{aligned}
F(x, y) &\in [0, 1] \\
F(x, -\infty) = F(-\infty, y) &= 0 \\
F(\infty, \infty) &= 1 \\
F(a, c) &\leq F(b, d) \quad \text{if } a \leq b \text{ and } c \leq d \\
P(a < x \leq b, c < y \leq d) &= F(b, d) + F(a, c) - F(a, d) - F(b, c) \\
F(x, \infty) &= F(x) \\
F(\infty, y) &= F(y)
\end{aligned}
$$

准备知识：高维密度函数

- **联合密度函数：** $f_{XY}(x,y) = \dfrac{\partial^2 F_{XY}(x,y)}{\partial x \partial y}$

$$
\begin{aligned}
F(x,y) &= \int_{-\infty}^{x} \int_{-\infty}^{y} f(z_1, z_1)\, dz_1\, dz_2 \\
f(x,y) &\geq 0 \\
\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dx\, dy &= 1 \\
P(a < x \leq b, c < y \leq d) &= \int_{c}^{d} \int_{a}^{b} f(x,y)\, dx\, dy \\
f(x) &= \int_{-\infty}^{\infty} f(x,y)\, dy \\
f(y) &= \int_{-\infty}^{\infty} f(x,y)\, dx
\end{aligned}
$$

准备知识：高维密度函数

- **边际密度和分布函数：**

$$f_Y(y)=\int_{-\infty}^{\infty} f(x,y)dx \qquad F_Y(y)=\int_{-\infty}^{y}\int_{-\infty}^{\infty} f(x,t)dxdt$$

$$f_X(x)=\int_{-\infty}^{\infty} f(x,y)dy \qquad F_X(y)=\int_{-\infty}^{y}\int_{-\infty}^{\infty} f(t,y)dydt$$

- **条件概率密度和条件概率分布**

$$f_{X|Y}(x\,|\,\mathrm{y})=\frac{f(x,\mathrm{y})}{f_Y(y)}$$

$$F_{X|Y}(X\,|\,Y)=\int_{-\infty}^{x} f_{X|Y}(t\,|\,\mathrm{y})\mathrm{dt}=\int_{-\infty}^{x}\frac{f(t,\mathrm{y})}{f_Y(y)}\mathrm{dt}$$

准备知识：独立性

- **随机变量X和Y独立：** $P(X \le x, Y \le y) = P(X \le x)P(Y \le y) \quad \text{for all } x, y$

- **随机变量X和Y独立：** $$\begin{aligned} F_{XY}(x,y) &= F_X(x)F_Y(y) \\ f_{XY}(x,y) &= f_X(x)f_Y(y) \end{aligned}$$

- **独立时的条件概率密度** $f_{X|Y}(x \mid y) = \dfrac{f(x,y)}{f_Y(y)} = f_X(x)$

准备知识：高维随机变量的相关函数

- **互相关函数矩阵：**

$$
\begin{aligned}
R_{XY} &= E(XY^T) \\
&= \begin{bmatrix} E(X_1Y_1) & \cdots & E(X_1Y_m) \\ \vdots & & \vdots \\ E(X_nY_1) & \cdots & E(X_nY_m) \end{bmatrix}
\end{aligned}
$$

- **互协方差矩阵：**

$$
\begin{aligned}
C_{XY} &= E[(X-\bar{X})(Y-\bar{Y})^T] \\
&= E(XY^T) - \bar{X}\bar{Y}^T
\end{aligned}
$$

准备知识：高维随机变量的协方差阵

- 自相关函数矩阵

$$
\begin{aligned}
R_X &= E[XX^T] \\
&= \begin{bmatrix} E[X_1^2] & \cdots & E[X_1X_n] \\ \vdots & & \vdots \\ E[X_nX_1] & \cdots & E[X_n^2] \end{bmatrix}
\end{aligned}
$$

- 自协方差矩阵

$$
\begin{aligned}
C_X &= E[(X-\bar{X})(X-\bar{X})^T] \\
&= \begin{bmatrix} E[(X_1-\bar{X}_1)^2] & \cdots & E[(X_1-\bar{X}_1)(X_n-\bar{X}_n)] \\ \vdots & & \vdots \\ E[(X_n-\bar{X}_n)(X_1-\bar{X}_1)] & \cdots & E[(X_n-\bar{X}_n)^2] \end{bmatrix} \\
&= \begin{bmatrix} \sigma_1^2 & \cdots & \sigma_{1n} \\ \vdots & & \vdots \\ \sigma_{n1} & \cdots & \sigma_n^2 \end{bmatrix}
\end{aligned}
$$

自协方差矩阵为半正定矩阵

$$
\begin{aligned}
z^T C_X z &= z^T E[(X-\bar{X})(X-\bar{X})^T] z \\
&= E[z^T (X-\bar{X})(X-\bar{X})^T z] \\
&= E[(z^T (X-\bar{X}))^2] \\
&\geq 0
\end{aligned}
$$

Note that $\sigma_{ij} = \sigma_{ji}$ so $C_X = C_X^T$.

准备知识：随机过程

随机过程：一列随机变量X(t)，t为时间

均为时变的函数：

$$F_X(x,t) = P(X(t) \le x)$$

$$F_X(x,t) = P\left[X_1(t) \le x_1 \text{ and } \cdots X_n(t) \le x_n(t)\right]$$

$$f_X(x,t) = \frac{d^n F_X(x,t)}{dx_1 \cdots dx_n}$$

$$\bar{x}(t) = \int_{-\infty}^{\infty} x f(x,t)\, dx$$

$$\begin{aligned} C_X(t) &= E\left\{[X(t) - \bar{x}(t)]\,[X(t) - \bar{x}(t)]^T\right\} \\ &= \int_{-\infty}^{\infty} [x - \bar{x}(t)]\,[x - \bar{x}(t)]^T f(x,t)\, dx \end{aligned}$$

$$R_X(t_1,t_2) = E\left[X(t_1)X^T(t_2)\right]$$

$$C_X(t_1,t_2) = E\left\{\left[X(t_1) - \bar{X}(t_1)\right]\left[X(t_2) - \bar{X}(t_2)\right]^T\right\}$$

离散时变线性系统的滤波问题

$$\begin{cases} X_{k+1} = A_k X_k + B_k u_k + w_k \\ Y_{k+1} = C_{k+1} X_{k+1} + n_{k+1} \end{cases}$$

$$k = 0,1,2,\ldots$$

$\{n_k\}$和$\{w_k\}$为随机过程

$$E(n_k) = 0, \forall k > 0$$

$$E(w_k) = 0, \forall k > 0$$

$$C_n(k, j) = \begin{cases} 0, \forall k \neq j \\ R_k, \forall k = j \end{cases}$$

$$C_w(k, j) = \begin{cases} 0, \forall k \neq j \\ Q_k, \forall k = j \end{cases}$$

离散时变线性系统的滤波问题

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$$C_w(k, j) = \begin{cases} 0, \forall k \neq j \\ Q_k, \forall k = j \end{cases}$$

目标：利用量测 $\{Y_i, 1 \le i \le k\}$获得 X_k 的线性最小方差估计

$$\hat{X}_k = \mathop{\arg\min}_{Z_k \in \{\text{linear function of } Y_i, 1 \le i \le k\}} E\left\{(Z_k - X_k)(Z_k - X_k)^T\right\}$$

概要

- 研究背景和问题描述
- 卡尔曼滤波算法及其主要特性
- 扩展卡尔曼滤波算法

卡尔曼滤波算法及参数设计

离散线性系统： $\begin{cases} X_{k+1} = A_k X_k + B_k u_k + w_k \\ Y_{k+1} = C_{k+1} X_{k+1} + n_{k+1} \end{cases}$

$k = 0,1,2,\ldots$

1930-2016

卡尔曼滤波(1960s, Kalman):

初始化过程：

$$\begin{cases} P_0 = E\left(X_0 - EX_0\right)\left(X_0 - EX_0\right)^T \\ \hat{X}_0 = EX_0 \end{cases}$$

参数选择：

$$\begin{cases} Q_k = E\left(w_k w_k^T\right) \\ R_{k+1} = E\left(n_{k+1} n_{k+1}^T\right) > 0 \end{cases}$$

$\bar{X}_{k+1} = A_k \hat{X}_k + B_k u_k$： 预测过程

$\bar{P}_{k+1} = A_k P_k A_k^T + Q_k$： 预测误差的协方差阵

$\hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}(Y_{k+1} - C_{k+1}\bar{X}_{k+1})$：更新过程

$K_{k+1} = (\bar{P}_{k+1} C_{k+1}^T)(C_{k+1}\bar{P}_{k+1}C_{k+1}^T + R_{k+1})^{-1}$： 更新增益

$P_{k+1} = \bar{P}_{k+1} - K_{k+1} C_{k+1} \bar{P}_{k+1}$： 估计/滤波误差的协方差阵

考虑线性定常系统:

$$\begin{cases} X_{k+1} = AX_k + Bu_k + w_k \\ Y_{k+1} = CX_{k+1} + n_{k+1} \end{cases}$$

$$k = 0,1,2,\ldots$$

$$\begin{cases} \hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}(Y_{k+1} - C\bar{X}_{k+1}) \\ \bar{X}_{k+1} = A\hat{X}_k + Bu_k \\ K_{k+1} = (\bar{P}_{k+1}C^T)(C\bar{P}_{k+1}C^T + R)^{-1} \\ P_{k+1} = \bar{P}_{k+1} - K_{k+1}C\bar{P}_{k+1} \\ \bar{P}_{k+1} = AP_kA^T + Q \end{cases}$$

定理：卡尔曼滤波满足如下特性：

1. 线性最小方差估计

2. $P_k = \mathbb{E}\left\{(X_k - \hat{X}_k)(X_k - \hat{X}_k)^T\right\}$

3.当(A,C)可观时，滤波误差均方有界

定理的证明需要下面3个引理

引理1. 设x和y分别是具有前二阶矩的随机向量，则利用y对x的线性无偏最小方差估计为

$$\hat{x}_y = \mathbb{E}\{x\} + R_{xy}R_y^{-1}(y - \mathbb{E}\{y\}),$$

x变量本身的期望　　增益　　y带来的新信息

其中

$$R_{xy} \triangleq \mathbb{E}\{(x - \mathbb{E}\{x\})(y - \mathbb{E}\{y\})^T\},$$

$$R_y \triangleq \mathbb{E}\{(y - \mathbb{E}\{y\})(y - \mathbb{E}\{y\})^T\}.$$

引理1的证明：由估计的线性无偏性可知$\hat{x}_y$具有如下形式：

$$\hat{x}_y = \mathbb{E}\{x\} + K(y - \mathbb{E}\{y\}).$$

由此易得

$$\mathbb{E}\left\{\left(\hat{x}_y - x\right)\left(\hat{x}_y - x\right)^T\right\}$$

$$= \mathbb{E}\{(x - \mathbb{E}\{x\} - K(y - \mathbb{E}\{y\}))(x - \mathbb{E}\{x\} - K(y - \mathbb{E}\{y\}))^T\}$$

$$= R_x - KR_{yx} - R_{xy}K^T + KR_yK^T$$

$$= \left(K - R_{xy}R_y^{-1}\right)R_y\left(K - R_{xy}R_y^{-1}\right)^T + R_x - R_{xy}R_y^{-1}R_{yx}$$

其中最后一个等式的第1项为非负定项，故欲令估计误差的协方差阵最小，当且仅当$K = R_{xy}R_y^{-1}$.

引理2. **设**u**为已知向量，**x**和**y**是具有前二阶矩的随机向量，记**y**对**x**的线性无偏最小方差估计为**$\hat{x}_y$**，则利用**y**对**$z \triangleq Ax + Bu + Cy$**的线性无偏最小方差估计为**$\hat{z}_y = A\hat{x}_y + Bu + Cy$**.**

引理2的证明：

无偏性需$\hat{z}_y$ 的形式为: $\mathrm{A}(\mathrm{E}\{x\} + \mathrm{K}(\mathrm{y} - \mathrm{E}\{\mathrm{y}\}) + Bu + \mathrm{Cy}$

最小方差需要：$\hat{z}_y = A\hat{x}_y + Bu + Cy$

引理3. 设x, $\mathcal{Y}$和y是具有前二阶矩的随机向量，记 $\mathrm{w} \triangleq [\mathcal{Y}^T \quad y^T]^T$, 若利用$\mathcal{Y}$对x和y的线性无偏最小方差估计分别为$\hat{x}_{\mathcal{Y}}$和$\hat{y}_{\mathcal{Y}}$，则利用w对x的线性无偏最小方差估计满足

$\mathbf{w} \triangleq [\boldsymbol{\mathcal{Y}}^T \quad \boldsymbol{y}^T]^T$：所有数据

$\boldsymbol{y}$：当前数据

$\boldsymbol{\mathcal{Y}}$：过去数据

$$\hat{x}_{\mathrm{w}} = \underbrace{\hat{x}_{\mathcal{Y}}}_{\text{利用过去数据得到的估计}} + \underbrace{\mathrm{K}}_{\text{新息增益}}\underbrace{(\mathrm{y} - \hat{y}_{\mathcal{Y}})}_{\text{当前数据带来的新息}},$$

其中

$$\mathrm{K} = R_{\mathrm{xy}|\mathcal{Y}} R_{\mathrm{y}|\mathcal{Y}}^{-1},$$

$$R_{\mathrm{xy}|\mathcal{Y}} \triangleq \mathbb{E}\{(\mathrm{x} - \hat{x}_{\mathcal{Y}})(\mathrm{y} - \hat{y}_{\mathcal{Y}})^T\},$$

$$R_{\mathrm{y}|\mathcal{Y}} \triangleq \mathbb{E}\{(\mathrm{y} - \hat{y}_{\mathcal{Y}})(\mathrm{y} - \hat{y}_{\mathcal{Y}})^T\}.$$

引理3的证明：

引入新变量：$e \triangleq x - \hat{x}_{\mathcal{Y}} - K(y - \hat{y}_{\mathcal{Y}})$, 其中$\mathrm{K} = R_{\mathrm{xy}|\mathcal{Y}} R_{\mathrm{y}|\mathcal{Y}}^{-1}$，则容易验证$\mathrm{E}\{e\} = 0$. 接下来分析e的进一步性质：

由引理1可得：

$$\hat{x}_{\mathcal{Y}} = \mathrm{E}\{\mathrm{x}\} + R_{x\mathcal{Y}} R_{\mathcal{Y}\mathcal{Y}}^{-1} (\mathcal{Y} - E\{\mathcal{Y}\})$$

$$\hat{y}_{\mathcal{Y}} = \mathrm{E}\{y\} + R_{\mathrm{y}\mathcal{Y}} R_{\mathcal{Y}\mathcal{Y}}^{-1} (\mathcal{Y} - E\{\mathcal{Y}\})$$

将上面两个式子带入到e的表达式中：

$$e = x - \mathrm{E}\{\mathrm{x}\} - R_{x\mathcal{Y}} R_{\mathcal{Y}\mathcal{Y}}^{-1} (\mathcal{Y} - E\{\mathcal{Y}\}) - R_{xy|\mathcal{Y}} R_{y|\mathcal{Y}}^{-1} (y - \mathrm{E}\{y\} - R_{y\mathcal{Y}} R_{\mathcal{Y}\mathcal{Y}}^{-1} (\mathcal{Y} - E\{\mathcal{Y}\}))$$

进一步再分析e与其它变量的相关函数，容易验证

$$\mathrm{E}\{e(\mathcal{Y} - E\{\mathcal{Y}\})^T\} = 0$$

此外还根据 $R_{xy|\mathcal{Y}} = R_{xy|\mathcal{Y}} R_{y|\mathcal{Y}}^{-1} R_{y|\mathcal{Y}}$ 和 $e \triangleq x - \hat{x}_{\mathcal{Y}} - R_{xy|\mathcal{Y}} R_{y|\mathcal{Y}}^{-1} (y - \hat{y}_{\mathcal{Y}})$ 得到

$$\mathrm{E}\{e(\mathrm{y} - \hat{y}_{\mathcal{Y}})\} = 0.$$

引理3的证明：

进一步由 $\mathrm{E}\{e(\mathcal{Y}-E\{\mathcal{Y}\})\}=0$ 和 $\mathrm{E}\{e(\mathrm{y}-\hat{y}_y)\}=0$ 以及

$$y-E\{y\}=\mathrm{y}-\hat{y}_y-R_{\mathrm{y}\mathcal{Y}}R_{\mathcal{Y}\mathcal{Y}}^{-1}(\mathcal{Y}-E\{\mathcal{Y}\}))$$

可得

$$\mathrm{E}\{e(\mathrm{y}-\mathrm{E}\{y\})\}=0,$$

从而有$R_{ew}\triangleq\mathbb{E}\{e(w-\mathbb{E}\{w\})^T\}=\mathbf{0}$. 设$\hat{e}_w$为w对e的线性最小方差估计，则根据引理1知$\hat{e}_w=0$.

根据e的定义有：

$$x=\mathrm{e}+\hat{x}_y+K(y-\hat{y}_y),$$

设$\hat{x}_w$为w对x的线性最小方差估计，则由引理2可得：

$$\hat{x}_w=\hat{e}_w+\hat{x}_y+K(y-\hat{y}_y)=\hat{x}_y+K(y-\hat{y}_y).$$

证毕。

定理的证明

记$\mathcal{Y}_k \triangleq \begin{bmatrix} Y_0^T & Y_1^T \cdots & Y_k^T \end{bmatrix}^T$对$X_k, X_{k+1}$和$Y_{k+1}$的线性无偏最小方差估计分别为$\hat{X}_k, \bar{X}_{k+1}$和$\bar{Y}_{k+1}$, 则根据引理2可得$\bar{X}_{k+1} = A\hat{X}_k + Bu_k$, $\bar{Y}_{k+1} = C\bar{X}_{k+1}$. 同时记$\hat{X}_k$和$\bar{X}_{k+1}$的估计误差协方差阵为$P_k$和$\bar{P}_{k+1}$, 则$\bar{P}_{k+1} = AP_kA^T + Q$. 进一步地，将**引理3中的$x$,$\mathcal{Y}$和$y$分别替换成$X_{k+1}$, $\mathcal{Y}_k$和Y_{k+1}**, 可得

$$\hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}(Y_{k+1} - C\bar{X}_{k+1})$$

$$K_{k+1} = \mathbb{E}\{(X_{k+1} - \bar{X}_{k+1})(Y_{k+1} - \bar{Y}_{k+1})^T\}\left(\mathbb{E}\{(Y_{k+1} - \bar{Y}_{k+1})(Y_{k+1} - \bar{Y}_{k+1})^T\}\right)^{-1}$$

$$= \bar{P}_{k+1}\, C^T\left(C\bar{P}_{k+1}C^T + R\right)^{-1},$$

$$P_{k+1} = (\mathbf{I} - K_{k+1}C)\bar{P}_{k+1}(\mathbf{I} - K_{k+1}C)^T + K_{k+1}RK_{k+1}$$

$$= \bar{P}_{k+1} - K_{k+1}C\bar{P}_{k+1}.$$

定理的证明（续）

$$\begin{cases} X_{k+1} = AX_k + Bu_k + w_k \\ Y_{k+1} = CX_{k+1} + n_{k+1} \end{cases}$$

$k = 0,1,2,\ldots$

记$Y_{k(n)} \triangleq \left[Y_{k-n+1}^T \quad Y_{k-n}^T \quad \cdots \quad Y_k^T\right]^T$, $M \triangleq$ $[C^T \quad (CA)^T \quad \cdots \quad (CA^{n-1})^T]^T$, 由于$(A,C)$能观，亦即$M$是列满秩的，从而**可构造如下估计**（先假设$u_k$ =0,≠0时的证明类似）

$$\tilde{X}_k = \begin{cases} A^k \mathbb{E}\{x_0\}, & k < n \\ A^{n-1}\left(M^T M\right)^{-1} M^T Y_{k(n)}, & k \geq n \end{cases}$$

易得**$\tilde{X}_k$的估计误差协方差阵$\tilde{P}_k$是有界的**。

由于$\tilde{P}_k$是最小方差的线性估计，从而$P_k \leq \tilde{P}_k$也是有界的。

概要

- 研究背景和问题描述
- 卡尔曼滤波算法及其主要特性
- 扩展卡尔曼滤波算法

非线性系统的滤波问题：

$$\begin{cases} X_{k+1} = f(X_k) + w_k \\ Y_{k+1} = g(X_{k+1}) + v_{k+1} \end{cases}$$

$$k = 0, 1, 2, \ldots$$

X_k : state, Y_k : measured output,

$(f(\cdot), g(\cdot))$: known C^1 functions

$(\{n_k\}, \{w_k\})$: uncorrelated zero-mean stochastic noise

Objective: linear minimum variance estimation, i.e.,

$$\hat{X}_k = \underset{Z_k \in \{\text{linear function of } Y_i, 1 \le i \le k\}}{\arg\min} E\left\{(Z_k - X_k)(Z_k - X_k)^T\right\}$$

对付非线性的主要方法：线性化

线性化
linearization

$+$

$\begin{cases} \text{Kalman Filter, Kalman.} \\ \text{H-infinity Filter, Banavar, etc.} \\ \text{Set-valued Filter, Zhu, etc.} \\ \ldots \end{cases}$

线性系统的卡尔曼滤波

$$\begin{cases} X_{k+1} = AX_k + BF(X_k, k) + w_k \\ Y_{k+1} = CX_{k+1} + n_{k+1} \end{cases}$$

$k = 0, 1, 2, \ldots$

X_k : state, Y_k : measured output

(A, B, C): known matrices

$F(X_k, k)$: nonlinear dynamics

卡尔曼滤波算法

$$\begin{cases} \hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}(Y_{k+1} - C\bar{X}_{k+1}) \\ \bar{X}_{k+1} = A\hat{X}_k + Bu_k \\ K_{k+1} = (\bar{P}_{k+1}C^T)(C\ \bar{P}_{k+1}C^T + R)^{-1} \\ P_{k+1} = \bar{P}_{k+1} - K_{k+1}C\ \bar{P}_{k+1} \\ \bar{P}_{k+1} = AP_k A^T + Q \end{cases}$$

Parameters design:

$$\begin{cases} Q_k = E\left(w_k w_k^T\right) \\ R_{k+1} = E\left(n_{k+1} n_{k+1}^T\right) > 0 \\ P_0 = E\left(X_0 - EX_0\right)\left(X_0 - EX_0\right)^T \\ \bar{X}_0 = EX_0 \end{cases}$$

Require the exact information of : linear model
noise's model
initial value's model

扩展卡尔曼滤波

$$\begin{cases} X_{k+1} = f(X_k) + w_k \\ Y_{k+1} = g(X_{k+1}) + v_{k+1} \end{cases}$$

$k = 0,1,2,\ldots$

X_k : state, Y_k : measured output,

$(f(\bullet), g(\bullet))$: known C^1 functions

$(\{v_k\},\{w_k\})$: uncorrelated zero-mean

$$A_k = \left.\frac{\partial f}{\partial X}\right|_{\hat{X}_k},$$

第k+1步：
在滤波值点的线性部分；

$$C_{k+1} = \left.\frac{\partial g}{\partial X}\right|_{\bar{X}_{k+1}}$$

第k+1步：
在预测值点的线性部分

扩展卡尔曼滤波

$$\begin{cases} X_{k+1} = f(X_k) + w_k \\ Y_{k+1} = g(X_{k+1}) + v_{k+1} \end{cases}$$
$$k = 0,1,2,\ldots$$

X_k : state, Y_k : measured output,

$(f(\bullet), g(\bullet))$: known C[1] functions

$(\{v_k\},\{w_k\})$: uncorrelated zero-mean

$$\begin{cases} \hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}\left(Y_{k+1} - g\left(\bar{X}_{k+1}\right)\right) \\ \bar{X}_{k+1} = F\left(\hat{X}_k, k\right) \\ K_{k+1} = (\bar{P}_{k+1}\bar{C}_{k+1}^T)(\bar{C}_{k+1}\bar{P}_{k+1}\bar{C}_{k+1}^T + R_{k+1})^{-1} \\ P_{k+1} = \bar{P}_{k+1} - K_{k+1}\bar{C}_{k+1}\bar{P}_{k+1} \\ \bar{P}_{k+1} = \bar{A}_k P_k \bar{A}_k^T + Q_k \end{cases}$$

$$A_k = \left.\frac{\partial f}{\partial X}\right|_{\hat{X}_k}, C_{k+1} = \left.\frac{\partial g}{\partial X}\right|_{\bar{X}_{k+1}}$$

Parameters design:

$$\begin{cases} Q_k = E\left(w_k w_k^T\right) \\ R_{k+1} = E\left(n_{k+1} n_{k+1}^T\right) > 0 \\ P_0 = E\left(X_0 - EX_0\right)\left(X_0 - EX_0\right)^T \\ \bar{X}_0 = EX_0 \end{cases}$$

Utilizing the linear part at the point of estimation value

$$\begin{cases} X_{k+1} = f(X_k) + w_k \\ Y_{k+1} = g(X_{k+1}) + v_{k+1} \end{cases}$$

$$k = 0, 1, 2, \ldots$$

Reif K., et al., Stochastic stability of the discrete-time extended Kalman filter, IEEE TAC, 1999

$$\begin{cases} \hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}\left(Y_{k+1} - g\left(\bar{X}_{k+1}\right)\right) \\ \bar{X}_{k+1} = F\left(\hat{X}_k, k\right) \\ K_{k+1} = (\bar{P}_{k+1}\bar{C}_{k+1}^T)(\bar{C}_{k+1}\bar{P}_{k+1}\bar{C}_{k+1}^T + R_{k+1})^{-1} \\ P_{k+1} = \bar{P}_{k+1} - K_{k+1}\bar{C}_{k+1}\bar{P}_{k+1} \\ \bar{P}_{k+1} = \bar{A}_k P_k \bar{A}_k^{\,T} + Q_k, \qquad A_k = \left.\frac{\partial f}{\partial X}\right|_{\hat{X}_k}, C_{k+1} = \left.\frac{\partial g}{\partial X}\right|_{\bar{X}_{k+1}} \end{cases}$$

AMSS

EKF算法的主要理论结果

Conditions:
- ➢ **Exact information of nonlinear model**
- ➢ **Sufficiently small noise and initial error**
- ➢ **Uniformly observable condition**

稳定性: $\sup_k \left\| E\left\{\left(\hat{X}_k - X_k\right)\left(\hat{X}_k - X_k\right)^T\right\}\right\| < \infty$

最优性？ 一致性？

谢谢！